# PACKAGE OF PRACTICES FOR *KHARIF* CROPS OF ASSAM

2025

অসম চৰকাৰ

সত্যমেব জয়তে

GOVERNMENT OF ASSAM

*Published jointly by*

**Assam Agricultural University, Jorhat &**

**Department of Agriculture, Govt. of Assam**

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**Atul Bora**
**Minister**
**Agriculture, Horticulture, Excise, Border Protection & Development, Implementation of Assam Accord.**

सत्यमेव जयते

**GOVERNMENT OF ASSAM**



# *MESSAGE*

With its rich natural potential and crop diversity, Agriculture plays a vital role in the Assam's State Economy, as 70% - 80% of the population is dependent on agriculture directly or indirectly for their livelihood, but Assam's economy is diversifying beyond agriculture into industry, services, tourism, and trade in recent times. In spite of various challenges, Assam has been witnessing steady progress in terms of economic, social, and infrastructural development in the recent years. Agricultural development in terms of diversification, and modernization through the implementation of sustainable, cost- effective and scientifically-driven agricultural practices, Assam has been witnessing tremendous improvement not only in increasing production and productivity of crops but also farm income, and rural livelihoods.

I am happy to know that the Assam Agricultural University, Jorhat jointly with the Department of Agriculture, Government of Assam is going to publish the "***Package of Practices for Kharif/Rabi/Horticultural Crops of Assam, 2025***" **along with the** "***Organic PoP for selected Crops of Assam, 2025*** **and** ***Climate Resilient Technologies Recommended for Assam, 2025***" with the inclusion of recently developed technologies pertaining to crop improvement, management and protection, which will provide impetus to the agricultural growth and development in the State.

I sincerely believe that this Package of Practices, 2025 will be a great help for our extension machineries and farming community to boost up production and productivity of major crops in the State.

I convey my best wishes on this auspicious occasion.

Date : 23rd February, 2026

**( Atul Bora )**

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## MESSAGE

Agriculture and allied sector is the backbone of our State economy and contributes approximately 22.10 % of total Gross State Value Added (GSVA) with a positive growth rate of 16.16% (FY 2022-23). Agriculture has been supporting the majority of the population for its livelihood and income generation. The fertile plains of the Brahmaputra and Barak valleys make the State naturally and highly suitable for agriculture, which is primarily dominated by cultivation of rice, tea, oilseeds and pulses alongside a number of horticultural crops. Thus, a rapid growth of agriculture could help not only in achieving self-reliance in food grains but also in attaining food security and reducing poverty in the State.

I am happy to know that the Assam Agricultural University, Jorhat jointly with the Department of Agriculture, Government of Assam is publishing the "*Package of Practices (PoP) for Kharif, Rabi, and Horticultural Crops of Assam, 2025*" along with the "*Organic PoP for selected Crops of Assam, 2025* and *Climate Resilient Technologies Recommended for Assam, 2025*" which includes 99 recently developed technologies pertaining to crop improvement, management and protection.

I am confident that the *Package of Practices 2025* shall serve as a useful handbook to empower our extension functionaries and farming community of the State towards enhanced agricultural production and productivity.

With best wishes,

**Smt. Aruna Rajoria, IAS**
Commissioner & Secretary to the Govt. of Assam,
Department of Agriculture
& Agriculture Production Commissioner, Assam
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(Recipient of Sardar Patel Outstanding Institution Award)



## Message

I am immensely happy to learn that as an outcome of the workshop on Package of Practice 2025, a total of 99 technologies for crop improvement, management, and protection were recommended for inclusion in the updated '*Package of Practices for Kharif, Rabi and Horticultural Crops, 2025, Organic Package of Practices, 2025 and Climate Resilient Technologies Recommended for Assam, 2025'*. The workshop was jointly organized on 6th September,2025 by Assam Agricultural University and the Department of Agriculture, Govt. of Assam.

With the inclusion of 99 technologies, the updated package of practices will boost the agricultural production in quality and quantity in different climatic conditions and adversaries. Besides the inclusion of new technologies, several existing technologies have also been refined so that farmers can rely upon them for climate resilient agriculture in particular.

I sincerely believe that the technologies so generated through the dedicated services of our scientists shall not only help in knowledge upgradation of the farmers, scientists, extension functionaries and other stakeholders, but also overall improvement in agricultural scenario of the state.

I record my sincere thanks and appreciation to the team of scientists associated with the preparation of the manuscript.

(Bidyut C. Deka)

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## IMPORTANT ABBREVIATIONS USED IN THIS HANDBOOK

**Agroclimatic Zones**

N = North Bank Plains Zone
U = Upper Brahmaputra Valley Zone
L = Lower Brahmaputra Valley Zone
C = Central Brahmaputra Valley Zone
H = Hill Zone
B = Barak Valley Zone

**Reaction to pests & diseases**

R = Resistant
MR = Moderately Resistant
S = Susceptible
MS = Moderately Susceptible
T = Tolerant

**Others**

FYM = Farmyard Manure
HYV = High Yielding Variety
*a.i.* =Active ingredient
RHSDRB = Rice Husk Saw Dust Rice Bran

# CEREALS

## Rice (*Oryza sativa*)

### Normal *Ahu* (Autumn rice) (Direct seeded)

**Varieties**:

| Varieties | Year of notification | *Agroclimatic zone | Duration (days) | Av. Plant height (cm) |
|---|---|---|---|---|
| **A. **Semi Dwarf** | | | | |
| IR-36 | 1982 | N,C,L,H | 110-120 | 85 |
| **B. Tall** | | | | |
| Banglami | - | N,L | 115-120 | 100-140 |
| Rangadoria | - | N,C,L | 115-120 | -do- |
| Maibee | - | H | 90-100 | -do- |
| Dimrou | - | H | 90-100 | -do- |
| Arun | - | All zones | 110 | 110-120 |

* Refer to page iv for full forms of agroclimatic zones

** Durations of semi-dwarf varieties are based on experiments conducted at Titabar. Durations of tall varieties are collected from the Field Trial Stations of respective zones.

*** Not recommended for blast endemic areas.

**Varieties recommended (for direct seeded, rainfed upland/*Jhum* land situations of Hills Zone only):**

| Variety | Year of release | Sowing time | Duration (days) | Fertilizer (N:P:K) dose (kg/ha) | Grain Yield (t/ha) | Reaction to insect pests & diseases |
|---|---|---|---|---|---|---|
| Maizubiron | - | April-May | 115 | 20:10:10 | 2.7 | Moderately resistant to blast and BLB and durable field resistance to stem borer, leaf folder, case worm and rice bug |
| Inglongkiri | 2017 | -do- | 110 | -do- | 3.4 | Resistant to blast and moderately resistant to stem borer, case worm and leaf folder |

| | | | | | | |
|---|---|---|---|---|---|---|
| Dehangi | 2017 | -do- | 115 | -do- | 3.3 | -do- |
| Rongkhang | 2017 | -do- | 115 | -do- | 3.3 | -do- |
| Haccha | 2012 | -do- | 115 | -do- | 3.5 | Moderately resistant to Leaf blast, Neck blast, Brown spot, Sheath rot, Stem borer, Leaf folder, Plant hopper under natural condition |

**Land selection**:

Areas having surface soil with medium to sandy texture overlying a silty clayey subsoil should preferably be selected.

**Seed Selection**:

Put the seeds in plain water and stir well. Select the sunken seeds rejecting the floating ones.

**Sowing Time**:

Seeds should be sown in March and April

**Dry Seed Treatment**:

Put seeds in a container and add any of the following fungicides according to the recommendation. Mix the fungicide thoroughly with seeds by agitating them for five minutes.

| **Fungicide** | **Dose (g/kg seed)** |
|---|---|
| Carboxin | 2.0 |

**Field Preparation**:

Prepare the land by 3 to 4 ploughings followed by laddering. Laddering is to be done properly to retain water uniformly in the field. FYM should be applied during initial field preparation.

**Manures and fertilizers**:

Compost or FYM @ 10 t/ha or 15 q/bigha should be applied.

| **Nutrient** | **Requirement (kg/ha)** | **Form** | **Fertilizer requirement** | |
|---|---|---|---|---|
| | | | **kg/ha** | **kg/bigha** |
| **A. Semi dwarf varieties** | | | | |
| N | 40 | Urea | 88 | 12 |
| $P_2O_5$ | 20 | SSP | 125 | 17 |
| $K_2O$ | 20 | MOP | 32 | 4 |

| **B. Tall varieties** | | | | |
|---|---|---|---|---|
| N | 20 | Urea | 44 | 6 |
| $P_2O_5$ | 10 | SSP | 62 | 9 |
| $K_2O$ | 10 | MOP | 17 | 3 |

Nutrient recommendation for semi-dwarf varieties is 30 : 30 : 20 kg/ha N, $P_2O_5$, $K_2O$ in the North Bank Plain Zone. Granulated mixed fertilizer at appropriate doses can also be applied.

Diammonium phosphate (DAP) in combination with rock phosphate or alone at the recommended level of nutrients (40 : 20 : 20) can be applied as a substitute for SSP and MRP or their combinations as an economic source of phosphate. For Hill Zone reduction of 50% chemical fertilizer by incorporating 10 t of FYM/ha is recommended.

**Time of Application of Fertilizers:**

1. Apply a full dose of phosphatic fertilizer at the time of final ploughing.
2. Apply half of nitrogenous and half of potassic fertilizer 15-25 days after germination or after first weeding.
3. The second top dressing with the remaining N and $K_2O$ is to be done 40-50 days after germination or after the second weeding

Apply 75% of RDF along with 1 t/ha vermicompost in each crop in direct seeded ahu rice-green gram-Toria cropping system in rainfed upland situation.

**Seed Rate**:

- **Line Sowing**: Sow seeds in lines with inter-row spacing of 20 cm @ 75 kg/ha (10-12 kg/ bigha)
- **Broadcasting**: Sow seeds @ 105-110 kg/ha (14 kg/bigha).

**Interculture**:

1. For weeding, use preferably wheel hoe or dry land weeder or *bindha* for laddering after 2 to 3 weeks from sowing. The second weeding should be done with wheel hoe or dry land weeder at 4-5 weeks from sowing. If wheel hoe or dry land weeder is not available, give manual weeding. Weeding should precede fertilizer application.
2. As chemical weed control measures apply Pretilachlor @ 0.75 kg/ha at 3 DAS followed by grubber at 30 DAS. For Hills Zone, application of Pretilachlor as pre-emergence weedicide @ 1.5 kg a.i./ha one day after sowing followed by two hand weedings at 35 and 50 days after sowing is recommended for rainfed upland rice.

**Irrigation in aerobic rice:**

In upland *ahu* rice, 5 cm irrigation should be applied in at 10-13 days interval.

**Plant Protection Measures**:

**A) Insect Pests:**

Plant protection measures should be adopted against insect pests at their economic threshold as given in Table 1.

To control rice pests, erect 50 'T'-perches per ha 2 ft (60 cm) above crop canopy as roosting site for insectivorous birds, which are to be removed before flowering in order to prevent activity of granivorous birds

**B) Root-knot Nematodes:** Apply *Pseudomonas fluorescens* @ 20 g/sq. m at the time of sowing.

**C) Diseases:**

**i) Blast:**

a) Grow tolerant varieties, such as Govind, Cauvery, IR 36

b) Treat the seeds as mentioned under seed treatment.

c) Monitoring of blast is important to schedule spraying. Observe the top five leaves and if 5% leaf area is damaged, resort to spraying. Normally three sprays can control the disease. Spray Hexaconazole 5EC @ 2g/*l* of water at tillering stage (40-55 days after sowing) and subsequently give two more sprays of ediphenphos @ 1 ml/*l* of water, one at panicle initiation stage and the other when the tip of the panicle just comes out.

**ii) Sheath blight disease**:

For control of sheath blight disease – spraying of two commercial plant-derived cymbopogon products *viz*. cymbopogon product 20EC @ 5 ml/*l* and neem product @ 3 ml/*l* is recommended for management of sheath blight disease of rice. The first spray should be given as soon as symptoms of the disease are observed in the field followed by a second spraying at a 10 days interval.

**iii) Bacterial leaf blight (BLB):**

a) Grow tolerant varieties, such as Govind and IR 36.

b) Avoid top dressing with nitrogen at panicle initiation stage if BLB has already appeared. Instead, top dress with 10 kg of $K_2O$ /ha or apply 5 kg of $K_2O$/ha in the form of foliar spray of 3% solution.

**iv) Brown spot disease:**

Dry (2 g/kg of seed) or wet (2 g/kg of seed/*l* of water) seed treatment with carboxin followed by one spraying of or ediphenphos @ 1 ml/*l* at initial symptom development stage.

**v) Bakanae Disease**:

a) Soak seeds for 24 hours with carboxin 2 g/kg seed/*l* of water.

b) Rogue out the infected tillers (elongated, from the infected hills in the main field).

**vi) Stem rot and Sheath rot**

Spray 2 rounds of tebuconazole 25.9 EC @ 1.5 ml/*l* at 15 days interval starting just after appearance of the disease to reduce Sheath Rot and Stem Rot incidence of rice.

**Pre-harvest Treatment on Standing Crop for Better Grain Quality**:

Harvest of *ahu* crop usually coincides with rainy weather. Dis-colouration as well as sprouting of seeds in the panicle itself or after harvest is due to high moisture content of the seeds. To overcome such problems, spraying of diquat 0.05% or paraquat 0.1% or common salt (NaCl) 10% should be done on the ear head @ 1000 liter/ha in terms of chemical solution at 20-25 days after 50% flowering. These chemicals enhance maturity by 5-7 days.

**Table 1. Damage identification and Economic Threshold Levels (ETLs) for different insect pests of rice and their chemical control**

| Crop stage and damage identification | Key pest | ETL | Insecticide (Technical Name) | Dose |
|---|---|---|---|---|
| **A. Nursery** | | | | |
| Curling of leaves, leaf blade rolls, yellowish to reddish discolouration | Thrips | Moderate to severe | Imidacloprid 70WG<br>Thiamethoxam 25WG | 24.5 g *a.i.* /ha or 0.3 g/*l*<br>25 g *a.i.*/ha or 0.03 g/*l* |
| Yellowing and withering of plants | Leaf & plant hoppers | Moderate to severe | Imidacloprid 70WG<br>Thiamethoxam 25WG | 24.5 g *a.i.*/ha or0.3 g/*l*<br>25 g *a.i.*/ha or 0.03 g/*l* |
| Presence of dead heart | Stem borer | Moderate to severe | Fipronil 5SC<br>Chlorantraniliprole 18.5 SC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>150 ml in 500 liters of water per ha (20 ml in 67 liters of water for 1 bigha) at ETL and 20 days after first spraying |
| **B. Main field (After transplantation)** | | | | |
| Upon unfolding, the edge of the central leaf shows discoloured (yellowish to whitish) Patches | Whorl maggot | More than 20% damaged hills upto 30 days after transplanting | Fipronil 5 SC | 50 g *a.i.*/ha or 1.5-2 ml/*l* |
| Leaf tissues scrapped in white parallel lines | Hispa | 1 adult or 1 damaged leaf/hill | Lambda-cyhalothrin 5EC<br>Emamectin benzoate 1.9EC | 12.5 g *a.i.*/ha<br>8.08 g *a.i.*/ha |
| Presence of dead heart | Stem borer | 1 egg mass/sq.m or 5% dead hearts | Fipronil 5 SC<br>Chlorantraniliprole 18.5 SC<br>Cartap hydrochloride 50 SC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>150 ml in 500 liters of water per ha (20 ml in 67 liters of water for 1 bigha) at ETL and 20 days after first spraying<br>2 ml/*l* of water at 30 and 50 days after transplanting |

| | | | | |
|---|---|---|---|---|
| Yellowing and withering of plants | Leaf and plant hoppers | 10 insects/hill<br>or 2 insects/hill in *tungro* endemic areas | Imidacloprid 70WG<br>Thiamethoxam 25WG | 24.5 g *a.i.*/ha or 0.3 g/*l*<br>25 g *a.i.*/ha or 0.03 g/*l* |
| Presence of silver shoots (galls) | Gall midge | 1 silver shoot/<br>sq.m in endemic areas or 5% silver shoots in non-endemic areas | Fipronil 5 SC<br>Lambda-cyhalothrin 5EC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>12.5 g *a.i.*/ha |
| **C. Active tillering stage** | | | | |
| Presence of tubular case by cutting the tips of the leaves, leaf tissues scrapped in white patches | Case worm | 1-2 cases/hill | Chlorantraniliprole 18.5SC | 30 g *a.i.*/ha |
| Leaves are folded along margins by webbing them together | Leaf folder | More than 1 damaged leaf per hill | Fipronil 5SC<br>Chlorantraniliprole 18.5 SC<br>Cartap hydrochloride 50 SC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>150 ml in 500 liters of water per ha (20 ml in 67 liters of water for 1 bigha) at ETL and 20 days after first spraying<br>2 ml/*l* of water at 30 and 50 days after transplanting |
| Presence of silver shoots (galls) | Gall midge | 1 silver shoot/sq.m. in<br>endemic areas or 5% silver shoot in non-endemic areas | Fipronil 5SC<br>or<br>Lambda-cyhalothrin 5EC | 50 g *a.i.*/ha or1.5-2 ml/*l*<br>or<br>12.5 g *a.i.*/ha |
| Leaf tissues scrapped in white parallel lines | Hispa | 1 adult or 1-2 damaged leaves per hill | Lambda-cyhalothrin 5EC<br>Emamectin benzoate 1.9EC | 12.5 g *a.i.*/ha<br>8.08 g *a.i.*/ha |
| Presence of dead heart | Stem borer | More than 5%<br>dead heart | Fipronil 5 SC<br>Chlorantraniliprole 18.5 SC<br>Cartap hydrochloride 50 SC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>150 ml in 500 liters of water per ha (20 ml in 67 liters of water for 1 bigha) at ETL and 20 days after first spraying<br>2 ml/*l* of water at 30 and 50 days after transplanting |
| **D. Boot leaf to heading stage** | | | | |
| Presence of partially or completely chaffy grains in the panicle | Rice bug | 1-2 bugs/sq.m | NSKE 1500 ppm | 3-5 ml/*l* |
| Presence of white Earhead | Stem borer | 1 moth/sq.m. | Fipronil 5 SC<br>Chlorantraniliprole 18.5SC | 50 g *a.i.* /ha or 1.5-2 ml/*l*<br>30 g *a.i.*/ha |

| | | | | |
|---|---|---|---|---|
| Larvae climb and cut the earhead | Ear cutting caterpillar | 1 larva/hill | Fipronil 5 SC<br>Chlorantraniliprole 18.5SC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>30 g *a.i.*/ha |
| Presence of shed grains or panicles at the base of the plant | Climbing cutworm | 1 larva/hill | Fipronil 5 SC<br>Chlorantraniliprole 18.5SC | 50 g *a.i.*/ha or 1.5-2 ml/*l*<br>30 g *a.i.*/ha |

# Rice *(Oryza sativa)*

## Transplanted Normal *Ahu* Rice (Autumn Rice)

**Varieties**

| Varieties | Year of notification | *Agroclimatic Zone | Duration (days) | Av. Plant height (cm) |
|---|---|---|---|---|
| **Semi-dwarf:** | | | | |
| IR 36 | 1982 | N,U,L,B,H | 130-140 | 105 |
| Lachit | 1992 | U | 125-130 | 95 |
| Chilarai | 1992 | U, B | 125-130 | 95 |
| Sonamukhi | - | B | 100-110 | 85 |
| Luit | 1997 | B | 100-110 | 110-115 |
| Haccha | 2022 | H | 115-120 | - |
| Dishang | 1998 | N,U,C, L,B | 90-95 | 110-115 |

* Refer to page iv for full forms of the agroclimatic zones
** Durations are based on experiments conducted at Titabar.
*** Not recommended for blast endemic areas.

**Land Selection:**

Areas with assured irrigation facilities only should be selected. Heavy to medium textured soils are preferred.

**Seed Selection:**

Seeds are put in plain water, stirred well, the sunken seeds are selected and the floated ones are to be rejected.

**Sowing in Nursery Bed:**

Seed should be sown in the nursery bed during March-April.

**Seed Treatment:**

**Wet Method:** After seed selection, the seeds should be soaked directly in any of the following fungicidal suspensions for 24 hrs. One liter of fungicidal solution is required to treat 1 kg of seed.

| Fungicide | Dose (g/kg seed) |
|---|---|
| Carboxin | 2.0 |

**Raising of Seedlings:**

a) **Preparation of seedbed:** Land has to be thoroughly puddled and seedbed of 10 m length and 1.25 m breadth is to be prepared with 30 cm gap in between beds.

**b) Manures and fertilizers:** In each seedbed, 20-30 kg cow dung/compost, 80 g urea, 80 g SSP and 40 g MOP are to be applied.

**c) Seed rate:** Well germinated seeds are to be sown @ 650 g to 1kg per bed. Seed requirement for transplanting one hectare of main field is 40 to 45 kg. In BPH endemic areas a lower seed rate is advocated.

**d) Water Management:** Irrigation water is to be supplied in furrows to maintain saturated condition in the surface soil of the nursery bed. However, standing water to a depth of 2-3 cm should be maintained at least 2-3 days before uprooting.

**e) Plant protection in seedbed:**

i) As soon as one or two blast spots are seen, or ediphenphos @ 1 ml/*l* of water is to be sprayed.

ii) Against root-knot nematode, apply, *Pseudomonas fluorescens* @ 20g / m2 at the time of sowing

iii) For control of nursery insect pests any one of the following insecticides is to be sprayed as and when necessary. Generally, an insecticidal spray at 5-7 days after sowing is effective against most pests.

| **Insecticide** | **Dose** |
|---|---|
| Imidacloprid 70WG | 24.5g *a.i.*/ha or 0.3g/*l* |
| Thiamethoxam 25WG | 25g *a.i.*/ha or 0.3 g/*l* |
| Fipronil 5SC | 50g *a.i.*/ha or 1.5-2 ml/*l* |

**High volume spray**: 40 ml of water/sq. m

**Low volume spray:** 13 ml of water/sq. m

**Field Preparation:**

1. One ploughing should be given at least 21 days prior to transplanting. An irrigation for land soaking should be applied before preparatory tillage.
2. Secondary irrigation should be applied at 10-12 days prior to transplanting followed by ploughing, laddering and puddling, accompanied by repairing and mud plastering of bunds. Thereafter, another irrigation should be applied for land submergence.
3. The final puddling should be done 4-5 days prior to transplanting. One irrigation should be applied before final puddling (depth of each irrigation: 5-7 cm).

**Manures and Fertilizers:**

Well rotten FYM or compost has to be applied @ 10 t/ha in addition to the fertilizers at rates given below in areas with moderate fertility level.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| **Semi dwarf varieties:** | | | | |
| N | 40 | Urea | 88 | 12 |
| $P_2O_5$ | 20 | SSP | 125 | 17 |
| $K_2O$ | 20 | MOP | 32 | 4 |

Above rates of nutrients will be valid for most of the rice growing areas of Assam. In case of poor soil, the rates of nutrients may be raised to the extent of 60 : 30 : 30 kg/ha N, $P_2O_5$ and $K_2O$ respectively. Granulated mixed fertilizers at appropriate doses can also be applied. Di-Ammonium Phosphate (DAP) in combination with rock phosphate or alone at the recommended level of nutrients (40 : 20 : 20) can be applied as a substitute for SSP and MRP or their combinations as an economic source of phosphate.

**Time of application of fertilizers:**

i) Only 1/3$^{rd}$ of the total urea, full doses of super phosphate and potash at the time of final puddling should be applied.

**i)** The second 1/3$^{rd}$ and third 1/3$^{rd}$ doses of urea should be applied at tillering and panicle initiation stages respectively. Top dressing of urea should be preceded by weeding.

**ii)** Super phosphate can also be incorporated into the soil at the active tillering stage (25-35 days after transplanting) along with the second dose of nitrogenous fertilizer.

**Use of Azolla to supplement nitrogenous fertilizers:**

Azolla can be used as supplement to nitrogenous fertilizers. About 300 kg fresh Azolla/ha is to be inoculated in the main field, pounded with 5-10 cm depth of water for about 2-3 weeks prior to final puddling. At the time of inoculation, super phosphate @ 8-10 kg/ha is to be applied in the field. Thereby, the corresponding quantities of phosphatic fertilizer should be reduced at the time of transplanting. Fresh Azolla @ 500 kg may also be applied in the standing water in a transplanted crop after the establishment of the seedling. There is no need for application of additional phosphatic fertilizer in the field at the time of inoculation with fresh Azolla after transplanting. Thereafter, Azolla will multiply to cover the field. Care should be taken to keep 5-10 cm depth of water for rapid multiplication. Normally sufficient quantities of biomass will be produced for two weeks.

Natural depression and ponds or tanks may be used for Azolla multiplication outside the main field and may be incorporated at the time of land preparation @ 6 t/ha of fresh Azolla.

Nitrogen dose can even be reduced up to 50% when Azolla is applied and incorporated in the field.

**Use of Biofertilizer:**

Application of organic manure @ 1 t/ha (on dry weight basis) along with mixed

inoculum of *Azospirillum amazonense* A-10 and *Bacillus megaterium* P-5 @ 4 kg/ha (0.4-0.5 kg/bigha), rock phosphate @ 10 kg $P_2O_5$per ha (rock phosphate 56 kg/ha or 7.5 kg/bigha) and muriate of potash @ 40 kg $K_2O$ per ha (muriate of potash 67 kg/ha or 9 kg/bigha) is recommended for rice in rice-rice, rice-legume-rice and sole rice sequences.

**Method of application of Biofertilizer:**

**A. Technique of application of the components if the INM package for transplanted rice:**

**Application of the organic manure and fertilizer component:** The required quantity of compost (approx. 1.5 q or 2 bullock cart loads per bigha should be applied at least a week before transplanting or at the time of final ploughing. Required quantity of potassium (40 kg $K_2O$ per ha or 5.5 kg per bigha) should be applied prior to transplanting. Rock phosphate component is used along with biofertilizer as slurry to treat the seedling roots.

**B. Treatment of rice seedling roots with the biofertilizer component and rock phosphate by slurry method:**

Prepare a pit in the corner of a plot on the morning of the transplanting day by simply raising a bund that touches the two already existing bunds to obtain a triangular shape. The size of the pit should be proportionate to place the quantity of seedlings required for a bigha of land. Excess water from the pit should be removed so that only mud remains. Now add 10 kg of dried compost to the required quantity of rock phosphate and mix with mud and then add the biofertilizer. The mixture of mud, compost, rock phosphate and biofertilizer should form uniform slurry. The rice seedling roots in the bundles should be free of adhered soil and dipped in the slurry mixture and kept for two hours. During the two hours time, the biofertilizer and rock phosphate adhere well to the seedling roots. Then remove the seedling bundles from the pit and keep on the raised bund. Care should be taken so that the seedlings are not swept in water of the plot as the Assamese farm women are habituated in occasional sweeping of the seedlings during transplanting. The biofertilizer on the root surface keep multiplying as the root grows bigger and by 20/30 days time they start supplying N, hormone and also soluble P to the roots in adequate quantity.

**Time of transplanting:** Transplanting should be done during April-May.

**Transplanting:**

Transplanting should be done with 3 seedlings per hill. Spacings should be 20 × 15 cm (33 hills /sq.m) for semi-dwarf and 20 × 20 cm (25 hills/sq.m) for tall traditional varieties. The wooden line marker of required spacing may be used for the same purpose. The depth of planting should be 4-5 cm.

**Gap-filling:**

Replanting of dead hills should be done within 7-10 days of transplanting with seedlings of same age.

**Water management:**

Application of 5 cm irrigation water 3 days after disappearance of ponded water is recommended in medium and heavy soils.

**Interculture:**

Preferably two weedings at 20 and 40 days after transplanting should be done. Alternatively, weeder can be used at the time of top dressing of nitrogenous fertilizer.

**Plant protection Measures in the field:**

**A) Insect Pests:**

Plant protection measures should be adopted against insect pests at their economic threshold level as given in Table-1.

To control rice pests, erect 50 'T'-perches per ha 2 ft (60 cm) above crop canopy as roosting site for insectivorous birds, which are to be removed before flowering in order to prevent activity of granivorous birds.

**B) Root-knot nematode:**

a) Apply, *Pseudomonas flourescens* @ 20g / sq. m at the time of sowing

b) Apply compost @ 1 ton/ha along with *Azospirillum* and PSB as root-dip treatment in rice

c) Apply *Bacillus subtilis* ($1x10^8$ cfu/g) @ 10 kg/ha before sowing to control rice root knot nematode (*Meloidogyne graminicola*). The bioagent should be enriched in well decomposed FYM (10 kg in 200 kg FYM) before application

**C) Diseases:** Same as in direct seeded *ahu*.

Pre-harvest treatment should be undertaken on standing crop for better grain quality (Same as in direct seeded normal ahu).

# Rice (*Oryza sativa*)

## *Sali* rice (Winter rice)

**Varieties**

| Varieties | Year of notification | *Agroclimatic zone | Duration (days) | Sowing time | Avg. plant height (cm) |
|---|---|---|---|---|---|
| **A. Transplanted normal *Sali*** | | | | | |
| **i) Semi dwarf** | | | | | |
| IR 36 | 1982 | U,C | 110-125 | July | 85 |
| AAU-Kmj Dhan-46 (Surma Dhan ) | - | N,U,C,L,B,H | 140-145 | July | 120 |
| Bahadur | 1994 | N,U,C,L,B,H | 150-155 | June | 114 |
| Piolee | - | N,U,C,L,B,H | 150-155 | do | 102 |
| Kushal | 1994 | N,U,C,L,B,H | 150-155 | do | 116 |
| Moniram | - | N,U,C,L,B,H | 150-155 | do | 105 |
| Ranjit | 1994 | N,U,C,L,B,H | 150-155 | do | 102 |
| AAU-TTB-Dhan 42 (Patkai) | 2023 | N,U,C,L,B,H | 150-155 | do | 120 |
| AAU-TTB-Dhan 43 (Shatabdi) | 2023 | N,U,C,L,B,H | 150-155 | do | 120 |
| AAU-TTB-Dhan 44 (Prachur) | 2023 | N,U,C,L,B,H | 150-155 | do | 125 |
| **ii) Sticky rice (*Bora-Chokuwa*)** | | | | | |
| **a) *Bora* rice** | | | | | |
| Bhogalee | 1994 | N,U,C,B | 155-160 | June | 99 |
| Aghoni | - | N,U,C,B | 160-165 | June | 105 |
| **b). *Chokuwa* rice** | | | | | |
| AAU-Diphu Dhan 6 (Diyung) | - | H | 140-145 | July | 105 |
| **iii) Multiple cropping situation** | | | | | |
| Satyaranjan | 1999 | N,U,L,B | 130-135 | do | 113 |
| Basundhara | 1999 | N,U,C,L | 130-133 | do | 107 |

| | | | | | |
|---|---|---|---|---|---|
| Mulagabhoru (TTB103-21-1) | - | N,U,C,L,B,H | 135 | do | - |
| Shraboni (TTB 404) | - | N,U,C,L,B,H | 135 | do | 125 |
| Numoli | 2020 | N,U,C,L,B,H | 130-135 | do | 125 |
| AAU-TTB Dhan 40 (Dholi) (submergence tolerant variety) | 2022 | N,U,C,L,B,H | 130-135 | June | 120 |
| **iv) Tall** | | | | | |
| Mahsuri | - | N,U,C,L,B,H | 140-145 | Mid-June | 130 |
| **v) Scented Rice** | | | | | |
| Keteki Joha | 2006 | N,U,C,B | 155-165 | June | 100-120 |
| Sugandhi | - | All zones | 160 | June | 130-140 |
| **vi) Purple Rice** | | | | | |
| Labanya | - | All zones | 145-150 | June | 115-120 |
| **vi) Drought tolerant Rice** | | | | | |
| Arun | - | All zones | 110 | June | 110-120 |
| Bina Dhan 17 | - | All zones | 125-130 | June | 100-105 |
| CR Dhan 802 | - | All zones | 150-155 | June | 105-110 |
| DRR Dhan 44 | - | All zones | 120-125 | June | 105-110 |
| **vii) Premium Quality Rice** | | | | | |
| DRR Dhan 69 (High zinc) | - | All zones | 135-140 | June | 105-110 |
| CR Dhan 311 (High protein and Zinc) | - | All zones | 135-140 | June | 120-125 |
| CR Dhan 310 (High protein) | - | All zones | 135-140 | June | 120-125 |
| **B. Post flood transplanted (late *Sali*)** | | | | | |
| Manohar Sali | 1969 | N,U,C,L,B | - | Mid Jul to1$^{st}$ week of Aug. | 130 |
| Solpona | - | N | - | do | 130-140 |
| Prasadbhog | - | N | - | do | do |
| Govindabhog | - | N | - | do | do |

| C. Direct Seeded late *Sali* | | | | | |
|---|---|---|---|---|---|
| Luit | 1997 | N,U,C,L,B | 90-95 | do | 110-115 |
| Kapilee | - | N,U,C,L,B | 90-95 | do | 110-115 |
| Dishang | 1998 | N,U,C,L,B | 90-95 | do | 110-115 |

* Refer to page iv for full forms of agroclimatic zones

**N.B**.

- IR 50 is recommended for double cropped medium land rice areas where Jaya is grown as *ahu* in Barak Valley Zone.
- Varieties with plant height more than 130cm are considered as tall varieties

**Rice varieties suitable for staggered planting in *sali* season with aged seedlings**

| Variety | Year of notification | Plant height (cm) | Duration (days) | Sowing time | Trans-planting time | Age of seedlings | Yield (t/ha) | Agroclimatic zone | Disease reaction |
|---|---|---|---|---|---|---|---|---|---|
| Prafulla | - | 134 | 150-160 | June | July-Sep | 30-75 | 4.5-5.0 | All except Hills zone | R to Sheath rot, MR to blast & BLB, S to sheath blight |
| Gitesh | 2017 | 130 | 150-160 | June | July-Aug | 30-60 | 5-5.5 | All except Hills zone | R to Sheath rot, MR to blast, T to BLB, S to sheath blight |

- Varieties with plant height more than 130cm are considered as tall varieties

**Rice varieties for waterlogged situation**

| Variety | Year of notification | Plant height (cm) | Duration (days) | Sowing Time | Transplanting time | Age of seedlings | Yield (t/ha) | Agroclimatic zone | Disease reaction |
|---|---|---|---|---|---|---|---|---|---|
| TTB 303-18-3 (Chakra Sali) | - | 135 | 138 | June | July | 30-35 | 4.76 | U,N, L,B | Resistant to brown spot; MR to blast & BLB; tolerant to sheath blight & stem borer |
| TTB 303-2-23 (Diphalu) | - | 140 | 137 | June | July | 30-35 | 5.11 | U,N, L,B | MR to blast, brown spot, BLB & Sheath blight; tolerant to stem borer |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| TTB 303-1-42 (Dhansiri) | - | 140 | 139 | June | July | 30-35 | 4.84 | U,N, L,B | MR to brown spot & sheath blight; tolerant to blast, BLB & stem borer |
| TTB 303-1-26 (Manah) | - | - | 138 | - | - | 30-35 | 4.66 | U,N, L,B | - |

**Submergence tolerant rice varieties for flash-flood situation**

| Variety | Year of notification | Plant height (cm) | Duration (days) | Sowing Time | Transplanting time | Age of seedlings | Yield (t/ha) | Agroclimatic zone | Disease reaction |
|---|---|---|---|---|---|---|---|---|---|
| Jalashree | 2017 | 133 | 150-155 | June | July | 30-35 | 4.0 | All except L& H | T to BLB & sheath blight MR to blast |
| Jalkunwari | 2017 | 138 | 150-155 | June | July | 30-35 | 4.0 | All except L & H | T to BLB & sheath blight MR to blast |
| Swarna Sub1 (IR 82809-237) | 2009 | 100 | 145-`150 | June | July | 25-30 | 4.5-5.0 | N,U,C,L, B,H | - |
| BR 11 Sub1 | - | - | 150 | - | - | - | 4.5-5.0 | N,U,C,L, B,H | - |
| Ranjit Sub1 | 2018 | 115 | 145-150 | June | July | 25-30 | 5.5-6.0 | N,U,C,L, B,H | Moderately resistant to neck blast; tolerant to stem borer and moderately resistant to BPH |
| Bahadur Sub1 | 2018 | 115 | 140-145 | June | July | 25-30 | 5.4-6.0 | N,U,C,L, B,H | Moderately susceptible to brown spot, bacterial leaf and sheath blight; tolerant to stem borer & moderately resistant to BPH |
| CR Dhan 802 | - | 105-110 | 150-155 | June | July | 25-30 | 4-4.5 | All zones | - |

- Varieties with plant height more than 130cm are considered as tall varieties
- *Refer to page iv for full forms of agroclimatic zone

**Rice varieties for low input situation:**

| Variety | Sowing Time | Duration (days) | Fertilizer ($N:P_2O_5:K_2O$) dose (kg/ha) | Grain Yield (t/ha) | *Agroclimatic zone | Reaction to insect pests and diseases |
|---|---|---|---|---|---|---|
| Gandhari (JR 1) | June 15-30 | 125 | 20:10:10 | 4.2 | C | R to blast, BLB, stem borer & rice bug; MR to sheath blight; MS to leaf folder |
| Mohan (JR 5) | -do- | 126 | -do- | 4.1 | C | R to blast, stem borer & rice bug; MR to sheath blight; MS to leaf folder |
| Srimanta (JR 2) | -do- | 138 | 20:10:10<br>40:20:20 | 4.4<br>5.1 | U,C,H | R to blast, stem borer; MR to BLB & sheath blight; MS to leaf folder |
| Bharati (JR 7) | -do- | 138 | 20:10:10<br>40:20:20 | 4.1<br>4.8 | U,C,H | R to blast, BLB & rice bug; MR to sheath blight & stem borer; MS to leaf folder |

**Rice varieties for organic situation**

| Variety | Sowing time | Duration (days) | Organic nutrients | Grain yield (t/ha) | *Agroclimatic zones | Reaction to insect pests & diseases |
|---|---|---|---|---|---|---|
| AAU Diphu Dhan 5 (Langpi) | June | 140-145 | As stated in organic package for rice | 5.8 | H | R to sheath blight & blast; MR to rice bug, stem borer & leaf folder |
| AAU DPU Dhan 07 (Amreng) | May-June- | 140-145 | As stated in organic package for rice | 5.8 | All zones | MR to sheath blight and blast disease and MR against gundhi bug, stem borer and leaf folder |

**Rice hybrids recommended**

| Varieties | Year of notification | Adaptation | Zone for which recommended | Grain yield (t/ha) | Disease & pest reaction |
|---|---|---|---|---|---|
| NK 5251 | 2012 | *Kharif* | UBVZ, NBPZ & LBVZ | 5.19 | Resistant to blast, GM; moderately resistant to SB |
| CRHR 5 | 2006 | *Kharif* | UBVZ, NBPZ & LBVZ | 6.10 | Moderately resistant to BL, BLB & SB |
| PAC 835 | 2009 | *Kharif* | UBVZ, NBPZ & LBVZ | 5.83 | Moderately resistant to BL, BLB & SB |

| | | | | | |
|---|---|---|---|---|---|
| US 312 | 2010 | *Kharif* | UBVZ, NBPZ & LBVZ | 5.90 | Resistance to blast, GM; moderately resistant to SB |
| PA 6444 Gold | 2015 | *Kharif* | UBVZ, NBPZ & LBVZ | 5.80 | Resistance to GM and BLB; moderately resistance to SB |

* Refer to page iv for full forms of agroclimatic zones

**Seed Selection**:

Put the seeds in plain water, stirred well, sunken seeds are selected and the floated ones are rejected.

**Seed Treatment**:

**A. Wet Method**: After selection, the seeds should be soaked directly in the following fungicidal suspensions for 24 hours. One liter of fungicide solution is required to treat one kg of seed.

| Fungicide | Dose (g/kg seed/liter of water) |
|---|---|
| Carboxin | 2.0 |

Treated seeds should be kept for incubation for 48hrs.

**B. Dry method for dry sowing:** As mentioned in case of Direct seeded normal *ahu* rice.

**RAISING OF SEEDLINGS**:

**A. Normal practice:**

**a) Preparation of seedbed**: Land is thoroughly puddled and seedbeds of 10 m length and 1.25m breadth are prepared with 30 cm gap in between the beds. The length of the bed may vary according to convenience.

**b) Manures and Fertilizers**: In each seedbed 20-30 kg cow dung/compost, 80 g urea, 80g SSP and 40g MOP are to be applied and mixed well with the soil.

**c) Seed rate**: Well germinated seeds are to be sown @ 650 g to 1kg per bed depending on grain size. Requirement of seed for transplanting one hectare of main field is 40-45 kg.

**d) Water Management**: Irrigation water should be applied in furrows to maintain saturated condition in the surface soil of the nursery bed. However, standing water to a depth of 2-3 cm should be maintained at least 2-3 days before uprooting.

**e) Plant protection in seedbed:**

- As soon as one or two blast spots are seen, ediphenphos @ 1 ml/*l* of water is to be sprayed.
- **Root knot nematode**: Apply, *Pseudomonas fluorescens* @ 20 g /sq. m at the time of sowing

**B. Mat nursery for mechanical transplanting**:

The success of mechanical transplanting (i.e. using self-propelled rice transplanter or

manually operated mechanical transplanter) depends upon the quality of seedling. Therefore, all care should be taken to prepare good quality mat nursery. Preparation of mat nursery is described below:

a. **Materials required:** Polythene sheet, wooden or iron frame, well pulverized & well-sieved soil, FYM, sprouted and treated paddy seed.

b. **Preparation seed:** Seeds are treated as per recommendation and kept immersed in water for one night in a jute bag. Seed bags are taken out of water and kept in open air for sufficient period till the length of sprouting becomes 1-2 mm. Lumps if any are broken gently to ensure uniform distribution.

c. **Dimension of plot to accommodate seedling:** A leveled plot of land free from any undulation and grass, etc. should be selected to accommodate seedling bed. Three numbers of beds, with effective length of 16 m and effective width of 1 m, each are needed to prepare seedlings for 1 ha of cropped area. The dimension of the plot should be selected as per the seedling requirement.

d. **Frame:** Removable frame is used to ensure uniform mat size. The dimensions of the mat frame should be equal to the dimension of the tray of the machine. Wooden bar or mild steel bar (2.5 cm × 0.3 cm) may be used as the boundary of the frame.

e. **Preparation of soil cow dung mixture:** The soil crushed and sieved (particle size less than 4 to 5 mm) is mixed with FYM in the ratio of 2 : 1. The FYM is also crushed to ensure homogenous mixture. The mixture should be free from stone or any other foreign objects.

f. **Preparation of seedling bed:** Polythene sheets (130 cm × 1650 cm) are placed on the selected plot of land. The frames are placed over the polythene sheet such that clear width remains 100 cm. Pulverized soil-FYM mixture is used to fill up the frame over the sheet to have a bed dimension 100 cm × 1600 cm × 1.5 cm. Wooden mallet may be used to level as well as to provide mild compaction to the soil-FYM mixture.

g. **Sowing of sprouted seeds:** Sprouted seeds are uniformly sown in the soil bed at the rate of about 1 kg/sq.m. Beds are sprinkled with water after sowing.

h. **Water management:** Beds are covered with light jute bags or straw and water should be sprinkled on the bed for 4 days under covered condition so that soil never dries. The covers are removed after four days. The sprinkling of water may need to be continued up to one week. After about one week seedling would attain some height and water may be applied by flooding. The depth of water should be half of the seedling height.

i. **Uprooting of seedlings:** Seedlings become ready for transplanting when their height becomes 15 cm with 3 to 4 leaves. The seedling mats are then cut into pieces to be used for transplanter.

j. **Field preparation**: Field should be prepared thoroughly by ploughing 4 to 5 times followed by harrowing and laddering. Ploughing should be started at least 21 days ahead of transplanting so that weeds are dried up/decayed.

**Land preparation by bullock drawn modified helical blade puddler:**

Modified helical blade puddler is an improved bullock drawn implement of puddling operation. The weight of the implement is about 26 kg and it covers about 50 cm in one pass. It can be operated by all types of bullocks available in Assam. One or two passes of puddler operation followed by one pass of mould board plough are sufficient for obtaining good quality puddle soil. The puddler can cover 2 to 2.5 bigha of land per day.

**Considerations for efficient use of the modified helical blade puddler**:

The well ploughed field should be covered with water at the time of puddler operation. To protect the sharpness of the blade the implement should never be operated over stones or hard ground. At the end of its operation the puddler should be properly cleaned and stored dry. Painting is advisable to avoid rusting.

**Manures and Fertilizers**:

Well rotten FYM or compost @ 10 t/ha or vermicompost @ 3-5 t/ha has to be applied during field preparation. In addition, the following nutrients are to be applied at rates given below in areas with moderate fertility level.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| **A. Semidwarf varieties** | | | | |
| N | 60 | Urea | 132 | 18 |
| $P_2O_5$ | 20 | SSP | 125 | 17 |
| $K_2O$ | 40 | MOP | 66 | 9 |
| **B. Tall varieties:** | | | | |
| N | 20 | Urea | 44 | 6 |
| $P_2O_5$ | 10 | SSP | 62 | 8 |
| $K_2O$ | 10 | MOP | 16 | 2 |

In case of poor soil, the rates of fertilizers may be required to increase to the extent of 60 : 30 : 30 kg/ha N, $P_2O_5$ and $K_2O$ respectively.

Diammonium phosphate (DAP) in combination with rock phosphate or alone at the recommended nutrient level (40 : 20 : 20) can be applied.

In monocrop *sali* areas of Barak Valley Zone, sowing of *dhaincha* is recommended as green manuring crop before *sali* rice.

The optimum doses for N-$P_2O_5$-$K_2O$ for HYV of *sali* rice is 60-20-40 and 60-20-20 for low and medium fertility classes of soil respectively in Hills zone.

**Use of Nano urea and nano DAP application**

Apply 1/3rd of recommended N with full dose of P K as basal and 1/3rd of N at active tillering stage followed by foliar application of Nano Urea @ 4 ml/liter at maximum tillering and PI stage in *kharif* rice.

Apply 50% Basal DAP (100% recommended NPK) with Root Dip Treatment (RDT) of Nano DAP @ 5 ml/lit followed by Foliar Spray Nano DAP @ 4 ml/lit at 30 DAT in *kharif* rice.

**Silicon application:**

Silicon is used as a growth promoter to improve plant growth, yield and increase the tolerance to both biotic (stem borer, leaf folder, grain discolouration, blast, etc.) and abiotic (moisture stress) stresses in rice. Silicon (Orthosilicic acid granule) is to be applied @ 20 kg/ha each at maximum tillering and at early flowering stages.

**Time of Application of Fertilizers:**

**a) For short duration varieties (100-120 days).**

**i)** Half of urea and whole of super phosphate and muriate of potash should be applied at the time of final puddling. In standing water, urea along with super phosphate and muriate of potash can be applied in pallet form.

**ii)** Second dose (half) of urea should be applied at the time of panicle initiation stage.

**b) For medium and long duration varieties (120-150 days):**

**i)** Half of urea and entire quantity of super phosphate and muriate of potash should be applied at the time of final puddling.

**ii)** Of the remaining part of urea, half at tillering stage i.e. 20-30 days after transplanting and other half at panicle initiation stage should be applied.

**iii)** For long duration varieties under rainfed lowland situation with water depth between 30-50 cm, basal incorporation of prilled urea at 30 kg N/ha is recommended.

**iv)** Split application of potassic fertilizer half as basal and half at maximum tillering stage is recommended for North Bank Plain Zone.

**N.B.**

i) Urea should be applied by mixing with moist soil in 1 : 10 proportion i.e. 1 part of urea with 10 parts of moist soil and incubate for 48 hours.

**i**) In case single super phosphate is not available diammonium phosphate (DAP) may be used with appropriate adjustment with urea.

iii) In the case of non-availability of single super phosphate in time, application can be delayed up to 30 days of transplanting.

iv) Standing water, as far as practicable, should be drained out before application of fertilizer.

v) In case bacterial leaf blight symptoms appear, stop top dressing of urea.

**OR**

**Mixed Fertilizers:** Granular mixed fertilizer of 15-15-15 grade can also be used to meet the N-$P_2O_5$-$K_2O$ requirement of rice crop. A total of 133 kg/ha (19 kg/bigha) of 15-15-15 grade mixed fertilizers will be equivalent to 20-20-20 kg N-$P_2O_5$-$K_2O$/ha (3-3-3 kg/bigha) which is to be supplemented by top dressing of 45 kg/ha urea to give 40-20-20 kg/ha (6-3-3 kg/ bigha) of N-$P_2O_5$-$K_2O$. Half the dose will give 20-10-10 kg/ha (3-1.5-1.5 kg/bigha) of N-$P_2O_5$-$K_2O$.

**Recommendation for use of Rock Phosphate** (If super phosphate is not used):

1. For raising two successive rice crops i.e. *ahu* followed by *sali* 60 kg $P_2O_5$/ha (300 kg/ ha MRP) should be applied at least 20 days ahead of *ahu* transplanting
2. For monocrop rice apply 30 kg $P_2O_5$/ha (150 kg as rock phosphate) at least 20 days ahead of ahu transplanting.

Application of FYM as per recommendation helps in rapid release of phosphorus from this source.

**Use of Azolla to Supplement Nitrogenous Fertilizer:**

Inoculate 500 kg of fresh Azolla/ ha in the field after transplanting. Allow it to grow for 2-3 weeks to cover the water surface. After coverage, either incorporate to the soil or allow it for self-decomposition.

To obtain 500 kg fresh Azolla, add 16 kg inoculum to an area of 400 cm water surface and allow it to grow for 3-4 weeks. Depending on the growth of Azolla, top dressing of nitrogen could be avoided.

**OR**

**Integrated nutrient management:**

Integrated nutrient management (INM) package of organic manure @ 1 t/ha (on dry weight basis) mixed inocula of *Azospirillium amazonense* A-10 and *Bacillus megaterium* P-5 @ 4 kg/ha (0.4 to 0.5 kg/bigha), rock phosphate @ 10 kg $P_2O_5$(56 kg/ha or 7.5 kg/bigha), MOP @ 40 kg $K_2O$/ha (67 kg Potash/ha or 9 kg/bigha) is recommended for rice in rice-rice, rice-legume-rice and sole rice sequence.

In low land *kharif* rice (Sali), incorporation of 45 days old dhaincha can substitute 50% of recommended N-$P_2O_5$-$K_2O$ in Hills Zone.

**Method of application of biofertilizer:** Same as Normal *ahu*.

**Potash Solubilizing Bacteria for K Nutrition in rice:**

Apply consortia of potash solubilizing bacteria (*Bacillus proteolyticus* + *Serratia liquifaciens*) @ 3.5 kg/ha or 500 g/bigha as seedling root dip treatment with N-$P_2O_5$-$K_2O$ @ 60-20-20 kg/ha

**Zinc Solubilizing Bacteria for Zn Nutrition in rice:**

Apply Zn solubilizing bacteria (biofertilizer) @3.5 kg/ha or 500 g/bigha as seedling root dip treatment in Zn deficient soils for rice along with recommended N-$P_2O_5$-$K_2O$.

**Application of boron and zinc:**

In rice–toria sequence, apply boron as basal @1.5 kg / ha (15 kg borax/ha or 2 kg borax/bigha) + Zinc @ 5 kg /ha (25 kg zinc sulphate heptahydrate/ha or 3 kg zinc sulphate heptahydrate/bigha) along with recommended N-$P_2O_5$-$K_2O$ for rice.

**Foliar application of boron:**

In late planted *sali* rice, spray 0.4 ppm boron (2.3 mg boric acid per liter of water) at the initial stage of flowering to reduce spikelet sterility and increase grain yield. For one hectare 100 liters of solution is required.

**FERTILIZER RECOMMENDATIONS AS PER FERTILIZER PRESCRIPTION EQUATIONS (FPE)**

Fertilizer prescription equations can be used to find out the amount of NPK fertilizers required to obtain a certain yield target of the crop based on soil test values for NPK. The FPEs can be used under cultivation practices where only chemical fertilizers are used and where chemical fertilizers + FYM/Vermicompost/Enriched compost etc. are applied. The FPEs are valid for different varieties of the same crop having not more than 15% variations in yield. The amount of NPK fertilizers will vary according to yield target and soil test values. Yield targets must not cross the potential yield of a particular crop.

**i) FERTILIZER PRESCRIPTION EQUATIONS (FPE) FOR HIGH YIELDING VARIETY OF WINTER RICE**

**Only Chemical Fertilizers**

**FN** = 5.22 x T - 0.75 x STVN

**FP** = 0.67 x T - 1.64 x STVP

**FK** = 5.12 x T - 1.70 x STVK

**Chemical Fertilizers + FYM**

**FN** = 5.22 x T - 0.75 x STVN - 0.11 x OM

**FP** = 0.77 x T - 1.64 x STVP - 0.10 x OM

**FK** = 5.12 x T - 1.70 x STVK - 0.30 x OM

where, FN, FP, FK-Fertilizer N, $P_2O_5$, $K_2O$; STVN, STVP, STVK-Soil Test Values for N, $P_2O_5$, $K_2O$, T-Targeted yield, OM- Organic component (FYM, Vermicompost, Enriched Compost etc).

**Ready Reckoners for two yield targets and two FYM levels for this equation are given in Appendix-XIV a & XIVb**

**ii) FERTILIZER PRESCRIPTION EQUATIONS (FPE) FOR SCENTED RICE VAR. KETEKI JOHA**

**a) Only chemical fertilizer**

**FN**= 6.80 * T – 0.79 * STVN

**FP**= 1.46 * T – 1.40 * STVP

**FK**= 5.20 * T – 0.87 * STVK

**b) Chemical Fertilizer + FYM**

**FN**= 6.80 * T – 0.79 * STVN - 0.50 * M

**FP**= 1.46 * T – 1.40 * STVP - 0.15 * M

**FK**= 5.20 * T – 0.87 * STVK - 0.44 * M

Where FN, FP and FK- Fertilizer N, P and K. T-Targeted yield, M-Organic components

**Ready Reckoners for two yield targets and two FYM levels for this equation are given in Appendix-XIV c & XIV d**

**iii) FERTILIZER PRESCRIPTION EQUATIONS (FPE) FOR WINTER RICE VAR. RANJIT-SUB 1**

**a) Only chemical fertilizer**

**FN**= 6.36 * T – 1.61 * STVN

**FP**= 1.35 * T – 2.65 * STVP

**FK**= 3.92 * T – 1.39 * STVK

**b) Chemical Fertilizer + FYM**

**FN**= 6.36 * T – 1.61 * STVN - 0.34 * M

**FP**= 1.35 * T – 2.65 * STVP - 0.45 * M

**FK**= 3.92 * T – 1.39 * STVK - 0.18 * M

Where FN, FP and FK- Fertilizer N, P and K. T-Targeted yield, M-Organic components

**Ready Reckoners for two yield targets and two FYM levels for this equation are given in Appendix-XIV e & XIV f**

**iii) FERTILIZER PRESCRIPTION EQUATIONS (FPE) FOR WINTER RICE VAR. NUMALI**

**c) Only chemical fertilizer**

**FN**= 6.46 * T – 1.64 * STVN

**FP**= 1.13 * T – 2.01 * STVP

**FK**= 2.41 * T – 0.78 * STVK

**d) Chemical Fertilizer + FYM**

**FN**= 6.46 * T – 1.64 * STVN - 0.56 * M

**FP**= 1.13 * T – 2.01 * STVP - 0.36 * M

**FK**= 2.41 * T – 0.78 * STVK - 0.47 * M

Where FN, FP and FK- Fertilizer N, P and K. T-Targeted yield, M-Organic components

**Ready Reckoners for two yield targets and two FYM levels for this equation are given in Appendix-XIV e & XIV f**

**Nutrient management in rice using Leaf Colour Chart (LCC):**

Apply urea as top dressing only when LCC reading is less than 4. If LCC is more than 4 then avoid top dressing to save urea and labour cost. Check LCC from 14 days after transplanting upto flowering at weekly interval.

**Nutrient management in hybrid rice (*Kharif*):**

Application of Zinc as basal @ 7.5 kg/ha ($ZnSO_4.7H_2O$ 35.7 kg/ha or 4.75 kg/bigha) and two foliar spray of 0.25 % Boron as Borax (Borax 1.5 kg in 600 liter (per spray)/ha or 200 g in 80 liter of water (per spray)/bigha at Panicle initiation and Milk stage along with the recommended dose of N : $P_2O_5$ : $K_2O$ (80:40:40 kg/ha).

**Details of the Proposed Recommendation:**

| **Nutrient** | **Requirement (kg/ha** | **Form** | **Requirement (Kg/ha)** | |
|---|---|---|---|---|
| | | | **Kg/ha** | **Kg/bigha** |
| N | 80 | Urea | 176 | 23.5 |
| $P_2O_5$ | 40 | SSP | 250 | 33.3 |
| $K_2O$ | 40 | MOP | 66.8 | 9.0 |
| Zn | 7.5 | $ZnSO_4$ $7H_2O$ | 35.7 | 4.75 |
| B | 3.00kg (0.25% Flair spray) at PI and Milk stage | Borax | 1.50 kg Borax in 600 liter (per spray) | 200 g Borax in 80 liter of water (per spray) |

The application method of Urea, SSP and MOP is same as the existing practice of *Sali* rice. The entire amount of Zn should be applied at the time of final puddling. The boron should be applied as foliar spray at Panicle initiation and Milk stage.

**Age of seedlings and spacing for Normal *sali***

| **Varieties** | **Age of seedlings** (days) | **Spacing (cm)** |
|---|---|---|
| 1. Short and medium duration varieties (upto 120 days) | 25 | 20 × 15 |
| 2. Medium long duration varieties (beyond 120-130 days) | 30-35 | 20 × 15 |
| 3. Long duration varieties (beyond 130 days)- | | |
| *Pankaj | 30-40 | 20 × 20 |
| Mashuri | 35-40 | 25 × 20 |
| Manohar Sali and other tall varieties | 35-45 | 25 × 25 |
| Ranjit | 35-40 | 25 × 20 |

| | | |
|---|---|---|
| Bahadur | 35-40 | 25 × 20 |
| Kushal | 35-40 | 25 × 20 |

*Transplanting of Pankaj seedlings can be delayed up to 60 days from the date of sowing (i.e. transplanting around 20th August) in BVZ.

**Age of seedlings and spacing for late *sali* (transplanted):**

| **Varieties** | **Age of seedlings (days)** | **Spacing (cm)** |
|---|---|---|
| 1. Short and medium duration varieties (upto 110 days) | 20-25 | 10 × 15/ 15×15 |
| 2. Medium long duration varieties (beyond 120-130 days) | 30-35 | 20 × 15 |
| 3. Long duration varieties (beyond 130 days)- | | |
| Mashuri | 35-40 | 25 × 20 |
| Manohar Sali and other tall varieties | 35-45 | 25 × 25 |
| Prafulla | 30-80 | 20×15 |
| Gitesh | 30-60 | 15×15/20×15 |

**N.B.:** For late transplanted crop fertilizers @ 40 : 20 : 20 kg/ha of N, $P_2O_5$ and $K_2O$ respectively should be applied as basal

**Number of Seedlings per Hill:**

i) 2-3 seedlings for normal planting (July – August)

ii) 4-6 seedlings for late planting (September).

**Depth of Planting:** 4-5 cm depth of planting should be maintained for all varieties.

**Interculture:**

i) Two weedings should be given with paddy weeder or hoe at 20 and 40 days after transplanting.

ii) For weed control, use Pretilachlor @ 0.75 kg/ha or Anilofos is to be applied @ 0.4 kg/ha at 3 days after transplanting.

iii) Apply Glyphosate 71% premixed Amonium sulphate or Glyphosate 41% + Amonium sulphate twice at an interval of two months for weed management in crop fallow situation

**Water Management:**

In s*ali* rice, application of 5 cm irrigation water 3 days after disappearance of ponding water is recommended in medium and heavy soils.

In rainfed *kharif* rice, height of bunds should be 30 cm to retain rainwater for higher yield of rice as well as to conserve residual moisture for better yield of succeeding relay crops.

**Plant Protection Measures in the Field:**

The most common pests are thrips (in the seedbed) stem borer, hispa (in endemic areas of Sivasagar, Cachar, Karimganj and Kamrup districts), swarming caterpillar and gall midge. Amongst the diseases, blast, sheath blight and bacterial leaf blight are most common. Close surveillance is necessary for timely control of the pests.

**A. Insect pests:**

i) Plant protection measures should be adopted against insect pests at their economic threshold levels as given in Table 1.

ii) Biocontrol measures against stem borer and leaf folder: 6-8 releases of *Trichogramma japonicum* @ 50,000 Nos./ha/week (6700 Nos. /bigha) starting from 30 days after transplanting gives significantly good control (30-60% parasitization) of rice stem borer and leaf folder. The performance of *Trichogramma* is at par with the chemical control in respect of control efficiency as well as economic benefits.

Since *Trichogramma* is an egg parasitoid, it's releases should have coincided with the egg laying activity of the pest. Timely releases are crucial. Releases should be made over the entire infested area throughout the egg laying period of the pests, which results in uniform and effective control**.**

**Method of application:** Each 'Trichocard' is to be cut into 6-12 pieces and distributed over the entire field by fixing them to the plants by using a stapler or adhesive. The parasitoids emerging from these disperse themselves.

To control rice pests, erect 50 'T'-perches per ha 2 ft (60 cm) above crop canopy as roosting site for insectivorous birds, which are to be removed before flowering in order to prevent activity of granivorous birds

**Biointensive IPM Package:** Biocontrol based IPM against foliar (Leaf folder, Rice hispa, Case worm), borers (Rice stem borer) and sucking pests (Green leaf hopper, Rice bug, Mealy bug, Thrips), the following measures is to be followed.

i) Two sprays of NSKE-5% (4 ml/*l*) at 30 and 55 DAT against foliar and sucking pest

ii) Erection of T-perches (bird perches) @ 50 Nos./ha (7 Nos/bigha). T-perches should be placed 2 ft (60cm) above crop canopy as resting site for insectivours birds, which are to be removed before flowering in order to prevent activities of grainivorous birds.

iii) Six releases of *Trichogramma japonicum* @ 50,000 Nos./ ha (6700 Nos. /bigha) at 10 days interval starting from 25 DAT against rice stem borer and leaf folder.

**B. Diseases:**

**i) Blast:**

**a)** Grow tolerant varieties, *viz*. Monohar Sali, Prasad, IR 36, Pankaj

**b)** Treat the seed as explained under seed treatment.

**c)** Monitoring of blast is most important to schedule spraying. Observe the top five leaves

and if 5% leaf area is damaged take resort spraying. Normally three sprays can control the disease. Spray hexaconazole 5EC @ 2 ml/*l* of water at tillering stage (30 days after sowing) and subsequently give two sprays of edifenphos @ 1 ml/ *l* of water at panicle initiation stage and when the tip of panicle just comes out.

**ii) Bacterial leaf blight**: Grow tolerant varieties, *viz*. Lakhimi and Mahsuri

**iii) Sheath blight:**

a) Two sprays of hexaconazole 5EC (2 ml/*l*) should be given, first at the appearance of symptoms and the other at 10 days after the first spraying.

b) Spraying of two commercial plant-derived Cymbopogon products @ 5 ml/*l* and Neem based fungicide @ 3 ml/*l* is recommended for management of sheath blight disease of rice. The first spray should be given as soon as symptom of the disease is observed in the field followed by a second spraying at 10-12 days interval.

c) Use *Trichoderma asperallum* (AAU-JivaShakti*) as seed treatment and seedling root dip treatment @ 5 mL/L followed by foliar application @ 5 mL/L at tillering and panicle initiation stage

**iv) Brown spot disease:**

Dry or wet seed treatment with Carboxin 2g/kg of seed followed by one spraying of ediphenphos @ 1 ml/*l* at initial symptom development stage is recommended for managing brown spot disease.

**v) Bakanae disease:**

a) Soak the seeds for 24 hrs. in a suspension of carboxin @ 2g/*l* of water.

b) Rogue out the infected tillers (elongated) from infected hills.

**Management of Direct Seeded Late *Sali*:**

a. Field should be prepared just after recession of flood by ploughing, cross ploughing and laddering to bring it to a puddle condition.

b. Sprouted seeds should be sown in lines 20 cm apart.

c. A seed rate of 75 kg/ha should be maintained.

d. Need based fertilizer application is advocated.

e. Pre-emergence herbicide pretilachlor @ 0.75 kg a.i./ha should be applied 2-3 days after sowing.

f. Wherever water management is possible, two irrigations at PI and flowering stages are recommended.

**C. Rodent Pest Management**

For rodent pest management in rice, baiting with bromadiolone (0.005%) at panicle initiation stage followed by local bamboo traps (*Endur chitap*) @ 30 traps/ha (4 traps /bigha) at milky stage should be used which gives up to 65-70% reduction in rodent population.

**D. Storage insect pest management:**

Application of neem oil @ 1.5 ml/kg gives effective protection of stored paddy seed against storage insect pests for up to 9 months without affecting germination.

**CONTINGENCY PACKAGE OF PRACTICES ON RICE FOR CHRONICALLY FLOOD AFFECTED AREAS**

**A. Raising Community Nurseries**: Community nurseries may be raised to meet the seedling requirement of flood affected areas. Community nursery may either be raised by the farmers of flood affected areas in non-flood prone or high lands or by other farmers from flood free areas to help distressed farmers.

**1. Varietal Selection**: Select recommended varieties for late planting having wide flexibility in respect of seedling age and transplanting time under the following situations.

a) In occasionally flood affected areas if flood recedes early and rice can be transplanted by mid-August, use varieties *viz*., Kushal, IET-6666, Pankaj, Biraj, Anderw Sali, Solpona, Gitesh and Prasadbhog.

b) In chronically flood affected areas where flood is expected to recede by the last part of August, varieties like Andrew Sali, Biraj, Monohar Sali, Kmj-1-19-1, Luit, Kapilee, Dum Sali, Gitesh and Prafulla should be selected. In absence of these varieties any traditional photoperiod sensitive coarse grain *sali* variety can be used.

c) In areas where transplanting is not possible before mid-September, extra early varieties such as Luit, Kapilee, Culture-1, Heera and Disang should be selected.

**2. Seedbed preparation:** Seedbed should be initially ploughed with addition of adequate FYM/compost. The Flat beds may be selected and a gap of 30 cm (1 ft.) may be left at an interval of every 1.25 m (4 ft.) as passage between the beds. Beds within the strip should be well levelled Chemical fertilizers @ 500 g urea, 500 g SSP and 250 g MOP for the area should be used to transplant one bigha of land should be applied at the time of final puddling. Strip beds are to be made after levelling the field.

**3. Seed Rate:** About 10 kg seeds are needed to transplant 1 bigha of land i.e. ten strips of 10 m × 1.25 m size @ 1 kg seed/bed.

**4. After care:** The gap of 30 cm may be converted into channels one week after sowing for draining out excess water during heavy showers and to supply water to the channels to keep the raised beds moist in the event of drought.

**5. Transplanting:** Transplanting may be done at a convenient time particularly soon after flood recession. It is to be noted that moisture stress is a common feature after flood recession in flood affected areas.

**6. Plant population:** Closer planting is essential in case of late planting since tiller development is checked due to emergence of non-effective tillers. In the case of modern varieties 36 hills/sq.m (20 cm × 15 cm) and in case of tall varieties 25 hills/sq. m (20 cm × 20 cm) are to be maintained.

7. **Seedlings/hill:** In late planting situations, the only option to have maximum panicles per unit area is from that of the main culm and primary tillers. Therefore, 6-8 seedlings per hill may only yield 18 panicles per hill, as the growth phase of the rice plant almost expires in the seedbed.
8. **Fertilizer:** In chronically flood affected areas where high silt deposition occurs; there may not be any need for fertilizer application. However, a basal application of fertilizer should be done in occasionally flood affected areas irrespective of varieties. The fertilizer doses are 40 : 20 : 20 or 20 : 20 : 40 N, $P_2O_5$, $K_2O$) as the case may be.

**B. Double Transplanting:** Double transplanting is a method for seedling multiplication under seedling scarcity condition. Traditional or improved varieties may be transplanted during July with closer spacing of 20 × 10 cm (50 hills/sq. m) and each tiller developed may be separated and planted once again, which fulfills the seedling requirement by 5-7 times.

1. **Varietal Selection:** Traditional or improved varieties with long duration (150 days or more) having photoperiod sensitivity, should be selected.
2. **Sowing:** Sowing should be done during the early part of June in well prepared seedbeds as indicated earlier.
3. **First transplanting:** Transplanting should be done in the early part of July with 25-30 days old seedlings. About 50 hills/sq. m (20 cm × 10 cm) is to be maintained during first planting. Fertilizers @ 20 : 10 : 10 kg/ha is to be applied for rapid and healthy tiller development.
4. **Second transplanting:** Uprooting should be done at 25-30 days after the first planting and each tiller may be separated and re-transplanted with a single tiller/hill. In case of further delay however, 3-4 tiller/hill may be planted with closer spacing. In flood affected areas, there is no need of any fertilizer application in second transplanting if planting is delayed beyond August and fertilizer is added only in first planting.

**C. Direct Seeding (Wet Sowing):** This is an effective and remunerative method of rice cultivation in flood affected areas after recession of flood. In general, very short duration or extra-early (less than 100 days), such as Luit, Kapilee, and any traditional photo period sensitive coarse grain varieties are suitable for wet sowing in the main field.

Seeds @ 75 kg/ha are to be soaked for 24 hours and incubated for 24-48 hours for sprouting. In the meantime, the field must be puddled with minimum tillage and levelled properly by laddering to ensure uniform moisture retention. Sprouted seeds are then broadcast uniformly on to the puddled and levelled field after application of basal dose of fertilizer @ 40 : 20 : 20 kg/ha. Nitrogen should be applied in 2 splits, *viz*., top dressing at 20 days after sowing and at 45-50 days after sowing. Only P and K are to be applied as basal dose.

Direct seeding of sprouted seeds on puddled soil can be done at any convenient time after flood recession up to September 10. The main drawbacks of this system are:

Water management as moisture stress is a common problem after flood recession. Late sowing may result in spikelet sterility if temperature falls early.

# SPECIAL PACKAGE FOR LOW AND DELAYED RAINFALL AREAS OF UPPER BRAHMAPUTRA VALLEY ZONE

**I. PRE-SOWING OPERATION**: Follow ploughing after harvest of rice in light soil either with tractor or M.B. plough for moisture conservation.

**II. VARIETAL SELECTION:**

**A. Direct Seeded *Ahu*:**

a) Traditional variety for early *ahu.*

b) Lachit, Chilarai, Luit, Kapilee, Govind, Rasi and other recommended varieties for regular *ahu*.

**B. Direct Seeded Rice (*Kharif*)**: Use modern recommended varieties.

**C. Transplanted *Sali* Rice**:

i) As per recommendation for normal situation.

ii) Varieties that can be grown up to August 15 with 45-50 days old seedlings: Pankaj, Kushal, Lakhimi

iii) Varieties that can be grown up to the end of August with 60 days old seedlings: Luit, Kapilee and traditional varieties.

**III. SEEDLING RAISING:**

i) Treat the seeds with 4% MOP (KCl) (600 ml/kg of seed) for 24 hours, dry in shade for 24 hours and sow.

ii) Prepare dry, well bunded, flat seedbed with adequate FYM (30 kg), 80 g urea, 80 g SSP and 80 g MOP per bed size of 10 m × 1.25 m.

iii) Under zero tillage condition, mulch with dry FYM powder up to 2 cm thickness and keep the beds moist.

**IV. APPLICATION OF FERTILIZERS:**

1. If drought situation appears, top dress additional quantities of MOP @ 5 kg/bigha and incorporate it.
2. Spray 2% muriate of potash solution on leaves if and when drought appears.
3. Top dressing of urea may be delayed up to heading if drought prevails at the stage of top-dressing. Other recommendations will be as per Package of Practices.

## PEST MANAGEMENT PRACTICES IN ENDEMIC AREAS WITH SPECIAL REFERENCE TO RICE HISPA

a) Destruction of initial population.
b) Destruction of alternate hosts (*Dol, Uridol* etc.)
c) Burning of stubbles after harvesting.
d) Deep ploughing during Feb- April.
e) Monitoring at regular intervals.
f) Avoidance of staggered planting.
g) Seedbed treatment or root dip treatment.
h) Clipping of leaf tips before planting
i) Use recommended insecticides at proper dosage and spray volume. Add 1% urea and 2% potash to spray fluid.
j) Planting of moderately resistant varieties.
k) Spray 1% neem seed oil using sticker (23 g/*l*).

For more details of pest management practices against rice hispa, refer to Page111.

# Rice *(Oryza sativa)*
# Wet Direct Seeded Sali Rice

**Varieties**: Varieties with good seedling vigor and rapid growth should be selected.

| Variety | Year of release | Duration (days) | Av. Plant Height (cm) | Grain yield (q/ha) | Disease and pest reaction |
|---|---|---|---|---|---|
| Ranjit Sub-1 | 2018 | 145-150 | 120-130 | 55-60 | Moderately resistant to neck blast, tolerant to stem borer and moderately resistant to BPH |
| Bahadur Sub-1 | 2018 | 140-145 | 120-130 | 55-60 | Moderately susceptible to brown spot, bacterial leaf and sheath blight, tolerant to stem borer and moderately resistant to BPH |
| Swarna Sub-1 | 2009 | 145-150 | 115-125 | 45-50 | - |
| Gitesh | 2017 | 150-160 | 120-130 | 50-55 | Resistant to sheath rot, moderately resistant to blast and tolerant to BLB and susceptible to sheath blight |
| Numoli | 2020 | 130-135 | 120-130 | 50-55 | Moderately susceptible to leaf blast, grain discolouration, neck blast, brown spot, sheath blight, BLB and sheath rot, moderately resistant to RTD, GLH, PH, stem borer and WM; tolerant to leaf folder, brown PH, gall midge and WBPH |
| TTB – 404 (Shraboni) | - | 135-140 | 120-130 | 50 | Moderately resistant to sheath rot, neck blast & RTD and moderately resistant to GLH, PH &Stem borer |

**Soil**

Typical soil, where direct seeded rice (DSR) can be grown, ranges from sandy loam to heavy clay. Medium land situations are preferred.

**Sowing time**

For early sali DSR, sowing should be done before the onset of monsoon i.e. in the last week of May to first week of June.

**Land preparation**

The field is to be ploughed thoroughly, first by tractor drawn disc plough during the month of April, followed by harrowing. Laddering is done for thorough pulverization and to

prepare fine seed-bed suitable for germination/emergence of rice. This is followed by puddling and levelling. Followings are the steps for land preparation:

- Irrigate the field with 2−3 cm of water. Impound water for about 3−7 days for making the soil soft for easy tillage.
- Drain the soil and wait till reasonable traction can be achieved for operation by plough/harrow.
- Use tractor/power tiller/animal power with MB plough/ disc plough /rotavator for primary tillage.
- Make 2 passes, first pass clockwise and second pass counter clockwise, both the time beginning at the edge of the field and works toward the centre of the field. Depth of tillage should be 10-15 cm.
- Again, keep the land flooded for 10-14 days.
- Use tractor/power tiller/animal power/rotavator/inclined plate puddler for 2 passes of secondary tillage/puddling.
- Do final secondary tillage 2 days before seeding to make the field levelled. For power tiller/animal power, use a plank leveller, for tractor, use rotavator and leveller. A laser leveller may also be used.
- Frequent bunds should be avoided initially. Once the seedlings establish, bunds may be erected with proper dyke height (30 – 40 cm). Repair the pre-existing bunds, destroy rat burrows, repair all holes and cracks, and re-compact the bunds.

**Land levelling**

Good land levelling is essential for success of DSR. Land levelling ensures water saving, uniform crop establishment, uniform distribution of irrigation water, improvement in nutrient use efficiency, better environmental quality and higher grain yields. Initial precision levelling by tractor drawn laser guided land leveller should be preferred. Land levelling may also be done by traditional leveller drawn by draft animal or by power tiller. This should be followed by laddering (planking).

**Seed rate and method of seeding**

For seeding with drum seeder, 25-30 kg seed/ha is sufficient. The method of sowing using drum seeder has been described in detail under ‘Use of Drum Seeder’.

**Seed treatment**

The seeds are treated with carboxin 75 WP @ 2.0 g/ kg seed/ l of water for 24 hours. The treated, soaked seeds are inoculated with bio-fertilizer (NPKZn consortia i.e. Azospirillum + PSB + KSB + ZSB consortia) @ 20 ml/kg of dry seed (10 ml/kg of untreated seed). The soaked seeds are incubated for 12-24 hours to enhance germination. Then the pre-germinated seeds are sown directly in the main field.

**Fertilizer Management**: Incorporation of organic manures @ 5 t/ha into soil during ploughing.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | Kg/ha | Kg/bigha |
| N | 60 | Urea | 132 | 18 |
| $P_2O_5$ | 20 | SSP | 125 | 17 |
| $K_2O$ | 40 | MOP | 66 | 9 |

**Method of fertilizer application:**

i) Entire quantity of super phosphate and muriate of potash should be applied at the time of final puddling.

ii) 1/3$^{rd}$ N after 15 days of crop emergence, 1/3$^{rd}$ at maximum tillering after 1$^{st}$ weeding and 1/3$^{rd}$ at Panicle initiation stage.

iii) Urea should be top dressed by mixing with 10 parts of moist soil and after 48 hours of incubation for better N use efficiency.

iv) When super phosphate is not available, Diammonium Phosphate (DAP) may be applied with appropriate adjustment with urea.

v) Standing water, as far as practicable, should be drained out before application of fertilizers.

vi) In case, bacterial leaf blight symptoms appear, stop top dressing of urea.

vii) Use of LCC for real time N management:

If LCC is used. 9 kg urea/bigha is to be applied as basal and then take LCC readings from 10 randomly selected topmost fully expanded leaves, starting at 21 days after the emergence of rice seedlings until flowering at 10 days interval. When the LCC value of six out of ten leaves fell below the critical level (LCC 4), 4.5 kg urea/bigha to be top dressed. If LCC is more than 4, then avoid top dressing to save urea and labour cost. LCC Readings are to be taken between 8-10 am when there is not much glare from the Sun. Every time the same person should take colour measurement at the same time of the day.

**Water management**

For the first 10-15 DAS, flush irrigate the field to keep the soil saturated but not flooded to enhance root development and anchorage if rain is not there. If rain is not there and irrigation facility is available adopt AWD (alternate wetting and drying) method of irrigation for water management with 5 cm irrigation depth.

**Weed Management**

**a) Chemical control:**

i. Apply Pretilachlor 50 % EC @ 750 g/ha within 3-5 days of sowing.

ii. Bispyribac sodium 10 SC @ 25 g/ha should be applied within 3-4 leaf stage of weeds for control of braod leaf weeds.

iii. Pyrazosulfuron-ethyl @ 0.015 kg/ha as pre-emergence followd by conoweeder at 40 days after sowing.

**b) Mechanical control:**

For wet DSR with 2-3 cm water in field, use of conoweeder (or any other rotary weeder or motorized rotary weeder) when emerged weeds are at 3-4 leaf stage. Repeat weeding two more times at 30-35 and 40-45 DAS. Remove the weeds near the plants manually.

**Herbicide spraying:**

Use clean fresh water. While using knapsack sprayer, maintain steady pressure, a steady walking speed and uniform spray application action. Use a multi nozzle (e.g., three) boom with flat-fan nozzles and slightly overlapping spray patterns at the soil surface. The overlap is achieved by holding the boom at approximately 50 cm above the target (for pre-emergence, the soil surface is the target, and for postemergence, weeds are the target). To minimize drifting by wind, use low pressures (< 2 bars), avoid spraying in very strong winds and spray perpendicular to the wind.

**Pests and diseases**

**i) Insect pests:**

To control yellow stem borers, leaf folder and other rice pests erect 'T'- bird perches (60-90 cm height) @ 50 nos./ha above the crop canopy as roosting sites for insectivorous birds, which are to be removed before flowering in order to prevent the activity of granivorous birds.

Install pheromone trap @ 8 traps/ha and also, place trichocards against yellow stem borer and leaf folder @ 50,000/ha at 25, 60 and 75 DAS.

Apart from these, chlorantraniliprole 18.5 SC @ 150 ml/ha can be sprayed at maximum tillering stage against stem borer and leaf folders. Use 500 l of water/ha for spraying insecticides. Need based chemical control can be used against other rice pests as per recommendation for transplanted Sali rice.

**iii) Diseases:**

| Disease | Control Measure |
|---|---|
| Blast: Observe the top five leaves and if 5% leaf area is damaged, take resort spraying. Normally three sprays control the disease. | Spray hexaconazole 5 EC @ 2 ml/l at tillering stage and subsequently give two sprays of edifenphos 50 EC @ 1 ml/ l of water at panicle initiation stage and next when the tip of panicle just comes out |
| Sheath blight | Two sprays of hexaconazole 5EC (2 ml/l) should be given, first at the appearance of symptoms and the other at 10 days after the first spraying. |
| Brown spot | Dry or wet seed treatment with carboxin 75 WP @ 2 g/kg followed by one spray with edifenphos 50 EC @ 1 ml/l at initial symptom development |

*Spray volume: 500-600 l/ha

**Use of Drum Seeder**

The following steps are to be followed for successful operation of the drum seeder to sow pre- germinated seeds:

1. Land preparation: Puddle the land and level thoroughly.
    a. A good tilth of the top soil up to 15-20 cm deep is desirable.
    b. Allow the water in the field and then puddle the field twice followed by laddering.
    c. Give a gap of at least 7-10 days between initial and final puddling for better weed control and nutrient availability.
    d. Perfect levelling ensures uniform seed distribution. levelling is also a must for better water management, weed control and healthy and even plant growth.
2. Water should be drained out at least 24 hrs before sowing, but the soil surface should not be allowed to dry.
3. At the time of sowing only paper-thin water level should be maintained in the puddled field.
4. Water should be flooded to the puddled field once in three days after sowing and drain out immediately. This practice should continue for 12 days. Thereafter depending upon the height of the seedlings water should be allowed to stand in the field as described under the head 'water management'.
5. Sow the seeds with drum seeder
    a. Do not fill drums more than 2/3 of full capacity.
    b. Walk at a steady pace (uniformity is essential).
6. Do not irrigate for 2-3 days to allow roots to anchor. Take care during rainy season as rainfall immediately after seeding may wash away the newly sown seeds.

**Seed Preparation for Drum Seeder**

- **Soaking**: Soak the treated seeds in gunny bag in water for 24 hours.
- **Incubation**: Keep the gunny bag with seed for 12 to 24 hours in wet condition. Avoid over as well as under germination. Inoculate the seeds with bio-fertilizers before incubation.
- **Drying**: Dry the seed in open air under shade for 10-15 minutes before sowing to facilitate separation of seeds.

**Operational procedure for Drum Seeder:**

- Assemble the paddy seeder and tighten all the bolts & nuts.
- Fill the drums with pre-germinated seeds. Remember only two-third of the drums are to be filled at a time.
- Close the doors of the drums and lock.
- Pull the seeder manually at a normal walking speed (1-2 km/h). Walk at a steady pace (uniformity is essential)

- The wheel impression in the first pass will serve as a marker.
- In the second pass, the wheel should pass on the same wheel impression of the previous pass to maintain the row-to-row spacing of 20 cm.
- Continue the seeding operation.
- Occasionally watch the dropping of the seeds through the holes of the seeder.
- Refill the drums when it reaches one fourth capacity.

**Advantages of using a plastic paddy drum seeder:**

- Labour cost of sowing is reduced drastically. Manhour saving over manual line sowing is 88.88 %
- Seed rate is reduced as only 25 kg/ha is required.
- Uniformity in seed sowing and better plant stand.
- Crop matures 10 days earlier.
- Light weight. Very easy to pull in wetland field. One person can easily cover 1 ha/day.

**Harvesting and threshing:**

Harvesting is done at physiological maturity, judged visually. The plants from the plot are cut by sickle, manually. The grains are properly cleaned, dried and weighed separately at 14% moisture content for storage.

# Rice *(Oryza sativa)*

## Bao Rice (Deep Water Rice)

**Varieties:**

| Varieties | Year of notification | *Agroclimatic zones |
|---|---|---|
| Maguri | - | N,U,C |
| Panikekua | - | N,U |
| **Padmapani | - | N |
| (PJNB) Panindra | 1999 | N |
| (PJNB) Padmanath | 1999 | N |
| Amona | - | U |
| Negheri | - | U,C,L |
| Kekua | - | C,L |
| Salibadal | - | B |
| Dholabadal | - | B |

**New deepwater rice varieties recommended**

| Variety | Duration | Yield (t/ha) | Adaptation |
|---|---|---|---|
| LPR 106 (Panchanan) | 186 (T), 240 (DS) | 3.0 | Deepwater (50-100 cm water depth) |
| KDML 105 [Padumani] | 180 (T), 230 (DS) | 2.6 | Deepwater (50-100 cm water depth) |

* Refer to page iv for full forms of agroclimatic zones

** This variety is early maturing (185 days) and escapes ufra disease.

**Land Selection:**

Areas having higher land submergence- unsuitable for normal rice are selected for growing of *bao* rice. Such areas are broadly classified into two types, *viz*.,

1) Water level up to 1 meter (deep water rice areas)
2) Water level beyond 1 meter (very deep-water rice areas).

**Land preparation:**

i) Stubbles of the previous crop should be burnt thoroughly to minimize nematode and pest infestation.

ii) Adequate ploughing and cross ploughing should be given.

**Seed Selection:** Plain water is used to select the seeds.

**Seed Treatment:**

**A. Wet method:** After selection, the seeds should be soaked directly in any one of the following fungicidal suspensions for 24 hours.

| **Fungicides** | **Dose (g/kg seed/liter of water)** |
|---|---|
| Carboxin | 2.0 |

One liter of fungicidal solution is required to treat one kg of seed.

**B. Dry method:** Seeds and any one of the following fungicides are put in a closed container and then agitated for five minutes for thorough mixing.

| **Fungicides** | **Dose (g/kg seed)** |
|---|---|
| Carboxin | 2.0 |

**Sowing Time:**

Sowing should be done during March – April.

**Method of Sowing:**

**i)** For *ahu* areas: *ahu* + *bao* mixtures in the ratio of 4:1 recommended (8 kg *ahu* + 2 kg *bao* seed/bigha).

**ii)** For *bao* areas: monocrop bao is recommended.

**Seed Rate:**

Seed rate for both the situations (mixture and monoculture) is 75 kg/ha.

**Fertilizers:**

Application of neem coated urea @ 30 kg N/ha in two equal splits as basal and at maximum tillering stage is recommended.

**Preparation of *neem* coated urea:**

*Neem* coated urea can be prepared by mixing 50 kg urea with 500 ml *neem* oil over a polythene sheet till uniform yellow colour is obtained.

As an alternative to neem coated urea, application of 4% urea solution as foliar spray at maximum tillering stage (150 DAS) @ 30 kg N/ha is recommended.

**i)** Wherever feasible "*bindha*" (rake) may be used.

**ii)** In the case of mixture, bao crop should be pruned along with harvesting of *ahu* crop.

**iii)** The field should be kept free from weeds like water hyacinth etc.

**Plant Protection:**

**Field rats:**

Apply dry poison baits in bait stations (broken earthen pot, coconut shell, bamboo cylinder, etc.). Pre-bait prepared out of 1kg cereal + 20g vegetable oil + 10g dried fish may be applied @ 10g/live burrow for 3 days. This treatment should be followed by addition of Zinc phosphide @ 25 g/kg bait on the 4$^{th}$ day.

**Bait formula:**

| | | |
|---|---|---|
| 1. | *Atta* or wheat bran or maize crush | 80g |
| 2. | *Gur* | 10 g |
| 3. | Fried fish (or dry fish) | 5g |
| 4. | Zinc phosphide | 5 g |

**N.B**. Sowing of *asra* (shallow water rice) under puddled condition is recommended for Barak Valley Zone.

***Ufra* disease control:**

1. Destruction of diseased stubbles and straw followed by several ploughing just after harvest
2. Delay sowing up to mid-April
3. Cultivation of early maturing bao variety like 'Padmapani'

# Maize

(*Zea mays*)

**Varieties**:

| Varieties | Duration (days) | Special features | Potential yield (q/ha) |
|---|---|---|---|
| **Hybrid:** | | | |
| Ganga 5 | 110-120 | - | 48 |
| Vivek Maize Hybrid 47 | 80-90 | Cob yellow with white cap, Semi flint and moderately resistant to MLB and C rust | 69 |
| Vivek Maize Hybrid 53 | 70-80 | Yellow, semi flint | 69 |
| Bio 9544 | 90-110 | Orange yellow, semi dent | 73 |
| **Composite:** | | | |
| Dhawal | 110-120 | - | 48 |
| Navjot | 90-100 | - | 48 |

**Note**:

a) Seed from hybrid maize should not be kept as the crop deteriorates due to segregation. The seeds from composite can be used for three to four years without much deterioration in yield.

b) All the above varieties are drought tolerant and resistant to most of the diseases to a fair degree.

c) Excepting Diara, all other varieties/hybrids are suitable for all the zones except Barak Valley Zone.

**Selection of Site and Land Preparation:**

Any well drained soil is suitable for maize. Sandy and sandy loam fertile soils are preferable. It requires a deep fine and firm tilth.

**Manures and Fertilizers:**

A combination of organic and inorganic fertilizer gives better results than inorganic fertilizer alone. Compost or FYM @ 5 t/ha should be applied.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 60 | Urea | 134 | 18 |
| $P_2O_5$ | 40 | SSP | 250 | 33 |
| $K_2O$ | 40 | MOP | 67 | 9 |

For Hills Zone, a dose of 90-40-40 kg N-$P_2O_5$-$K_2O$ per hectare is recommended under rainfed condition.

**FERTILIZER RECOMMENDATIONS AS PER FERTILIZER PRESCRIPTION EQUATIONS (FPE)**

Fertilizer prescription equations can be used to find out the amount of NPK fertilizers required to obtain a certain yield target of the crop based on soil test values for NPK. The FPEs can be used under cultivation practices where only chemical fertilizers are used and where chemical fertilizers + FYM/Vermicompost/Enriched compost etc. are applied. The FPEs are valid for different varieties of the same crop having not more than 15% variations in yield. The amount of NPK fertilizers will vary according to yield target and soil test values. Yield targets must not cross the potential yield of a particular crop.

**FERTILIZER PRESCRIPTION EQUATIONS (FPE) FOR HYBRID MAIZE**

**A. NPK Alone for Hybrid Maize**

**FN=** 5.18 * T – 0.63 * STVN

**FP=** 1.66 * T – 1.43 * STVP

**FP=** 3.20 * T – 0.62 * STVK

**B. Integrated, NPK + IPNS for Hybrid Maize**

**FN=** 5.18 * T – 0.63 * STVN - 0.83 * M

**FP**= 1.66 * T – 1.43 * STVP - 0.30 * M

**FP=** 3.20 * T – 0.62 * STVK - 0.42 * M

**Ready Reckoners for two yield targets and nutrient levels for this equation are given in Appendix-XVA, XVB. XVC & XVD**

**Method of Fertilizer Application:**

FYM or compost should be applied during land preparation. The entire quantity of SSP and MOP and half of the total urea is to be applied in furrows (8-10 cm deep) and covered with 4-5cm of soil. Sowing of seed should be done at least 2 days after fertilizer application.

There remaining half of urea should be top dressed in two equal doses followed by earthing up; in each case. The first ¼ at 35 days after germination or when the plants are at knee high stage and the second ¼ at the tassel initiation stage of the crop i.e. 45- 60 days after germination or at the time of elongation of the flag leaf whichever is earlier. The fertilizer should be applied on both sides of the row and weeds should also be removed from the field during this operation. However, before application of the fertilizer a light hoeing is to be given between the lines 2-3 days ahead.

Apply 75% of RDF along with vermicompost 1 t/ha in each crop in maize-green gram-rajmah cropping system in rainfed upland situations.

Apply Zinc @ 2.5 kg/ha ($ZnSO_4.7H_2O$) @12 kg/ha as basal along with the recommended dose of NPK (60:40:40 kg/ha) and 3 foliar spray of 0.50% $ZnSO_4.7H_2O$ (3 kg $ZnSO_4.7H_2O$ in 600 liter water/ha per spray) at knee high, tasseling and silking stage

**Seed Rate:**

Depending upon the seed size about 22.5-25.0 kg of seeds/ha (3-3.5 kg/ bigha) are required.

**Seed Treatment:**

Seeds should be dressed with carboxin @ 2 g/kg of seed

**Sowing Time:**

| *Agroclimatic zone | L | U | C | N | B | H |
|---|---|---|---|---|---|---|
| **Sowing time** | Last week of January to March | Last week of January to March | Last week of January to March | Feb-March | March - May | **April-May |

* Refer to page iv for full forms of agroclimatic zones

**July and August for late planting in hills.

**Method of Sowing:**

Seeds should be dibbled at a depth of 3-4 cm in rows 65-75 cm apart and at a distance of 20-25 cm from seed to seed within the rows. Two seeds per hill are to be sown followed by thinning10 days after germination retaining one seedling/ hill. Depending upon the fertility status of the soil, a wider spacing can be used. Varieties like Diara can be sown at closer spacing because of their short stature and shorter duration.

**Weeding and Inter-culture:**

Light hoeing and earthing up should be done as and when necessary. For further and adequate control of weed, simazine should be applied @ 0.5 – 1.0 kg a.i/ha in 1000 liters of water a pre-emergence spray.

**Irrigation and Drainage:**

Maize is primarily grown as rainfed crop; but application of irrigation at the time of soil moisture deficit ensures optimum yield. Silking stages and grain filling period are the two critical stages of the maize crop.

Excess rainwater can be drained out through surface drainage channel of 25 cm width, 15 cm deep spaced at 6 m distance.

**Water Management**

Maize should be sown in paired rows of 50 cm between pairs and 70 cm gap is to be given between two pairs. Two rows of black gram should be sown in between two pairs of maize

**Plant Protection:**

When infestation of stem borer is noticed, fipronil 5 SC @ 1.0 kg/ha in 200-250 liters of water should be sprayed with power sprayer or in 700-800 liters of water with hand sprayer.

In areas where banded sclerotial disease (*Rhizoctonia solani*) is noticed, the plants

should be sprayed with hexaconazole 5EC @ 2 ml/*l* of water at an interval of 12-15 days.

Cobs can be protected from graminivorous birds by wrapping the cobs with adjacent leaves of the same plant at vulnerable grain maturity stage in outer 3 border rows. Erect Shining Reflecting Ribon Stripes 1 feet above the crop canopy in North-South direction with twisting to reflect the sunlight for better protection

**Harvesting:**

Harvesting should be done as soon as the husk covers turn brown and the silks are completely dry. Harvesting immediately after a shower should be avoided. Ears (cobs) should be thoroughly dried in the sun before shelling or storing unshelled.

# Finger Millet (*Marua*)

*(Eleusine coracana)*

**Varieties:**

INR1008, TANU-9 promising varieties collected from BHU, Banaras and local varieties.

**Plan type**: Erect, medium tall and significantly yielding higher the local variety

**Duration:** 80 – 90 days.

**Grain yield:** 18-20q/ha (transplanted)

**Other Characters:** Average length of fingers/panicle 6-7 cm

**AAU-GSG-Maruadhan 1 (Gossaigngaon Maruadhan):** First high yielding (33 q/ha) finger millet variety of Assam. It is a stress tolerant variety that grows well in rainfed upland areas, where *Sali* paddy does not perform well. It is also a nutri-cereal, rich in protein (10.67%), crude fibre (11.75%), iron (8.91 mg/100 g) and calcium (300 mg/100 g).

**Soil Type:** Sandy loam

**Time of sowing:** One month old seedlings are to be transplanted in the 1st week of September.

**Seedbed:**

For seed sowing a raised (10 – 25 cm) seedbed of size 10 × 1.25 m with 30cm gap between the beds should be prepared; 20-30 kg cow dung/bed should be applied and mixed thoroughly with the soil. 150 g of seeds should be sown per bed.

**Seed rate:** 12 kg transplanting one hectare

**Spacing:** 25cm × 15 cm

**Manures and Fertilizers:** Apply FYM/Compost @ 5 t/ha or 6q/bigha.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 40 | Urea | 88 | 12.00 |
| $P_2O_5$ | 20 | SSP | 125 | 16.50 |
| $K_2O$ | 20 | MOP | 32 | 4.25 |

Half of N and full doses of FYM, $P_2O_5$ and $K_2O$ should be applied as basal and the remaining half of N to be top dressed 30 days after transplanting.

**Interculture:**

First weeding at 20 25 days after transplanting and subsequent weeding a sand when necessary.

**Plant Protection:**

**1) Blast:**

- Early transplanting (1st week of August to 4th week of August) of one month old seedlings.
- Spraying of tricyclazole/ediphenphos @ 0.1% significantly controls the disease
- Two sprays of propiconazole 10.7% + tricyclazole 34.2% @ 0.1% or propiconazole @0.1% just after the initiation of blast symptoms at 10 days interval and 3rd spray just after the flowering stage

**Harvesting:**

Harvesting should be done when 75-80% of the panicles turn yellowish.

**Use:**

It is used as a substitute of rice by the farmers. It is also used for making porridge, *sattu, akhoi* and preparation of flour and beverage.

# PULSES AND OILSEEDS

## Black gram

*(Vigna mungo* L.*)*

**Varieties:**

| Varieties | Year of notification | Duration (days) | Yield (q/ha) | Disease reaction | Protein content (%) | Season and area of adaptation |
|---|---|---|---|---|---|---|
| Saoniamah (SB 121) | - | 70-80 | 10-12 | -do- | 24.00 | Suitable CBVZ of Assam |
| Manas (SB 32-13) | - | 80-90 | 10.00 | Resistant to CLS, YMV and WB | - | Suitable for all zones of Assam except BVZ |
| SBC 40 | - | 75-85 | 11.89 | Resistant to CLS & YMV, moderately resistant to WB | 25.20 | Suitable for both *kharif* and summer for all zones of Assam except BVZ |
| SBC 47 | - | 75-80 | 14-16 | Resistant to CLS and YMV | 25.50 | Suitable for both *kharif* and summer for all zones of Assam except BVZ |
| SBC 50 (Baibhav) | - | 78-80 | 14-16 | Resistant to YMY, web blight and CLS. | 25.80 | Normal and late kharif season, suitable to all zones of Assam |
| PU-31 (Notified) | 2008 | 70-75 | 12-14 | Resistant to mungbean Yellow Mosaic Virus and *Cercospora* Leaf Spot | 25.00 | *Kharif*, Suitable to UBVZ, LBVZ, CBVZ and NBPZ |
| AAU SHN Urd 02 (Lakhi) | 2023 | 75-80 | 14-16 | Highly resistant to Mungbean yellow mosaic virus and *cercospora* leaf spot, moderately resistant to web blight. Moderately resistant to leaf roller, pod borer and storage pests under field conditions | 25.20 | *Kharif,* suitable to all zones of Assam |

**Suitable for all zones, except Saoniamah, which is recommended for Central Brahmaputra Valley Zone only*.

**Black gram varieties recommended for delayed sowing (up to 30 September)**

| Varieties | Duration (days) | Yield (q/ha) | Disease reaction |
|---|---|---|---|
| Beki (SB 27-3) | 80-85 | 12-13 | Resistant to CLS, YMV and WB |

| | | | |
|---|---|---|---|
| Kolong (SB 25-19) | 80-85 | 12-13 | Resistant to CLS, YMV and WB. |
| Sonkush (SB 23-5) | 80-90 | 9.17 | Resistant to CLS, YMV and WB |

**Sowing Time:** 15th August to 30th September

Saoniamah: Mid July to mid-August.

**Soil Type:**

Black gram can be grown on a wide range of soils, but sandy loam soil is preferable.

**Land Preparation:**

The land is to be ploughed 2-3 times followed by leveling.The stubbles are to be removed. Surface drains should be provided to facilitate quick removal of excess water from the field.

**Liming:**

Requisite amount of lime is to be applied after soil test in order to bring the soil pH around 6.0.

M**anures and Fertilizers:** Compost or FYM @ 1 t/ha or1.3 q/bigha should be applied

| **Nutrient** | **Requirement (kg/ha)** | **Form** | **Fertilizer requirement** | |
|---|---|---|---|---|
| | | | **kg/ha** | **kg/bigha** |
| **Without Rhizobium culture*** | | | | |
| N | 15 | Urea | 32 | 4 |
| $P_2O_5$ | 35 | SSP | 220 | 30 |
| $K_2O$ | 15 | MOP | 25 | 3.5 |
| **With Rhizobium culture** | | | | |
| N | 10 | Urea | 22 | 3 |
| $P_2O_5$ | 35 | SSP | 220 | 30 |
| $K_2O$ | 15 | MOP | 25 | 3.5 |

The quantity of N is to be reduced proportionately to the quantity of N added in the form of FYM (each ton of FYM contributes about 5 kg N). Diammonium phosphate (DAP) 75 kg/ha or 10 kg/bigha should be applied in lieu of urea and SSP in non-inoculated crops.

If sulphur free fertilizer (DAP) is used, apply @ 20 kg S/ha as basal through gypsum (133 kg/ha or 18 kg/bigha) in addition to the recommended dose of N-$P_2O_5$-$K_2O$.

Apply Boron as basal @ 1.0 kg/ha (10 kg borax/ha or 1.30 kg borax/bigha along with the recommended dose of N-$P_2O_5$-$K_2O$ for all zones of Assam

**Seed Inoculation:**

For seed inoculation with *Rhizobium* cultures of Shillongani, Ahatguri, Kamalabari or any other suitable strains should be used. Seeds are inoculated with 50 g/kg of seeds. However, detailed instructions are available in each packet of the culture. Also inoculate seeds with PSB @ 50 g/kg seed along with rhizobium.

**Seed Rate**:

Sow black gram seeds @ 22.5 kg/ha or 3 kg/bigha

**Spacing:**

- Row to row : 30 cm
- Plant to plant : 10 cm

**Interculture:**

One weeding at 20-25 days after sowing is to be done.

Apply propaquizafop 2.5% + imazethapyr 3.75% (ready mix) @ 125 g /ha (commercial product @ 2 l / ha) at 15-20 DAS as post emergence

**Plant Protection:**

**A. Diseases:**

a) **Leaf spot** (*Cercospora* spp.): As soon as disease appears spray copper oxychloride @ 0.3% (1.8 – 2.0 kg/ha in 600-700 liters of water) at an interval of 7-10 days. Alternatively, Tebuconazole 25 EC @ 600-700 ml (0.1%) mixed with 600 – 700 liter of water/ha (80-90 ml in 80 – 90 liters of water/bigha) should be applied.

b) **Web Blight** (*Rhizoctonia solani*): Spray tebuconazole 25 EC @ 0.1% (600 ml -700 ml mixed with 600 – 700 liters of water/ha) three times starting from the appearance of the disease at 10 - 15 days interval.

*Rhizoctonia solani* induced damping off, root rot and seedling blight can be effectively managed by seed treatment with slurry method using commercial formulations of *Trichoderma* spp. @ 5 g/kg of seeds.

**B. Insect Pests:**

a) Against aphids, jassids, flea beetle, pod borers, pod bug sand leaf folder**,** spray chlorantraniliprole 18.5 SC @ 20 g *a.i.* /ha in 500-700 liters of water.

b) The spread of yellow mosaic virus (YVM) can be checked by controlling white fly (*Bemisia tobacci*) by applying thaimethoxam 25 WG @ 0.3g/*l*

**C. Nematode:**

For control of nematode *Meloidogyne incognita,* seed soaking with carbosulfan 25EC @ 0.01% for 6 hrs should be done.

**Harvesting:**

Harvesting is to be started when 75% of the pods mature indicating full darkish pod and

brittle on slight pressure.

**Protection against Storage Pests:**

Properly dried black gram seeds should be mixed thoroughly with black pepper seed powder @ 3 g/kg of seed, against bruchid infestation during storage. Treated seeds should be kept in poly bags with outer covering of gunny bags.

# Green gram

*(Vigna radiata* L.*)*

**Varieties:**

| Varieties | Duration (days) | Yield (q/ha) | Disease reaction | Protein content (%) | Cooking quality | Adaptation |
|---|---|---|---|---|---|---|
| SGC 16 | 65-68 | 12.52 | Resistant to CLS, YMV & moderately resistant to WB | 24.50 | Very good | Suitable for both kharif and summer for all zones of Assam except BVZ |
| SGC 20 | 65-68 | 12-13 | Resistant to CLS & YMV | 24.40 | Very good | Suitable for all zones of Assam except BVZ |
| Sonai (SG 21-5) | - | 9.93 | Resistant to CLS, YMV and MR to WB | 24.70 | | *Kharif* season |
| AAU SHN Mung 02 (Buroi) | 65-68 | 13-14 | Highly resistant to Mungbean yellow mosaic virus and *cercospora* leaf spot, moderately resistant to web blight. Moderately resistant to leaf roller, pod borer and storage pests under field condition | 24.40 | - | *Kharif season, s*uitable to all zones of Assam |

**Sowing Time:**

Sowing time is 15$^{th}$ August to 30$^{th}$ September. August is the suitable month for sowing in Central Brahmaputra Valley and Hill Zones. Sowing can be delayed up to 1$^{st}$ week of September without seed inoculation with PSB and up to 2$^{nd}$ week of September with seed inoculation with PSB @ 50 g/kg of green gram seed in Central Brahmaputra Valley Zone.

**Soil Type:**

Well drained sandy loam soil is preferable.

**Land Preparation:**

The land is to be ploughed 2-3 times followed by leveling. Stubbles should be removed completely. Surface drains should be provided to facilitate quick removal of excess water from the field.

**Liming:**

Requisite amount of lime is to be applied after soil test to bring soil pH around 6.0.

**Manures and Fertilizers:** Compost or FYM @ 1 t/ha or 1.3 q/bigha should be applied.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| **Without Rhizobium culture*** | | | | |
| N | 15 | Urea | 32 | 4 |
| $P_2O_5$ | 35 | SSP | 220 | 30 |
| $K_2O$ | 15 | MOP | 25 | 3.5 |
| **With Rhizobium culture** | | | | |
| N | 10 | Urea | 22 | 3 |
| $P_2O_5$ | 35 | SSP | 220 | 30 |
| $K_2O$ | 15 | MOP | 25 | 3.5 |

The quantity of N is to be reduced proportionately to the quantity of N added in the form of FYM (each tonne of FYM contributes about 5 kg N).

Diammonium phosphate (DAP) @ 75 kg/ha or 10 kg/bigha may be applied in lieu of urea and SSP in non-inoculated crop.

Soil application of 0.4 % (w/w) biochar (80 kg/bigha) produced from weed biomass or crop residue along with recommended dose of fertilizers.

**Seed Rate:** 20 kg/ha or 2.75 kg /bigha

**Spacing:**

- Row to row : 30 cm
- Plant to plant : 10 cm

**Seed inoculation with Rhizobium and PSB culture:**

For seed inoculation with rhizobium culture, either Majuli 10 or any other suitable strains may be used. Seeds should be inoculated with 150 g/3-4 kg seeds. Detail instructions area available in each packet of culture. Also inoculate seeds with PSB @ 50 g/ kg seeds along with Rhizobium.

**Interculture:**

One weeding at 20-25 days after sowing is to be done.

**Drainage:**

Excess rainwater can be drained out through surface drainage channel of 25 cm wide; 15 cm deep spaced at 6 m distance.

**Plant Protection:**

**A. Diseases:**

i) **Leaf spot** (*Cercospora* spp): As soon as disease appears spray copper oxychloride @

0.3% (1.8-2.0 kg in 600-700 liters of water/ha) at an interval of 7-10 days. Alternatively, after appearance of *Cercospora* leaf spot, tebuconazole 25EC @ 600-700 ml (0.1%) mixed with 600 – 700 liters of water/ha (80-90 ml in 80 – 90 liters of water/bigha) should be applied.

ii) **Blight** (*Rhizoctonia solani*): Spray tebuconazole 25 EC @ 0.1% (600 ml -700 ml mixed with 600 – 700 liters of water/ha) three times starting from the appearance of the disease at 10 - 15 days interval.

*Rhizoctonia solani* induced damping off, root rot and seedling blight can be effectively managed by seed treatment with slurry method using commercial formulations of *Trichoderma* spp. @ 5 g/kg of seed.

**B. Insect Pests:**

i) Spraying of spiromesifen 22.9 SC@1.25 ml/*l* of water at vegetative and reproductive stages of mung bean is effective against sucking pests.

ii) Against flea beetle, pod borers, and leaf folder, spray chlorantraniliprole 18.50 SC @ 20 g *a.i.* /ha ha or lamda-cyhalothrin 5EC@ 150 -250 ml/ha in 500-700 liters of water.

iii) The attack by yellow mosaic virus (YMV) can be checked by controlling white fly (*Bemisia tabaci*) as follows: 2-3 sprayings of fipronil 5 SC @ 1.5-2 ml/*l* of water or lamda-cyhalothrin 5 EC@ 150 -250 ml/ha is to be given, first spray at 10 days after germination and subsequent sprays at 15 days interval based on insect population.

**C. Nematode**:

**For management of *Meloidogyne incognita-***

- Seed soaking with carbosulfan 25EC @ 0.01% for 6 hrs.
- Alternatively, treat the seeds with Neem Seed Kernal Powder (NSKP) @ 5 g/kg of seeds
- Alternatively, treat the seeds with *Trichoderma viride* @ 5 g/kg of seeds

**Harvesting:**

Harvesting is to be done when 75% of the pods turn darkish in colour and brittle on slight pressure.

**Protection against Storage Pests:**

Properly dried green gram seeds should be mixed thoroughly with black pepper seed powder @ 3 g/kg of seed, against bruchid infestation during storage. Treated seeds should be kept in poly bags with outer covering of gunny bags.

# Horse gram

*(Macrotyloma uniflorum)*

**Varieties:**

| Crop | Variety | Plant Height (cm0 | Branch / plant | Cluster / plant | Seeds/ pod | Pods/ plant | 1000 seed weight (g) | days to maturity | Av. yield (q/ha) |
|---|---|---|---|---|---|---|---|---|---|
| Horse gram | Local (Kulthi) | 75-85 | 7-9 | 16-19 | 6-8 | 85-95 | 28.20 | 120-130 | 10-15 |

**Selection of site:** Well drained soil is suitable. Sandy and sandy loam fertile soils are preferable. Crop cannot thrive in waterlogged condition. Heavy rains during the initial stages of crop growth affect nodule formation owing to poor aeration in the soil

**Land preparation:** Land is to be ploughed 2-3 times followed by laddering. Cleaned the land by removing stubble. To facilitate removal of excess water surface drain should be provided

**Seed rate** : 30 kg/ha (4.0 kg per bigha)

**Spacing** : 30cm X 10cm

**Manure and fertilizer:** Compost or FYM @ 2 t/ha or 2.67 q/bigha should be applied

**Seed inoculation:** Seed should be inoculated with PSB culture @ 50 g/ kg of seeds along with *Rhizobium* @ 50 g/kg of seeds

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 15 | Urea | 32 | 4 |
| $P_2O_5$ | 35 | SSP | 220 | 30 |
| $K_2O$ | 15 | MOP | 25 | 3.5 |

**Time of sowing:** Mid-August to mid-September

**Weed Management:** Due to luxuriant growth an early weeding/hoeing at 20-25 days after sowing is enough

**Plant protection**

**A) Insect Pest:**

**i) Aphids**: Chlorantriniprole 18.5 SC @ 1 ml/3 liters of water or imadachloropid 17.8 SL@ 3 ml/10 liters of water

**ii) Pod borer**: Chlorantriniprole 18.5 SC @ 1 ml/3 liters of water or lamda-cyhalothrin 5 EC @ 150-250 ml/ha

**B) Disease:**

**iii) Yellow Mosaic (Virus vector-white fly)**: 2-3 spray of chlorantriniprole 18.5 SC @ 1ml/3 liters of water or lamda-cyhalothrin 5 EC @ 150-250 ml/ha at 15 days interval starting from 10 days after germination

**iv). Root rot**: Seed treatment with *Trichoderma viride* @ 5-10 g/kg of seed, avoid early sowing in infested areas and spot drenching with propiconazole @ 1 ml/lit

**Harvesting:** At maturity all the pod and leaves turn yellow. Harvesting is done by cutting the plants with sickle

# Cow pea

*(Vigna unguiculata)*

**Varieties:**

**RC19:** Recommended for all zones. Plant type erect and height is 80-120 cm. It bears 6-12 pods per plant and pods are 10-12 cm long. Average100 seed weight is 10g. Average yield 10-14 q/ha.

**TVX 644-02E**: Recommended for all zones. Plant type erect and height is100-120cm. Pods are long containing 10-14 seeds per pod. Seeds are creamy in colour and bold. Average 100 seed weight is 15 g. Average yield 12-15 q/ha.

**Sowing Time:** August

**Soil Type:** It can be grown on all types of soils. But well drained sandy loam soil is most suitable.

**Land Preparation:** The land is to be ploughed 2-3 times followed by leveling.

**Seed Rate:** The seed requirement is 30-35 kg/ha or 4.0-4.5 kg/bigha.

**Spacing:** A spacing of 30cm between rows and 10-15 cm between plants is to be given.

**Manures and Fertilizers:** Compost or FYM @ 4-5 t/ha or 6q/bigha is to be applied.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 15 | Urea | 32 | 4 |
| $P_2O_5$ | 35 | SSP | 220 | 30 |
| $K_2O$ | 0 | MOP | - | - |

**Interculture:** One weeding is to be done at 20-25 days after sowing.

**Plant Protection:**

**A. Disease:** No serious disease has so far been observed.

**B. Insects**: Three sprays of *Lecanicillium lecanii* $1x10^8$ cfu @ 5 ml/liters of water can be applied to manage aphid (*Aphis craccivora*). First spray should be done at 15-20 days after germination followed by second and third sprays at 15 days interval.

**Harvesting:** Cow pea for grain purpose is harvested by picking the mature pods. Threshing can be done by beating the dried pods.

**Uses:** Cowpea grains contain 24% protein and a rich source of carbohydrate, calcium, phosphorus and iron. The grain can be used as a pulse, either as whole or as *dal* and also as flour after husking or with husk. The pods can be used as vegetables when tender.

# Pigeon pea *(Arhar)*

(*Cajunus cajan* L.)

**Variety:**

**T-21** is recommended for all zones. It is a semi spreading, medium tall (170-185 cm in height) variety. Crop duration is 170-200 days.

**Time of Sowing:** June – July.

**Soil Type:** Well drained sandy loams to loamy soil are most suitable.

**Land Preparation:** The land is to be ploughed 3-5 times followed by harrowing and removal of weeds.

**Seed Rate:** 12-15 kg/ha or 1.6-2.0 kg/bigha.

**Spacing:** 75cm between rows and 30cm between plants.

**Manures and Fertilizers:** Compost or FYM @ 4.5 t/ha or6q/bigha is to be applied.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| **Without Rhizobium culture*** | | | | |
| N | 15 | Urea | 32 | 4 |
| $P_2O_5$ | 40 | SSP | 250 | 35 |
| $K_2O$ | 0 | MOP | - | - |
| **With Rhizobium culture** | | | | |
| N | 10 | Urea | 22 | 3 |
| $P_2O_5$ | 40 | SSP | 250 | 35 |
| $K_2O$ | 0 | MOP | - | - |

**Interculture:** One weeding is to be done at 20-25 days safter sowing.

**Use of honeybee in pigeon pea field for yield enhancement:** Installation of five (5) *Apis cerena* colonies/ha (1 colony/bigha) gives effective yield in pigeon pea.

**Plant Protection:**

**A. Diseases:**

**Wilt:** The pathogen inhabits the soil and perpetuates. Rogueing and burning of diseased plants partly help in controlling the disease.

**Average Yield:** 12-15q/ha.

**Inter cropping of *Arhar* with Green gram, Black gram and Sesamum:**

Intercropping of *arhar* with green gram/black gram and sesame for the hill slope (20-

25%) either in single/ double row or broadcasting in between rows of arhar is recommended. The crops are to be sown in lines across the slope. A basal dose of 15 kg N and 40 kg $P_2O_5$/ha are to be applied. To get optimum yield from both main and inter crop, the crops should be sown within the last week of July to first week of August.

# Soybean

*(Glycine max)*

**Varieties:**

| Varieties | *Agroclimatic zones | **Duration (days) | **Grain yield (q/ha) |
|---|---|---|---|
| KDS753 (Notified: 2020) | All zones | 110-115 | 6.0 |
| MAC1407 | All zones | 110-120 | 9.8 |
| JS20-116 | All zones | 100-115 | 8.0 |
| Dsb32 | All zones | 115-125 | 8.0 |

* Refer to page iv for full forms of agroclimatic zones
** Duration and grain yield are based on Jorhat center.
# These varieties can be sown even during first fortnight of September for seeds.

**Sowing Time:**

- **May-June:** North Bank Plains Zone, Upper Brahmaputra Valley Zone '& Central Brahmaputra Valley Zone
- **Mid-May-Mid June**: Hills Zone
- **June-April**: Lower Brahmaputra Valley Zone & Barak Valley Zone

**Soil:**

Soybean can be grown on wide range of soils but it grows best on fertile well drained loamy soils. Water logging is injurious to the crop. In acidic soils where pH is below 6.0, lime (200 mesh powder form) should be added @ 2 t/ha, at least 7days before sowing. Liming is necessary once in three years.

**Land Preparation:**

Soybean should be grown on well pulverized bed. Land should be prepared by giving deep ploughing followed by laddering. There should be provision of surface drains with gentle slope for good drainage.

**Fertilizers:**

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 20 | Urea | 45 | 6 |
| $P_2O_5$ | 60 | SSP | 375 | 50 |
| $K_2O$ | 40 | MOP | 70 | 10 |

Potash is required for healthy development of seeds. Foliar application of MOP is recommended in soils with low potash content.

Fertilizer should be placed in furrows at a depth of 5 to7 cm below the seeds and should be covered with a layer of 3-4cm soil and make it level with the field. In case placement is not possible, the fertilizers should be evenly broadcast and incorporated in the soil.

**Foliar-Spray of MOP:** MOP should be sprayed at the flower initiation stage between 30 and 35 days after sowing with hand sprayer for proper grain filling in pods.

Preparation of MOP solution for spraying with hand sprayer:

| Quantity of MOP (kg) | Quantity of water (l/ha) | Strength of MOP solution (%) | No. of spray | Type of sprayer |
|---|---|---|---|---|
| 12.00 | 400 | 3 | 1 | Hand Sprayer |

**Seed Inoculation:**

The seed should be moistened with clean water with care so as to avoid excessive wetting. Rhizobium culture should be mixed with seeds @ 15g/kg so that a thin coat of inoculum is deposited on each seed. The inoculated seeds should not be exposed to the sun. One packet (150 g approx.) of AAU culture which is available in the Production Unit of AAU is sufficient to inoculate seeds required for one bigha of land.

**Seed Rate and Time of Sowing:**

For branched type soybean varieties (Bragg, Alnkar, Ankur, Silajit, PK 271, and PK262) the seed rate is 60 kg/ha or 8.5 kg/bigha when sown early, normal and late to obtain a plant population of about 0.2 million/ha. For less branching type soybean varieties (JS2, Moti, Kalitur and Pusasoy bean) the seed rate is75 kg/ha or 10.5 kg/bigha for the same period of sowing to obtain a plant population of 0.4 million/ha. For late sown crop, seed rate is 80 kg/ha or 11.5 kg/bigha for a population of 0.6 million/ ha. Germination is affected if it rains within 72 hours of sowing.

Seeds are sown in lines 45 cm apart at a distance of 5 to10 cm between seeds. Depth of seedling should not exceed 4cm. The depth can further be reduced to 2 to 3 cm if moisture content in soil is high. The rows should be covered with a thin layer of paddy straw for better germination.

**Interculture Operations:**

Two manual weedings at 15 and 36 days after sowing should be done.

**Plant Protection:**

**A. Diseases:** Soybean seeds should be treated with Carboxin 2g/kg of seed.

*Rhizoctonia solani* induced dry root rot in soybean can be effectively managed by seed treatment with *Trichoderma*s spp by slurry method @ 5 g/kg of seed.

Seed Treatment with thiophanate methyl + pyroclostrobin (Combo) 500 FS @ 2 ml/kg seed and thiomethoxam 600 FS @ 2 ml/kg seed for management of root rot complex, pod blight disease and insect pests of soybean

**B. Insect Pests:** Spraying of flubendiamide 39.35 SC @ 72 g *a.i.* /ha should be done against hairy caterpillars, semi looper, leaf roller and flea beetle.

**Harvesting:**

At maturity all the leaves turn yellow and fall off and the stalks stand only with pods. Harvesting can be done at this stage by cutting the plants with sickles or by breaking at the base.

**Seed Storage:**

The grains should be dried before storage to ensure that the moisture content of seeds does not exceed 10 percent. Seeds can be stored in dry bins or polythene bags (250-gauge thickness) in airtight condition. Bins should be kept on wooden racks so that it does not come in direct contact with the ground. Treatment of seeds with Carboxin @ 2 g/kg of seed is absolutely necessary when stored for seed purposes.

**Drenching:**

Drenching of soybean seeds should be done before any kind of soybean preparation. For this purpose, 0.5% sodium carbonate solution, be prepared and mixed with seed @ 2 liters/ kg of seed and boiled for 20 minutes. Then the boiled seeds should be washed with plain water 2-3 times to remove the trypsin inhibitor.

**Use of Soybean:**

Soybean contain 40% protein and 20% high quality oils. Besides this it is also a source of vitamin B complex, thiamine and riboflavin. Several food products can be prepared from soybean.

**Soybean Milk:**

1. Seeds should be soaked in water for 12 hours.
2. Seed coats should be removed by rubbing the soaked seeds between palms.
3. A paste should be made by grinding the peeled seeds.
4. The paste should be boiled in water which should be about three times of its weight and the milk should not be over boiled. Otherwise, its nutritive value may be lost. A few drops of ginger juice and a few bay leaves should be added for removing the beany odour.

Soybean milk is useful against intestinal disorder. It prevents summer diarrhoea. It is easily digestible and can be used as diet for old debilitated and convalescents. Soybean can also be used for preparation of *Chapati, Kachuri, Malpowa, Bhujia, Ladoo, Chutni, Bor,* Cake etc.

**Soybean Curd:**

To prepare curd, soy milk should be poured in an appropriate container while it is moderately hot.

**Green Beans:**

Green soybeans are widely used as much in the same way as the pea is used. When the bean reaches its full size, but yet green and tender, it is collected from the plant. It is usually boiled and taken as green vegetable. Green beans are also used in salads. In order to secure the

best flavour, the beans should be cooked when they still remain affixed in the pods. In such cases, they are soaked in boiling water for a few minutes to make it easy to shell.

**Poultry Feed:** Soybean and maize mixture are an excellent poultry feed.

# Sesamum

*(Sesamum indicum)*

**Varieties:**

| Variety | Year of notification | Agro. Zone | Time of sowing | Duration (days) | Yield (q/ha) | Oil content (%) | Seed Colour | Pest & Disease reaction |
|---|---|---|---|---|---|---|---|---|
| AST-1 (AAUDT 9304-14-4-1, IC 551415) | 2020 | All zone | June to 1st Fortnight of August | 80-90 | 8-10 | 46 | Black | MR to Root Rot, Phyllody, Bacterial Blight and Mite |

**Soil:** Well drained sandy loam soils are preferable.

**Land preparation:** Land should be prepared to a fine tilth by 3-4 ploughings followed by laddering.

**Time of Sowing:** July to 31st August for *Phytophthora* blight endemic areas

**Spacing:** 30 cm between rows 15 cm between plants

**Seed Rate:** 4 kg/ha for both broadcasting and line sowing

**Manures and Fertilizers:** Compost or FYM should be applied @ 10 t/ha. Fertilizer requirements are as follows:

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 30 | Urea | 65 | 9 |
| $P_2O_5$ | 20 | SSP | 125 | 17 |
| $K_2O$ | 20 | MOP | 33 | 5 |

Apply the whole quantity of fertilizer at the time of sowing.

Apply N @ 30 kg/ha, P @ 20 Kg/ha, K @ 20 Kg/ha, B @ 1.5 Kg/ha, Zn @ 10.5 Kg/ha as basal along with FYM @ 3 t/ha and biofertilizer @ 50 gm/Kg seed as seed treatment (Azospirillum, PSB and KSB each)

**Drainage:**

Provide surface drains of 15 cm depth and 25 cm width spaced at 6 m distance connected to a collector drain by the side of the field.

**Weeding and Interculture:**

One weeding should be done at 20 days after owing by *khurpi* or hoe. The crop should be thinned out simultaneously to maintain the required spacing within the rows.

**Use of honeybee for yield enhancement:**

Five (5) *Apis cerena* colonies/ha (1 colony/ bigha) should be installed for effective pollination of Sesame and optimization of productivity.

**Plant Protection:**

Seeds should be treated with metalaxyl or carboxin @ 2 g /kg of seed against stem rot as well as phytophthora blight. Alternatively, seeds should be treated with a commercial formulation of *Trichoderma* spp. @ 5 g/kg of seeds against stem rot and phytophthora blight.

In phytophthora blight endemic areas, sesamum is to be rotated with cereals at every two years. In the case of phyllody disease, the infected plants should be rouged out.

Seed treatment with imidacloprid 70 WG @ 6 g/kg of seed followed by three sprays of thiomethoxam 25WG @ 2 g/10 liters of water at 20, 35 and 50 days after germination of sesame seeds

**Harvesting:**

The crop should be harvested when the leaves and stems turn yellow and defoliation starts, or one or two capsules at the bottom start cracking from the tip.

# Summer Sesamum

*(Sesamum indicum)*

**(For Hills Zone only)**

**Varieties:**

| Varieties | Year of notification | Duration (days) | Grain colour | Yield (q/ha) | Oil content (%) |
|---|---|---|---|---|---|
| AAU SHL TIL 1 (Champawati) | 2021 | 80 | Black | 9.04 | 46.0 -49.0 |
| AST-1 | - | 80-90 | Black | 8-10 | 46 |

**Sowing Time:** First week of March to first week of April.

**Spacing:** 30 cm between rows 15cm between plants

**Fertilizers:** Fertilizer requirement are as follows:

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 38 | Urea | 85 | 11 |
| $P_2O_5$ | 25 | SSP | 156 | 21 |
| $K_2O$ | 25 | MOP | 42 | 6 |

**Other practices are the same as in *kharif* sesamum**

# Groundnut

*(Arachis hypogaea)*

**Varieties:**

| Varieties | Plant height (cm) | Duration | Oil content (%) | Yield q/ha (nuts) |
|---|---|---|---|---|
| TG-37A (Notification: 2004) | 35-50 | 114 days | 48-51 | 13.8-21.7 |

**Soil Type:** Well drained sandy loam soil is suitable.

**Land Preparation:** Land should be prepared to a fine tilth by 3-4 ploughings followed by laddering. Deep ploughing should be avoided. Ploughing depth should not exceed 15 cm.

**Time of sowing:** June – July.

**Seed Rate:** 65-70 kg kernel/ha.

**Seed Treatment:** Treat the selected kernels with carboxin @ 2 g /kg of kernel.

**Method of Sowing:** Seed should be sown in line at 5 cm depth.

**Spacing:**

- Row to Row: 35 cm
- Plant to Plant: 15 cm

**Manures and Fertilizers:** Compost or FYM @ 10t/ha should be applied

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 20 | Urea | 44 | 6 |
| $P_2O_5$ | 40 | SSP | 250 | 33 |
| $K_2O$ | 30 | MOP | 50 | 7 |

**Weeding and earthing up:** One weeding should be given at 30-35 days (i.e. flower initiation stage) after sowing the seeds followed by earthing up to promote easy penetration of pegs into soil.

**Plant Protection:**

1) Spray chlorothalonil 75 WP or hexaconazole @ 2 g/*l* of water to control leaf spot (*Tikka*) disease. The first spray should be given as soon as the initial symptoms are detected followed by a second spray after 20 days of the first spray.
2) Seed treatment with imidacloprid 70WG @ 6 g/kg of seed followed by three sprays of thiomethoxam 25WG @ 2 g/10 liters of water at 20, 35 and 50 days after germination of sesame seeds

**Harvesting:** The stage of harvesting is attained when the vine begins to turn yellow and leaves start shedding. Harvesting is done by digging out the pods.

# Rice bean

*(Vigna umbellata)*

**(For Central Brahmaputra Valley Zone only)**

**Varieties:**

| Variety | Sowing time | Duration (days) | Grain yield (q/ha) | Disease reaction |
|---|---|---|---|---|
| RBL 4 | Mid-August to mid September | 88-93 | 19-21 | Resistant to YMV and Cercospora leaf spot |
| RBL 5 | -do- | 90-95 | 19-21 | -do- |
| Beziamah | -do- | 90-95 | 20-22 | -do- |

**Sowing Time**: Mid-August to mid-September

**Soil Type:** Preferably sandy loams oil

**Land Preparation:** Fine tilth is not necessary for cultivation of rice bean. The land is to be ploughed 2-3 times followed by leveling. The weeds should be removed completely. Surface drains should be provided to facilitate quick removal of excess water from the field.

**Liming:** Requisite amount of lime is to be applied after soil test in order to bring soil pH around 6.0.

**Manures and Fertilizers:** Compost or FYM should be applied @ 4-5 t/ha.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 20 | Urea | 44 | 6 |
| $P_2O_5$ | 40 | SSP | 250 | 33 |
| $K_2O$ | 0 | MOP | 0 | 0 |

**Seed Rate:**

| Variety | Kg/ha | Kg/bigha |
|---|---|---|
| RBL 4 and RBL 5 | 22-27 | 3.0-3.5 |
| Beziamah | 15-20 | 2.0-2.7 |

**Spacing:**

- Row to row : 30cm
- Plant to Plant : 10cm

**Interculture:** One weeding at 20-25 days after sowing

**Harvesting:**

Harvesting is to be started when 75% of the pods mature indicating full darkish pod, brittle on slight pressure.

## SUGARCROPS

# Sugarcane

## *(Saccharum* sp*)*

**Varieties:**

| Varieties | Maturity | Harvesting time (%) | Cane yield (t/ha) | *Gur* recover (%) | Sucrose content (%) | Reaction to insect pests | Special feature |
|---|---|---|---|---|---|---|---|
| Doria | Early | Dec-Jan | 70-80 | 10-12 | 18-19 | Moderately susceptible to top and Plassey borer | Profuse tiller, good ratooner and Nonflowering |
| Kalang | -do- | -do- | -do- | -do- | -do- | -do- | -do- |
| Barak | -do- | -do- | -do- | -do- | -do- | -do- | Profuse tiller, good ratooner, tolerant to lodging and non-flowering |
| Co 997 | Early as well as mid | Dec-Jan | 70-80 | 10-12 | 18-20 | -do- | Profuse tiller, good Ratooner & drought resistant |
| Dhansiri | -do- | -do- | -do- | 10-12 | 18-20 | -do- | Profuse tiller, quick growing & tolerant to waterlogging |
| Lohit | -do- | Dec-March | -do- | 10-12 | 18-20 | -do- | Profuse tiller and good ratooner |
| CoBln 94063 (Nambor) | Early | Dec-Jan | 70-90 | 1-12 | 18-19 | Moderately resistant to borers and red rot | Profuse tiller, good ratooner, tolerant to waterlogging of 10-30cm depth from June to Sept. and flood of 5-6ft depth during July/Aug |
| CoBln 9006 (Kapilipar) | Mid late | Feb-Mar | 70-85 | 10-12 | 20-21 | -do- | -do- |
| CoBln 02173 (Doiyang) | -do- | -do- | 70-80 | 11-13 | 20-21 | -do- | Profuse tiller, good ratooner |

**Selection of Setts:**

Top tender portion of the cane (sett) is especially suitable for planting. Late shoots and late planted canes also provide good planting materials. Each sett should be 3or 4 budded. Setts

should be free from red rot and borer infestation. It is necessary that planting materials are collected from fields/areas free from red rot.

For obtaining good quality planting materials in large quantities, it is advisable to raise a separate seed crop with extra care. Late planted crop *i.e*. planted in June-July provides good quality setts for planting in next spring. Besides the usual fertilizer dose of 135 : 70 : 60 kg/ha N : $P_2O_5$ : $K_2O$) an extra dose of 65 kg N (145 kg urea) should be top dressed to the crop in October. Adequate plant protection measures should also be taken to keep the crop free from pests/diseases.

For planting late in season (April-May) use of pre-germinated materials is beneficial. For this purpose, top portions of cane are to be germinated first in nursery by keeping them horizontally under light earth cover for a month and made to shorter cuttings with 3 buds in each, for planting. The leaves of the shoots are clipped off and planting should be done in rainy or cloudy weather.

**Treatment of Setts:**

Setts are to be treated before planting by dipping them in 0.2% carboxin or 0.1% azoxystrobin. In the process of treatment of setts when the solution gets reduced by about 50% it should be brought to the original volume by adding a solution of equal strength.

**Selection of site and Land preparation:**

Uplands and areas free from water stagnation should be selected. Trenches/furrows (25 cm width and 20 cm depth) should be made in well prepared soil at a distance of 75-90 cm. Bottom of the trenches should be loosened by light hoeing before application of manures and fertilizers.

**Manures and Fertilizers:** Compost or FYM has to be applied @ 10 t/ha intrenches/ furrows before planting cane. Besides, the following fertilizers are to be applied.

| **Nutrient** | **Requirement (kg/ha)** | **Form** | **Fertilizer requirement** | |
|---|---|---|---|---|
| | | | **kg/ha** | **kg/bigha** |
| N | 135 | Urea | 300 | 40 |
| $P_2O_5$ | 70 | SSP<br>or<br>MRP | 440<br><br>350 | 60<br><br>50 |
| $K_2O$ | 60 | MOP | 100 | 15 |
| **Alternatively, the following fertilizers may be used:** | | | | |
| N | 135 | Urea | 235 | 34 |
| $P_2O_5$ | 70 | DAP | 150 | 20 |
| $K_2O$ | 60 | MOP | 100 | 15 |

Granulated mixed fertilizer may also be used instead of the above fertilizers. The per hectare requirement of mixed fertilizer of 15 : 15 : 15 grade is 450 kg (64 kg/bigha) which should be applied in trenches/furrows at planting followed by topdressing of urea @ 150 kg/ha (21 kg/bigha).

**Time and Method of Application of Fertilizers:**

Entire quantity of phosphatic and half of potassic fertilizers are to be applied in furrows/trenches and mixed well with the soil before planting the setts. Nitrogenous fertilizers are to be applied in two splits, 1/3$^{rd}$ at planting and 2/3$^{rd}$ at first earthing up. The remaining half of the potassic fertilizer may be top dressed along with urea. Application of nitrogenous fertilizer should be completed within 90-100 days of planting.

**Method of Planting:**

Three or four budded setts should be planted by end-to-end method in trenches/furrows. Setts should be covered lightly with about 5cm of soil.

**Seed Rate:**

About 45,000 to 52,000 setts (6.5 to 7.5 t) are required for planting one hectare. The requirement per bigha would be about 6,000 to 7,200 setts weighing about 1 tonne.

**Time of Planting:**

March is suitable for planting sugarcane (spring planting). Early planting (late January-February) is advisable where irrigation facilities are available. When autumn planting (October planting) is done it is necessary to put a companion crop of mustard (single row) in between cane rows.

**Mulching:**

Planting setts in furrows and subsequently mulching the ridges with 50-micron plastic film and mulching the furrows with sugarcane trashes conserves soil moisture and gives higher cane yield.

**Weed Control:**

One weeding should be given within 30-35 days of planting followed by another within 60-90 days of spring planted crop. In October planted crops weeding should be done as and when necessary.

**Irrigation:**

Apply three irrigations of 6 cm depth during April, October and November at alternate furrows.

**Earthing Up:**

A light earthing up should be given to fill trenches/furrows within 1½-2 months after planting. The second earthing up should be done 1½ -2 months after the first earthing up. The second earthing up will transform the trenches/furrows into ridges. The furrows, thus made in between the ridges will facilitate drainage of excess water during high rainfall.

**Stripping and Propping:**

During the growth period of cane, the old and dry leaves should be removed.

The canes should be provided mechanical support to prevent lodging.

**Plant Protection:**

**a) Termites, red ants and white grubs:** Apply bifenthrin 10EC @ 100 g a.i. /ha

Drenching of termite mound soil with entomopathogenic nematode, *Hetrorhabditis bacteriophora* @ $5.0X10^9$ IJS/mound for three times at one month interval

**b) Borers:** Spread of stem borer attack in May-July can be checked by burying/burning of infested canes. Any of the following insecticides may also be used against these pests. Two to three rounds of fortnightly spraying starting from the rush of egg laying should be given.

| **Insect pests** | **Insecticides technical names** | **Quantity ( g *a.i.* /ha)** |
|---|---|---|
| Early shoot borer & Root borer | Fipronil 5SC | 75-100 |
| Early shoot borer, Root borer | Fipronil 0.3GR | 75-100 |
| Early shoot borer | Flubendiamide 20WG | 75 |
| Early shoot borer, Top borer | Chlorantraniliprole 18.5SC | 75 |

Spraying of chlorantraniliprole 18.5 SC @ 75 g *a.i.*/ha (8 ml/10 *l* of water) at 45 days after germination or initiation of dead heart symptoms followed by 2nd spraying after 30-45 days for managing borer pests

**c) Red Rot** (*Colletotrichum falcatum*): Red rot infested canes dry up and ultimately die. The canes become shriveled; the leaves and the leaf sheaths dry up and when the stem is split open characteristic reddening of internal tissue with white transverse and are observed.

Disease free setts should be used to prevent the spread of red rot. Setts from diseased cane or diseased fields should be avoided. Water stagnation in the field should be avoided and roguing of the affected plants should be practiced. Canes of the disease affected field should be harvested early and stubbles should be burnt. Field should be newly planted after 4-5 months. Ratooning should be discouraged.

**d) Wilt** (*Cephalosporium sacchari*): Measures recommended for red rot should be adopted.

**Harvesting:**

Sugarcane should be harvested at the ground level. Late suckers should be removed at the time of harvesting. Sugarcane is ready for harvest when the desired level of sucrose is attained in juice of different varieties.

In non-flowering varieties, the maturity is indicated by cessation of growth characterized by leaves appearing to emerge from a single point. In the case of flowering varieties, the harvesting has to be completed within two months of flowering.

**Sett Preservation:**

During dry months (December to April) setts can be preserved by adopting "deep trench trash-cover" method in which three-budded setts keeping in narrow trenches. Setts are to be covered with dry trash and water should be sprinkled at least twice a month.

**Ratoon Management:**

Field should be properly cleaned after harvesting and ridges should be broken down by hoeing or ploughing. Stubbles should be cut with a sharp knife at ground level for uniform establishment of the ratoon.

Gaps in ratoons need to be filled up by planting pre-germinated materials within a month of harvesting of the crop. One three-budded pre-germinated piece of sett for each 25 cm gap is sufficient.

Cow Dung/compost should be applied immediately after breaking of ridges. Nitrogen, Phosphorus and Potash @ 150, 70 and 60 kg per ha, respectively should be top dressed in two splits. Half of fertilizers should be applied at first earthing up and the remaining half should be applied at second earthing up, within 60 days of stubbles having.

For better control of weeds in ratoon, bispyribac sodium 10 SL @1 kg *a.i*/ha should be used as pre-emergence, followed by the same as post emergence application after 3 weeks of stubble shaving.

Other cultural practices are the same as in plant cane. Special attention should be paid to plant protection measures.

The ratoon crops can be raised profitably by proper management practices. Ratooning for more than two years usually makes the crop liable to greater damage by insect pests and diseases. Ratooning of red rot or heavily insect pest infested crops should be avoided.

## FIBRE CROPS

# Jute

## *(Corchorus sp.)*

**Varieties:**

| Variety | Year of notification | Sowing time | Fibre yield (q/ha) | Reaction to stem rot | Remarks |
|---|---|---|---|---|---|
| **A. *C. capsularis*:** | | | | | |
| JRC 212 (Sabujsona) | 1974 | Mar-April | 22-26 | Moderately resistant | Suitable for mid and high land |
| Apeswaree (*C capsularis*) | - | Mid Feb to April | 25.5 | Resistant to stem and root rot | Suitable for upland, medium and low land situation, better with high Fibre Tenacity, 25 g/tex |
| Kkhyati (AAUCJ-2) | 2017 | Mid Feb to April | 28.0 | Resistant | Suitable for all jute growing areas |
| **B. *C. olitorius* :** | | | | | |
| JRO 524 (Navin) | 1977 | Mid-March to May | 25-30 | Resistant | Suitable for medium low land, resistant to premature flowering |
| Tarun | 2007 | End of Mar to May | 31.50 | Resistant to Stem rot and root rot | Suitable for all agroclimatic zones |

**Soil:** Loam or sandy loam is preferable.

**Land Preparation:**

Jute requires a clean and fine seedbed. Land is to be ploughed and laddered 5-6 times to obtain a fine seedbed.

In soils where pH is below 5.5, lime should be applied at least 21 days before sowing @ 20-30 q/ha once every 3 years.

Seed Rate:

**A) For Line sowing:**

- *C. capsularis* : 6-7 kg/ha or 800-900 g/bigha.
- *C. olitorius* : 5-6 kg/ha or 660-800 g/bigha.

**B) Broadcast:**

- *C. capsularis* : 10-11 kg/ha or 1300 – 1500g/bigha.
- *C. olitorius* : 6-7 kg/ha or 800 - 900g/bigha.

**Seed inoculation:** Inoculate seeds with biofertilizers i.e. azotobacter @ 50 g/kg seed and PSB @ 50 g/kg seed.

**Spacing:**

For line sown crop, the spacing is 22-25 cm between rows and 5-7 cm between plants. Optimum plant stand should be 60-70/sq.m after thinning.

**Manures and Fertilizers:**

Wherever possible cowdung or compost should be applied @ 5 t/ha during land preparation; and the amount of nutrients thus supplied will be reduced by corresponding reduction from the recommended dose of fertilizer. One ton of cowdung or compost supplies approximately 5 kg N, 2-5 kg $P_2O_5$and 5.0 kg $K_2O$.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| **A. For *Capsularis*:** | | | | |
| N | 40 | Urea | 88 | 12 |
| $P_2O_5$ | 25 | SSP | 156 | 20 |
| $K_2O$ | 30 | MOP | 50 | 7 |
| **B. For *Olitorius*:** | | | | |
| N | 30 | Urea | 66 | 9 |
| $P_2O_5$ | 25 | SSP/MRP | 156/125 | 20/15 |
| $K_2O$ | 25 | MOP | 42 | 6 |

**Note**: In case of phosphatic fertilizers "Mussorie phos" can be substituted for SSP in *Olitorius* jute at least 3 weeks ahead of final land preparation.

**Mode of Application:**

Apply 50% and 50% $P_2$-$O_5$ and 100% $K_2O$ as basal dressing at the time of final land preparation and the remaining fertilizers at 4-6 weeks after sowing (CBZ, BVZ & NBPZ).

Top dressing of fertilizer mixture grade 8 : 10 : 12 at 15 DAS and MOP with subsequent foliar spray of urea is recommended.

**Foliar Spray of Urea:**

Where top dressing is not possible, foliar spray of urea is advocated. Urea (11.5 kg N/ha) is sprayed between 40-60 days after sowing. The first dose of urea should preferably be given 40-50 days after sowing with a low volume power sprayer (Micronette) and the second spray is to be given after 10-15 days. With Aspee Bolo sprayer, three sprayings are required, the first spray being followed by two more sprays at an interval of 10 days. With hand operated (high volume) sprayer, it will be necessary to spray 2 times; the first one (40-45 DAS) being followed by another spray at 55-60 days after sowing.

**Preparation of Urea Solution for Foliar Spray:**

| Amount of Urea (kg) | Amount of water (lit) | Strength of urea solution (%) | Nos. of spray | Type of sprayer |
|---|---|---|---|---|
| 12.50 | 90 | 14% | 2 | Micronette power sprayer |
| 8.50 | 85 | 10% | 3 | Aspee Bolo power sprayer |
| 12.50 | 420 | 3% | 2 | Hand sprayer |

**INM for Jute (*Olitorius*)**

***Zones**:* CBVZ, NBPZ & LBVZ

Apply N-$P_2O_5$-$K_2O$ @ 7.5-13-25 kg/ha as basal + Biofertilizer *Azospirillum* and PSB @ 50 g each/kg of seed. Apply 7.5 kg N/ha at 4-6 weeks after sowing.

| Nutrients | Requirement (kg/ha) | | Source | Fertilizer requirement | | | |
|---|---|---|---|---|---|---|---|
| | | | | At sowing | | After 4-6 weeks | |
| | At sowing | After 4-6 weeks | | kg/ha | kg/bigha | kg/ha | kg/bigha |
| N | 7.5 | 7.5 | Urea | 16.3 | 2.2 | 16.3 | 2.2 |
| $P_2O_5$ | 13 | - | SSP | 81.2 | 10.8 | - | - |
| $K_2O$ | 25 | - | MOP | 41.8 | 5.6 | - | - |

**Interculture:**

Thinning of the plants should be done 3-4 weeks after emergence. One hand weeding is to be given at the time of thinning. Wheel hoeing between rows within 4-5 weeks after sowing suppresses weeds considerably.

In broadcast crop *bindha* should be used 2-3 times at 2-3 weeks of crop age. Thereafter, thinning should be done to maintain a spacing of 12-13 cm between plants to ensure a population of about 6-7 lakh/ha (60-70 plants/sq. m).

**Chemical Weeding:**

Fluchloralin @ 3 *l*/ha in 1000 liters of water is to be applied as pre- emergence spray 3 days before sowing.

**Harvesting and Retting:**

Jute is to be harvested at the small pod stage to get optimum yield and good quality fibre. If jute is succeeded by rice, it can be harvested even at 100-120 days to facilitate timely ploughing for rice crop. Plants are to be cut close to the ground.

The harvested plants should be made in bundles of 15-20 cm diameter and left in the field in standing position for 2-4 days not only for shedding of leaves but also to facilitate the retting process. Thin and thick plants must be bundled separately.

Retting is the most important factor responsible for producing fine quality of jute fibre.

The bundles are to be arranged side by side in water and tied together so as to form regular platform (*Jak*) and if possible, a second layer of bundles may also be placed on top and covered with materials like water hyacinth, *keturi* leaf, paddy straw thatch etc. The *'Jak'* should be submerged by putting weight of concrete slabs, bricks, stone or dry log, use of mud clods, banana stems and freshly cut logs as materials for weight should scrupulously be avoided.

Optimum temperature of water for retting ranges from 32-34°C. Depending upon the temperature condition the retting process takes about 10-30 days to complete when the fibre separates easily from the wood. Over retting and under retting should be avoided.

To accelerate the process of retting one or two sticks of *Dhaincha* or Sunhemp should be inserted in each jute bundle before steeping. This provided substratum for quick microbial activity leading to multiplication of microbes and intensification of retting process.

Apply CRIJAF sona microbial consortium equally in all the three layers of bundles of jute in the retting jak @ 4.0 kg for the produce of one bigha (**Zones***:* CBVZ, NBPZ & LBVZ)

The fibre is to be extracted from each jute stalk separately and the stripped fibre should be washed thoroughly in clean water and dried in mild sunlight over a bamboo bar for 2-4 days.

Apply 37.5 kg of bacterial consortium formulation (PATSHAKTI) in the JAK of jute harvested from 1 hectare, to reduce the ratting duration and to improve fibre quality.

**Crop Rotation:**

Crop rotation in the jute field is most important to maintain soil fertility and also to minimize the incidence of insect pests and diseases. The sequences of jute-rice- rapeseed and jute-rice-wheat have been found profitable and are recommended for adoption wherever possible.

**Jute based cropping system for upland rainfed situation:**

A cropping sequence of Jute (JRO 524 or JRO 7835) – Black gram (SB-121 or Saonia mah) - *Toria* (TS-38 or M-27 or TS-36) has been recommended for upland rainfed conditions.

**Plant Protection:**

**A. Insect Pests:**

**i) Jute hairy caterpillar, Semi looper, yellow mite, Stem weevil:**

(a) Hairy caterpillar egg masses and caterpillars are to be handpicked and destroyed by putting them in kerosenized water. For semilooper, perches may be fixed in jute fields or nearby to facilitate predatory birds to sit on.

(b) Two sprays of chlorantraniliprole 18.5 SC @ 0.3 ml/*l* or lambda cyahalothrin 5 EC @ 0.6 ml/*l* of water is effective against jute hairy caterpillar (*Spilosoma obliqua)* and semilooper (*Anomis sabulifera).* First spray should be done at the appearance of the pest followed by second spray after 15 days.

**B. Diseases:** Root rot, stem rot, seedling blight and anthracnose:

a) Soil pH should be raised to 5.5 by application of lime.

b) Application of potash should be increased up to 50 kg $K_2O$/ha.

c) Jute-Rice crop rotation should be followed.

d) Moderately resistant varieties like JRC 212, JRC 7447, UPC 94, JRO-524 and resistant variety Tarun should be grown.

e) For managing damping off, apex rot and seedling blight: For managing these diseases caused by soil borne *Macrophomina phaseolina,* seed treatment with slurry method using commercial formulation of *Trichoderma* spp. @ 5 g/kg seed is effective.

f) Seed treatment with a combination of azoxystrobin 18.2 + difenoconazole 11.4 SC @ 1.0 ml/kg seed followed by spray @ 0.75 ml/*l* at 40-45 days of crop age for the management of induced disease complex in jute (seedling blight, stem rot, root rot) caused by *Macrophomina phaseolina.*

**C. Integrated pest and disease management module for *olitorius* jute:**

- Soil application of *Trichoderma viride* @ 2.5 Kg /ha (mixed with 150 Kg FYM, covered with moist gunny bag and incubated for 48 hours in shade) in soil at the time of sowing.
- Manual weeding at 3-4 weeks after sowing
- Hand picking and destruction of egg masses and larvae of Bihar hairy caterpillar
- Erection of bamboo perches @ 40 Nos./ha
- Two sprays of neem oil @ 4 ml/liter of water at 2$^{nd}$ week of June and 1$^{st}$ week of July.
- One spray of recommended insecticide (if necessary).

# Jute (Seed Crop)

(*Corchorus sp.*)

**Land preparation:**

Land is to be ploughed, cross ploughed and laddered 5-6 times. Weeds and stubbles are to be removed to obtain a fine and clean seedbed.

**Soil:**

Loamy and sandy loam in upland situation

**Time to Sowing:**

June for Central Brahmaputra Valley Zone July and August for other zones

**Seed Rate:**

**For line sowing**

*Capsularis* : 6 kg/ha or 800 g/bigha

*Olitorius* : 4 kg/ha or 550 g/bigha

F**or broadcasting:**

*Capsularis* : 10 kg/ha or 1.5 kg/bigha

*Olitorius* : 7 kg/ha or 1.0 kg/bigha

**Spacing:**

**Line sowing**: 30 cm × 15 cm (ensuring plant population about 2.2 lakh/ha)

**Manures and Fertilizers:**

Compost or FYM @ 5 t/ha or 7 q/bigha should be applied. The quantity of nitrogen is to be reduced proportionately. Each tonne of FYM /compost supplies 5 kg of nitrogen.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 20 | Urea | 44 | 6 |
| $P_2O_5$ | 20 | SSP | 125 | 16 |
| $K_2O$ | 20 | MOP | 32 | 4.5 |

The whole quantity of P and K fertilizers should be applied as basal dressing and a full dose of N fertilizer should be applied as top dressing at 3-4 weeks of crop age. *Musoori phos* can be substituted for SSP.

**Interculture:**

Usually, two manual weedings are to be given, first at 3-4 weeks along with thinning operation and second weeding at 5-6 weeks of crop age

**Plant Protection:**

**A. Diseases:** For control of stem and root rot, azoxystrobin 18.2SC + difenoconazole 11.4SC @ 0.75 ml/*l* is to be applied for effective control of diseases.

**Harvesting:**

Harvesting is to be done at full maturity but before the pods start shattering. The jute plants should be well dried on the threshing floor and seeds can be expelled by beating. Seeds are then winnowed, sun dried and kept for storage.

**Average Seed Yield:**

*Capsularis* : 4-5 q/ha

*Olitorius* : 3-4 q/ha

# Mesta

(*Hibiscus* spp.)

**Varieties:** HC 583 and AMC 108

**Time of Sowing:** April – mid May

**Soil:** Well drained loam to clayey loam soil is suitable.

**Land Preparation:**

Land should be ploughed and laddered 5-6 times to obtain a fine tilth.

**Manure and Fertilizers:**

FYM or compost @ 7-8 t/ha should be applied during land preparation. The quantity of N is to be reduced from the fertilizer on the basis of 5 kg N per tonne of FYM/compost applied in the field.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 40 | Urea | 90 | 12 |
| $P_2O_5$ | 20 | SSP | 125 | 16 |
| $K_2O$ | 20 | MOP | 32 | 5.2 |

**Mode of Application of Fertilizers:**

The entire quantity of phosphatic and potassic fertilizers should be applied at the time of final land preparation. Nitrogenous fertilizer should be top dressed in two equal splits at 4-5 and 7-8 weeks after emergence.

**Seed Rate:**

**a) Line sowing** : 13-15 kg/ha (1.8-2.0 kg/bigha)

**b) Broadcasting** : 18-20 kg/ha (2.5-3.0 kg/bigha)

**Spacing:** 30 cm between rows

15 cm between plants

**Weeding and Thinning:**

One hand weeding at 20-30 days after emergence is to be given and plants are thinned out simultaneously to maintain requisite plant population. In line sown crop, wheel hoe at 2 and 5 weeks after emergence should be used.

**Retting and extraction of fibre:**

The process of retting and extraction of fibre are similar to that of jute.

**Fiber Yield:** 22-25 q/ha.

# Cotton

(*Gossypium* spp.)

**Varieties:**

| Varieties | Plant height (cm) | No. of branches/plant | No. of balls/plant | Seed cotton yield (q/ha) | Ginning (%) |
|---|---|---|---|---|---|
| **A. Long staple**: | | | | | |
| MCU 4 | 94 | 7.7 | 19.0 | 10.50 | 34 |
| MCU 1 | 105 | 5.7 | 31.3 | 12.26 | 33 |
| Laxmi | 120 | 7.7 | 19.0 | 8.12 | 31 |
| Buri | 102 | 7.4 | 29.9 | 12.23 | 31 |
| J 34 | 130 | 11.6 | 44.3 | 12.21 | 32 |
| **B. Short staple** : | | | | | |
| Garo | 171 | 3.0 | 41.6 | 6.29 | 28 |
| Mizo Khaki | 183 | 3.0 | 38.6 | 6.53 | 23 |
| Dambuk | 147 | 3.0 | 39.6 | 5.68 | 35 |
| Phelopi | 155 | 3.7 | 25.7 | 3.71 | 24 |

**Soil:**

Cotton can be grown on a variety of soils. It requires well drained soil as it cannot tolerate water logging.

**Land Preparation:**

Cotton does not require fine seedbed preparation. Deep tilled coarse seedbed is required; 3-4 ploughings followed by laddering are sufficient.

**Time of Sowing:** July

**Treatment of Seed:**

Seeds are rubbed with a paste consisting of a mixture of cowdung and moist earth and then dried for some time in shade.

**Method of Sowing:** Dibbling of 4-5 seeds/hole in line

**Seed Rate:** 20 – 25 kg/ha

**Spacing:** 45 cm between rows and 30 cm between plants.

**Depth of Seedlings:** 1.5 cm to 2.0 cm.

**Manure and Fertilizers:** Compost or FYM should be applied @ of 10-20 t/ha.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 60 | Urea | 130 | 18 |
| $P_2O_5$ | 30 | SSP | 187 | 27 |
| $K_2O$ | 30 | MOP | 50 | 7 |

Compost or FYM is to be incorporated with the soil during first ploughing. The entire quantity of P and K and half of N fertilizers should be applied as basal dressing during final land preparation before sowing. Remaining half of N should be applied 40 days after sowing at the flowering stage.

**Interculture:**

One hand weeding along with thinning is to be given at 15 days after sowing.

**Picking:**

The cotton is ready for first picking by the end of November. Generally, 3-4 pickings are required. Picking in the morning hours is better to avoid an admixture of dried leaves and bracts. Seed cotton is dried for a few hours after picking and then stored.

**Yield:**

Average yield 8-12 q/ha (seed cotton) for long staple varieties and 4-6 q/ha (seed cotton) for short staple varieties.

# Ramie

(*Boehmeria nivea*)

**Varieties:**

R 1411, R 1449, R 1452, R 67-34 (Kanai)

**Type of Soil:**

Sandy loam to loam having a pH value of 5.5-6.0

**Climate:**

Warm (25-31$^0$C) and moist

**Altitude:**

Below 300 m above mean sea level.

**Propagation:** Mainly by rhizome, seed and stem cuttings.

**Time of planting:** May – September.

**Spacing:** Row to row 60 cm and plant to plant 30 cm

**Depth of planting:** 5-6 cm

**Time of staging:** 3rd week of April in the next year

**Fertilizer:** N: P: K (kg/ha) 30 : 15 : 15 kg/per cutting

**Requirement/ha:**

| | | | |
|---|---|---|---|
| a) | Rhizome | : | 375-400 kg |
| b) | Lime | : | 3.5 – 4 t (if acidic) |
| c) | Urea | : | 65 kg/cutting |
| d) | SSP | : | 94 kg/cutting |
| e) | MOP | : | 26.0 kg/cutting |
| f) | Ramie compost | : | 20.0 t/year |
| g) | Cowdung | : | 10-12 t/year. |

**Harvesting:**

Four cuttings are taken per year. 1st cutting 50 DAP, 2nd cutting 45 days after 1st cutting, 3rd cutting 45 days after 2nd cutting, and 4th cutting 50 days after 3rd cutting.

**Yield:**

**Fibre yield**: 16-22 q/yr/ha

**Rhizome yield:** 10-12 t/ha after 5th year

# FODDER CROPS

## Dinanath grass

(*Pennisetum pedicellatum*)

**Varieties:**

JP-12, Pusa 3 and Pusa 19

**Land Preparation:**

Well drained land should be selected. One ploughing by mould board plough followed by 3 to 4 harrowings and planking are sufficient for land preparation. Growth of seedlings is very slow at the early stage i.e. up to 35 to 40 days. Therefore, it is very important to keep the field free from weeds at this stage.

**Time of Sowing:** April to June.

**Seed Rate:** 8 kg/ha

**Spacing:**

Row to row: 30 cm

Plant to plant: 10 cm

**Manures and Fertilizers:** Apply compost or FYM @ 4-6 t/ha or 6-8 q/bigha

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 60 | Urea | 132 | 18 |
| $P_2O_5$ | 30 | SSP | 186 | 25 |
| $K_2O$ | 0 | MOP | - | - |

30 kg N and a full dose of $P_2O_5$ should be applied as basal dressing i.e. at the time of final land preparation and rest 30 kg N as top dressing after first cutting.

**Number of Cuttings:**

Two cuttings may be done. First cutting at 60 days after sowing and the 2nd cutting is to be done at 45 days after the first cutting.

**Green Fodder Yield:** 50-60 t/ha in two cuttings

# Cowpea (Fodder) (*Vigna unquiculata*) and Rice Bean (Fodder)

(*Vigna umbellata*)

**Varieties:**

Cowpea : No 10, No. 42/1, EC 4216

Rice bean : K-1

**Rice bean varieties recommended:**

| Variety | Sowing time | Duration upto 0% flowering stage (days) | Green forage yield (q/ha) | Dry matter yield (q/ha) | Disease & insect pest incidence | Agroclimatic zone |
|---|---|---|---|---|---|---|
| Bidhan rice bean-1 | July-Aug | 70-80 | 250-290 | 50-60 | Less | All zones except Hills zone |
| Bidhan rice bean-2 | -do- | 70-80 | 250-300 | 50-60 | Less | -do- |
| Shyamalee-1 | -do- | 70-80 | 300-320 | 50-65 | Less | All zones |

**Land Preparation:**

Well drained land should be selected. One ploughing by mould board plough followed by 2 to 3 harrowings is sufficient to prepare the land.

**Time of Sowing:** April – May.

**Seed Rate:** Cowpea : 50 kg/ha

Rice bean : 25 kg/ha

**Spacing:** Row to row : 35 cm

Plant to plant : 10 cm

**Manures and Fertilizers:** Apply compost or FYM @ 3-4 t/ha or 4-6 t/bigha.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 20 | Urea | 44 | 6 |
| $P_2O_5$ | 40 | SSP | 248 | 35 |
| $K_2O$ | 20 | MOP | 32 | 5 |

**Time of Cutting:** One cutting at 50% flowering stage (after 70 days of sowing)

**Green Fodder Yield:** 20 – 25 t/ha.

# Maize Fodder

(*Zea mays*)

**Varieties:**

Ganga 5, Vijoy and African Tall

**Land Preparation:**

For maize crop, a good seedbed is very much essential. The field should be initially ploughed and harrowed 4-5 times followed by planking.

**Time of Sowing** **:** Before June

**Seed Rate** **:** 60 kg/ha

**Spacing** : Row to row : 30 cm

Plant to plant : 10 cm

**Manures and fertilizers:** Apply compost or FYM @ 3-4 t/ha or 5-6 q/bigha.

| **Nutrient** | **Requirement (kg/ha)** | **Form** | **Fertilizer requirement** | |
|---|---|---|---|---|
| | | | **kg/ha** | **kg/bigha** |
| N | 60 | Urea | 132 | 18 |
| $P_2O_5$ | 30 | SSP | 186 | 27 |
| $K_2O$ | 30 | MOP | 82 | 7 |

30 kg N, the whole of $P_2O_5$ and $K_2O$ should be applied as basal dressing at the time of final land preparation and 30 kg N as top dressing after 40 days of sowing.

**Irrigation:**

Normally the rainy season crop does not require irrigation. When there is no rain and soil becomes dry the crop should be immediately irrigated.

**Number of Cuttings:**

One cutting in a year after 75 days of sowing is recommended. To get optimum yield, maize should be cut for green fodder at the silking stage.

# Teosinte

*(Euchlaena maxicana)*

**Varieties:** Sirsa, TL 16, TL 2

**Soil Type:** Well drained sandy loam soil is preferable.

**Land Preparation:**

One ploughing by mould board plough followed by 4-5 harrowings and plankings are sufficient for land preparation. Stubbles and weeds are to be removed completely. Surface drains should be provided for immediate removal of excess water from the field.

**Time of Sowing:** April to June

**Seed Rate:** 40 kg/ha

**Spacing** **:** Row to row : 40 cm

Plant to plant : 15 cm

**Manures and Fertilizers:** Apply compost or FYM @ 3-4 t/ha or 5-6 q/bigha.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 60 | Urea | 132 | 18 |
| $P_2O_5$ | 30 | SSP | 188 | 25 |
| $K_2O$ | 20 | MOP | 33 | 5 |

Apply 40 kg N, the whole of $P_2O_5$ and $K_2O$ as basal dressing at the time of final land preparation and 20 kg nitrogen as top dressing after 40 days of sowing.

**Interculture:**

One weedings at 25-30 days of sowing is sufficient.

**Irrigation:**

Normally the rainy season crop does not require irrigation. When there is no rain and soil becomes dry the crop should immediately be irrigated. The first irrigation should be given before sowing of seeds to get optimum soil moisture for proper germination of seeds. Subsequent irrigation will depend upon rainfall.

**Time of Cutting:**

One cutting is to be taken at 70-80 days of sowing. To get optimum yield, teosinte should be cut at the silking stage.

**Green Fodder Yield**: 30 t/ha

# FORAGE CROP (PERENNIAL)

## Hybrid Napier

*(Pennisetum purpureum)*

**Varieties:** NB 21, IGFRI-6, CO-2, CO-4, CO-5

**Soil Type:** Well drained soils should be selected.

**Land Preparation:** One ploughing by mould board plough followed by 2-3 harrowings and puddlings are sufficient to prepare the land.

**Time of Planting:** April to June

**Spacing:** Row to row : 50 cm

Plant to plant : 50 cm

**Manures and Fertilizers:** Apply compost or FYM @ 8-10 t/ha or 1-1.5 t/bigha.

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 120 | Urea | 265 | 35 |
| $P_2O_5$ | 50 | SSP | 310 | 45 |
| $K_2O$ | 30 | MOP | 48 | 7 |

Nitrogenous fertilizer should always be applied in 4 splits. Entire quantities of phosphatic and potassic fertilizers along with the first split of nitrogen are to be applied as basal dressing and the other 3 splits at the time of intercultural operation and after alternate cutting.

Fertilizer savings to the tune of 50% could be possible by application of vermicompost and FYM @ 2.5 t/ha each along with 50% recommended dose of fertilizer.

**Irrigation:**

First irrigation should be given immediately after planting. Subsequent irrigation will depend upon the amount of rainfall.

**Number of Cuttings:**

During the first year 4-6 cuttings may be done. First cutting is done 3 months after planting and subsequent cuttings are done at an interval of 5-7 weeks.

**Green fodder yield:** 80 – 100 t/ha

## *Setaria* grass

(*Setaria sphacelata*)

**Varieties** **:** Kazungula, Nandi and PSS-1

**Soil Type** **:** Well drained upland soils rich in organic matter should be selected.

**Land Preparation:**

One ploughing by mould board plough, followed by 2-3 harrowings and plankings are sufficient to prepare the land.

**Time of Planting:** April to June

**Spacing** **:** Row to row: 50 cm

Plant to plant: 50 cm

**Rooted Slips Requirement**: 40,000/ha (5,000/bigha)

**Manures and fertilizers:** Apply compost or FYM @ 5 t/ha or 7 q/bigha

| Nutrient | Requirement (kg/ha) | Form | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 120 | Urea | 265 | 35 |
| $P_2O_5$ | 50 | SSP | 310 | 45 |
| $K_2O$ | 30 | MOP | 48 | 7 |

Nitrogenous fertilizer @ 40 kg/ha should be applied as basal and the rest amount of nitrogenous fertilizer should be applied @ 20 kg/ha after each cut. The entire quantities of phosphatic and potassic fertilizers are to be applied along with the basal dose of nitrogen (40 kg/ha).

Fertilizer savings to the tune of 50% could be possible by application of vermicompost and FYM @ 2.5 t/ha each along with 50% recommended dose of fertilizer.

**Interculture:**

Intercultural operation in the form of light hoeing is necessary at 30-35 days of planting and also after each cut.

**Irrigation:**

First irrigation should be applied immediately after planting if a dry spell prevails. Subsequent irrigation will depend upon rainfall.

**Number of cuttings:**

First cutting is to be taken at 60-65 days after planting and subsequent cutting at an interval of 45-60 days.

**Management of old stand of setaria grass:**

Stubble shaving at ground level or burning of stubbles in the month of February along with intercultural operation after each cut is recommended.

**Green Forage Yield:** 800 to 1000 q/ha

# Congosignal grass

(*Brachiarea ruziziensis*)

***Zones:*** All Zones

**Varieties**: DBRS-1, DBRS-7

**Soil Type**: Well drained upland soils rich in organic matter should be selected

**Land Preparation**: One ploughing by mould board plough, followed by 2-3 harrowing and planking are sufficient to prepare the land

**Time of planting**: As congosignal is a perennial grass the first year is an establishment year. Planting can be started from March with the onset of pre-monsoon showers and can be continued up to July to get good yield in the first year. The crop can be continued up to five years.

**Spacing**: Row to row: 50 cm; Plant to plant: 50 cm

**Rooted slips requirement:** 40,000/ha (5000/bigha)

**Manures and fertilizers:** Apply compost or FYM @ 5t/ha or 7q/bigha

| Nutrient | Requirement (kg/ha) | Form of fertilizer | Fertilizer | |
|---|---|---|---|---|
| | | | kg/ha | kg/bigha |
| N | 120 | Urea | 264 | 35 |
| $P_2O_5$ | 50 | SSP | 313 | 42 |
| $K_2O$ | 30 | MOP | 50 | 7 |

Nitrogenous fertilizer @ 40 kg/ha should be applied as basal and the rest amount of nitrogenous fertilizer should be applied @ 20 kg/ha after each cut. The entire quantities of phosphatic and potassic fertilizer along with the first split of nitrogen are to be applied as basal dressing. 25 % chemical fertilizer can be saved when both FYM + vermicompost is applied @ 2.5 t/ha.

**Intercultural Operation:**

Intercultural operations in the form of light hoeing are necessary at 30-35 days of planting and also after each cut.

**Irrigation:**

First irrigation should be applied immediately after planting if a dry spell prevails. Subsequent irrigation will depend upon rainfall.

**Number of cuttings:**

First cut is to be taken at 60-65 days after planting and subsequent cut at an interval of 40-45 days.

**Green Forage Yield:** 110-120 t/ha

# OTHER CROPS

## Broom grass

*(Thysanolaena maxima)*

**Zones*:*** All Zones

**Variety:** GBGC-1

**Soil type:**

It is grown on marginal lands and wastelands. It also grows well on a wide range of soils from sandy loam to clay loam and is free from Waterlogged condition of soil.

**Land Preparation:**

Land should be clean, cross plough the soil and well level the field.

**Time of Planting**:

Mid of March - mid of May is the optimum time of planting.

**Time of harvesting:**

Harvesting starts from Mid-February and continues till March end.

**Spacing and Plant Population:**

Row to Row spacing 1.5 m and plant to plant spacing 1.5 m to maintain the tussock or hill population 4, 444 tussock or hill /ha

**Fertilizer and Manure:**

| Nutrients | Requirement (kg/ha) | Source | Fertilizer requirement | |
|---|---|---|---|---|
| | | | kg/ha | g/pit |
| N | 60 | Urea | 130 | 29.25 |
| $P_2O_5$ | 45 | SSP | 281 | 63.23 |
| $K_2O$ | 30 | MOP | 50 | 11.25 |

Planting the broom grass slips/rhizome in fertile soil ensures the best yield. For first year, Farmyard manure 5-10 t/ha or 1-2 kg FYM + 14.62 gm Urea + 63.23 gm SSP + 11.25 gm MOP fertilizer per pit are mixed into the pits before planting, remaining 14.62 gm Urea should be applied before flowering initiation stage in ring method.

From the second year, the whole amount of 29.25 gm Urea + 63.23 gm SSP + 11.25 gm MOP fertilizer should be applied in ring method after harvesting of 1$^{st}$ crop and follow the second-year doses in the subsequent year.

**Intercultural operation:**

In order to grow broom grass the slips/ Rhizome in the planting site must be clear of weeds and debris. Intercultural operation in the form of light hoeing is necessary at 30-35 days of planting and also after each cut. Application of weedicide Glycides @ 2.5 ml/*l* is

recommended to control weeds.

**Gap filling:**

Gap filling should be done 1 month after transplanting to ensure uniformity of plant population.

**Irrigation:**

Generally, irrigation is not required during planting time as it is transplanted during monsoon season. If a dry spell prevails irrigation should be applied immediately after planting. Depending upon soil condition during inflorescence initiation stages, single irrigation is required.

**Harvesting method:**

The mature panicles which turn light green or red are harvested in the winter season from Mid of February to end of March. The timing of the harvest is essential as if the plant is harvested prematurely (5–7 days) their production declines, while if it is harvested late it will begin to wilt. The panicles are either harvested by cutting above the soil separating the panicle for stem or pulling the panicles out by hand. It is important to make sure the Harvesting young sprouts are not damaged or the plants uprooted during harvest. The yield will be the lowest in the first and fifth year with the highest in the third. The ratoon of this grass can be taken up to $5^{th}$ or $6^{th}$ year of planting. Generally, the centre of tussock is devoid of new culms during the $4^{th}$ and $5^{th}$ year of planting as they arise from the periphery. To obtain high yield from broom grass, fields are burnt during April after harvesting the crop in $5^{th}$ or $6^{th}$ year. The Rhizome is again planted as per the method described earlier.

**Duration:**

$1^{st}$ year takes 11-month duration from planting to harvesting and from $2^{nd}$ year onward it takes 10 months duration.

# OTHERS

## IMPROVED PRACTICES IN *JHUM* CULTIVATION

The Hill Zone of Assam occupies an area of 15.222 sq km in the districts of Karbi Anglong and North Cachar Hills. The hilly areas of Karbi Anglong and North Cachar Hills are 66 percent and 85 percent of the total areas of the districts, respectively. As such *Jhum* cultivation is practiced widely in the whole zone except in the flat land in Karbi Anglong district which is about 34 per cent of the land area of the district. Karbi Anglong district has 5452 sq km and North Cachar Hills has 2597 sq km area under Jhum cultivation which is more than 50 per cent of the total geographical area of the zone. This is a mixed cropping system under slash and burn agriculture. The uneconomical agriculture, heavy soil erosion and continued depletion of soil fertility are the major problems of the zone.

In order to achieve improvement over the traditional practices of Jhum cultivation in the zone, a model of improved practices has been recommended with 20-25 percent slope with an annual rainfall of 1000-1200 mm.

i) Replace traditional crop varieties with improved recommended varieties.

ii) Grow pumpkin, cucumber, pineapple, turmeric, ginger and colocasia in the border with their respective recommended cultural practices.

iii) Grow field crops like rice, maize, cotton and sesamum by placing in the middle portion of the cultivated area with a common fertilizer dose of 40 : 20 : 20 of N, $P_2O_5$ and $K_2O$ kg/ha. Sow maize and cotton at 2 m row distance in alternate rows. In between maize and cotton rows, sow rice and sesamum in alternate strips.

iv) Practice line sowing/planting across the slope.

v) Grow blackgram in *kharif* with recommended cultural practices after harvest of rice, maize, sesamum and pumpkin grown in the preceding summer season.

vi) Both guinea (*Panicum maximum*) and citronella (*Cymbopogon wintaroanus*) are recommended as strip crops in the 20% hill slopes. The width of the grass strips would be 75 cm to be grown along the contour. Rice (summer) followed by green gram (*kharif*) are to be grown as per Package of Practices in between grass strips at a width of 4-5 metres. Grass strip cropping also helps in controlling soil erosion, run off and formation of terrace naturally.

# TECHNOLOGIES FOR INTENSIVE CROPPING IN DIFFERENT AGROCLIMATIC ZONES OF ASSAM

**Situation: Upland**

**A) Upper Brahmaputra Valley Zone**

**i) Annual crop**

Sugarcane (spring) + Cowpea Sugarcane (autumn) + *Toria* Pigeonpea + Ginger/turmeric Summer rice/vegetables

Blackgram/greengram (Sept)/Pea/*Rajmah* (Oct)/Vegetables (Chilli + French bean + Cucumber)

Cowpea/maize (fodder) Soybean/*Toria*

**ii) Perennial crop**

Arecanut + Banana + Pineapple

Arecanut + Black pepper + Banana + Pineapple

**B) North Bank Plains Zone**

Summer greengram - Groundnut/Potato/Pea/*Toria* Summer rice - Vegetables/lentil/Pea/*Rajmah*

**C) Central Brahmaputra Valley Zone:**

Summer greengram - *Kharif* vegetables Lentil/pea/*Toria*

Summer rice - *Kharif* vegetables/*Rabi* vegetables Vegetables - *Kharif* vegetables/*Rabi* vegetables Rice - *Kharif* vegetables/*Rabi* vegetables Sesamum + Blackgram (1: 1)

Pigeonpea + Sesamum

**D) Lower Brahmaputra Valley Zone:**

**i) Annual**

1) As in CBV and Rice-Blackgram-Potato Rice-Sesamum-Wheat.
2) Vegetables+vegetable; Vegetables+flower intercropping.

Brinjal+Frenchbean/Knolkhol/*Methi* Pea + Tuberose

Coriander + marigold

**ii) Perennial crop**

1) Coconut + ginger + Turmeric

Coconut + betelvine + Assam lemon + banana + Pineapple + Ginger

**E) Hills Zones (Slopes of 20-25%)**

Rice (DS) - Blackgram/greengram Rice (DS) - Cotton

Rice- Greengram – *Toria* Pigeonpea + Sesamum/blackgram/greengram Rice (DS) + Greengram/Blackgram (2 : 1)

**Situation: Medium/Medium Lowland:**

**A) Upper Brahmaputra Valley Zone**

| **i) March-June** | **June-Oct** | **Nov- Feb/March** |
|---|---|---|
| | Early rice)T) Early rice(T) | *Toria*/Niger/lLinseed Potato/Pea/Vegetables |
| **(for light textured soils)** | | |
| Rice (DS) | Rice (T) | *Toria*/Relay pea/Pat (F) |
| **ii) April-July** | **July – Nov** | **Nov/Dec-Feb.** |
| Rice (T) | Rice (T) | Oat (F)/Wheat (irrigated) |
| Vegetables | Rice (T) | Vegetables (Chilli / Radish + French bean/Tomato / Potato |
| | Maize | Tomato |

First rice crop of sequence is of 100-110 days duration *viz*., Luit, Kapilee, second rice of 130-140 days duration, *viz*., Mahsuri, Jaya, Satya.

**B) North Bank Plain Zone**:

| **March-July** | **July/Aug-Nov** | **Nov/Dec –Feb** |
|---|---|---|
| Green gram | Rice (T) | Potato |
| | | *Toria* (with low input level) |
| | Rice (T) | *Rajmah*/pea |
| | Rice (T) | Wheat/vegetables |
| | Maize | Rajmah |
| | Rice (T) | Oat (fodder) |

First rice crop of 100-110 days duration, 2nd one of long duration 150 days in two crop sequences.

**C) Central Brahmaputra Zone**:

| **March-June** | **July – Nov** | **Nov-Feb/March** |
|---|---|---|
| Jute | Rice (T) | *Toria*/Wheat |
| Summer moong | Rice (T) | Pea/vegetables |
| | Early Rice (T) | Vegetables |

| | | |
|---|---|---|
| Cow pea | Rice (T) | *Toria* |
| (Fodder) | Rice (T) | Rice (T) (Irrigated Lowland) |
| Lady's finger | Rice (T) | *Toria* |
| Jute | - | Wheat/*Toria*/Pea/Potato |

**D) Lower Brahmaputra Valley Zone:**

| **March-July** | **July/Aug-Oct/ Nov** | **Oct/Nov-Feb** |
|---|---|---|
| Rice (DS) | Rice (T) | Potato |
| Greengram | Rice (T) | *Toria* |
| Rice (T) | | Vegetables/Pea |
| Jute | Rice | Potato |

**E) Barak Valley Zone:**

| | | | |
|---|---|---|---|
| Rice (T) | Rice | - | |
| Early rice (T) | - | Pea/*Toria*/Niger Rice (T) | Rice (T) |
| | *Toria*/Pea | | |

Post-rice mustard can be sown till late November, while niger/linseed can be sown till December.

**F) Hills Zone:**

| | | |
|---|---|---|
| Maize | | Cabbage |
| Maize | Green gram/Black gram/Pigeon pea | |
| Rice (DS) | Green gram | *Toria* |
| Rice (DS) | Black gram | Wheat |

**SITUATION: FLOOD PRONE AREAS**

Flood prone areas are spreading along the river systems are found in all the districts of the state. Though the advent of flood differs in different areas and the intensity differs from year to year in the same area, the features of the areas are almost identical; hence the recommended technologies apply to all the areas.

| **Feb-May** | **Late Aug-Nov** | **Dec- Feb** |
|---|---|---|
| Early summer rice (DS) | Late winter rice (T) | - |
| Summer vegetables | -do- | Pea-potato |
| Summer vegetables | -do- | Groundnut/Pea |
| Summer rice/ | Fallow | Potato/Vegetable/ |

| | | |
|---|---|---|
| Summer pulse | | *Toria*/ Wheat/Pea |
| Groundnut/melons | Fallow | Early pulse/Vegetables |
| Summer rice (Feb-June) | - | Sweet potato (Sept-Feb) |

For late planting seedlings of 50-55 days of age can be used in closer spacings of 15 cm × 15 cm and 20 cm × 15 cm for semi-dwarf and tall varieties, respectively.

Rice varieties of short duration for pre and post flood situations are – Luit, Kapilee, (100-110 days duration) and also Lachit, Govind (120 days duration), Kalinga 3, Sonamukhi, Heera. In post flood situation sprouted seeds to be sown in wet condition.

**Technologies for Management of Specific Systems**

**Rice (T) –Wheat**:

Rice transplanting – Mid July, Wheat sowing – Mid Nov Recommended fertilizer for both the crops

Need based irrigation for wheat Rice (T) – *Toria*/*Rajmah*/Pea Rice-Mid duration variety

*Toria*- Sowing to be completed by early November, varieties –TS-36, TS-38, M-27

Application of organic manure at 5 t/ha and 25% of recommended level of fertilizers to both the crops increase yields.

*Rajmah*/Pea – Sowing to be over by early November.

**Rice – relay pea**

Pea to be sown 7-10 days before rice harvested with 25-50% higher seeding rate.

Residual soil nutrients available from preceding sali rice grown with the application of vermicompost (1 t/ha) and FYM (2.5 t/ha) under low input (N : $P_2O_5$ : $K_2O$ @ 20 : 10 : 10 kg/ha) condition to produce significant higher pea yield without any additional doses of fertilizers when pea is relayed with *sali* rice and one irrigation of 4 cm depth in pea at flowering stage.

(In Upper Brahmaputra Valley Zone, pea to be harvested by the end of February)

**Rice (T) – Oat (fodder)**

Oat can be sown after rice till late December. Two cuttings can be taken from early sown oat.

- Cutting of rice stubbles at ground level.
- Application of minimum tillage with one cross ploughing.
- Application of recommended dose of fertilizer @ 20 : 20 : 20: : N : $P_2O_5$ : $K_2O$ kg/ha
- Seed inoculation with *Azospirilum* @ 40 g/kg seed and PSB @ 40 g/kg seed.
- Sowing of seeds behind the plough with row spacing 25 cm
- Top dressing N@ 20 kg/ha after the first cut at 50 DAS.

**Rice (T) – Rice (T)**

Substitution of 25-50% of inorganic N in one of the crops through Azolla/FYM or straw helps cutting down fertilizer requirements. FYM/cut residue is to be applied @ 270-540 kg/bigha at the time of puddling. Azolla @ 22-44 kg/ha is to be inoculated one week after transplanting.

Closer spacing of 15 × 15 cm or 15 × 10 cm for short duration varieties like Luit/Kapilee.

Second rice var. is to be of long duration (150-160 days) for low land, *viz*., Ranjit, Bahadur, Kushal, Moniram to be transplanted in July.

Preemergence application of pretilachlor @ 0.75 kg a.i./ha at 3-5 days after transplanting (DAT) is recommended for autumn rice.

**Intercropping of pulse crops with sesame:**

- Intercropping system
  - (a) Sesame + black gram in 1 : 1 row proportion
  - (b) Sesame + green gram in 2 : 2 row proportion
- Sowing time: Mid-August – Mid September
- Row spacing: 30 cm
- Fertilizers: 30-20-20 of N-$P_2O_5$-$K_2O$/ha

**Intercropping of lathyrus with oats after Sali rice:**

- Cutting three rows of rice stubbles at ground level leaving next three rows without cutting
- Application of FYM @ 2 t/ha along with 10-20-20 kg/ha N-$P_2O_5$-$K_2O$ as basal
- Broadcasting the seeds of Lathyrus to the strips with rice stubbles using seed rate 25 kg/ha.
- Sowing of seeds of oat in lines behind the plough to the three row strips without stubbles with seed rate 50 kg/ha and row spacing of 25cm.
- Top dressing of N @ 10 kg/ha to be done after first cut at 50 DAS.

**Rice – Pulses/Oilseed:**

- Green manuring with *S. aculeata, S. rostrata* or green leaf manuring for the rice crop is beneficial in the establishment of *rabi* crops after rice.

- Use 60 kg $P_2O_5$ as rock phosphate 20 days ahead of planting the 1st rice crop and no phosphatic fertilizer for the second crop.

**Jute-*Toria*:**

- 75% of recommended dose of N-$P_2O_5$-$K_2O$ through chemical fertilizers and 25% N through water hyacinth compost in jute, and 50% of recommended dose of fertilizer in *toria* are recommended.

**Crop sequences for Hill Zone:**

- On 25-40% hill slopes of Hill Zone, following crop sequences under rainfed conditions:
- Direct seeded ahu rice (var. Inglongkiri) – Toria (var. TS 38)
- Direct seeded ahu rice (var. Inglongkiri) – Black gram (var. KU 301)

**Green gram - Wheat:**

- Seed inoculation in green gram with rhizobium @ 50 g/kg seed and PSB @ 50 g/kg seed.
- Basal application of 40 kg $P_2O_5$ kg/ha in green gram along with recommended N and K
- Basal application of 30 kg $P_2O_5$ kg/ha in wheat along with recommended N and K

## CROPPING SYSTEM FOR INCOME GENERATION

- Ridge gourd – broccoli – lady's finger

# CROPPING SYSTEM FOR FORAGE CROP PRODUCTION

**A) Sequential cropping**

***i) Intensive forage production.***

1) Cowpea-Teosinte/maize/Dinanath grass-oats
2) Cowpea/Ricebean-Cowpea/Ricebean-Oats/maize.
3) Maize + Cowpea-Teosinte + Cowpea-Oats
4) Maize + Cowpea-Dinanath grass-Oats

***ii) Year-round fodder production***

1) Teosinte/maize–Oat–Cowpea (fodder)

***iii) With food/Commercial crops:***

1) Rice-Oats
2) Cowpea-Rice-Oats
3) Rice(S)-Cowpea/Ricebean-Maize
4) Rice+Ricebean/Cowpea-Rice-Oats/Maize
5) Green gram-Cowpea (Fodder)-Rapeseed
6) Maize+Cowpea(Fodder)-Sesamum-Rapeseed

**B) Mixed/intercropping:**

***i) Forage crop mixture***

1. Oats + Pea
2. Oats + *Khesari*
3. Maize + Pea/*Lathyrus*
4. Maize + Cowpea/Ricebean
5. Teosinte + Cowpea/Ricebean
6. Sorghum + Cowpea

***ii) Food-Forage mixture***

1. Pea (Grain) + Oat (F)
2. Rapeseed/Niger + Oat (F)
3. Lentil/gram + Oat (F)
4. Rice + Cowpea (F)

***iii) With perennial grass:***

1) Guinea/Seteria + Ricebean/Cowpea–Dinanath grass/Cowpea/Ricebean–Oat/Pea/*Khesari*
2) Hybrid Napier + Cowpea-Cowpea-Oat

**iv) Growing of guinea grass in Agro-forestry system:**

Growing of guinea grass in agro-forestry system, i.e. under shade tree with 50% higher dose of recommended fertilizer (80 : 40 : 40 :: N : P : K kg/ha).

**C. Fertilizer Management in Forage based Cropping System:**

Application of lime (20% of L.R) + recommended dose of fertilized + VAM to rice bean crop during *kharif* is recommended in Rice bean- Oat cropping sequence.

# APICULTURE

Honeybees are essential for pollination of many cross-pollinated crops, and also for production of honey which generates income to the farmers. There are four major honeybee species *viz*., Rock bee (*Apis dorsata*), little bee (*Apis florea*), Indian bee (*Apis cerena*) and Western bee (*Apis mellifera*). Out of them, the former two are wild and the latter two are domesticated species. Indian bee is a brownish black, locally available domesticated Asiatic species. The beekeeping practice of North-East India is mainly based on this species. This is indigenous to India with average honey yield of 12 kg per hive per annum and the foraging range is 0.8-1 km. Western bee is an exotic golden yellow species most widely and commercially reared in the world. This is larger in size than Indian bee having average honey yield of 35-40 kg per annum with a foraging range of 2 km. The species is successfully introduced in Northern India and effort is being made to introduce it in Assam including North East India.

**Apiary site:**

An apiary is a place where honeybee colonies are reared. Site with proper sunlight, air circulation and shade should be selected for the apiary. Good apiary location is one with abundance of nectar and pollen producing plants. Commercial beekeepers generally migrate their colonies from one place to another for flora and produce more than one api crop of several types. Some of the important bee foraging crop-plants are rapeseed- mustard, niger, buckwheat, sunflower, safflower, sesamum, bottle gourd, pointed gourd, pumpkin, ridge gourd, sponge gourd, maize, soyabean, *jamun, litchi*, mango, drumstick, guava, citrus, coconut, areca nut, date palm, pummelo, ironwood, tamarind, mayflower, pomegranate etc.

**Beehive and other equipment:**

Honeybees are reared in the modern beehive based on the principle of bee space. Beehive is composed of brood chamber and honey chamber. The brood chamber is meant for rearing progeny and the super or honey chamber is used for secreting honey. The following beekeeping equipment are required for honeybee rearing.

i) Beehive**:**
   a) ISI A type 8 frames for Indian bee, ISI B type 10 frame for Indian bee.
   b) Langstroth 10 frame for Western bee.

ii) Smoker to produce smoke for effective handling of the colony

iii) Bee veil to prevent bee stinging during handling of the colony

iv) Hand gloves for effective handling of the colony

v) Swarm-catcher for collection of colony from natural source

vi) Honey extractor for extraction of honey

vii) Uncapping knife

**Rearing Season:**

Seeds or colonies are normally available during spring i.e. February to March as this is

a peak-breeding season for the honeybees. Colonies can be collected from the natural sources or may be procured from the beekeepers. Rearing can be started with three or four frame worker bees having one-year mated queen. During winter (December to January) and in spring (February to March) are the best seasons to start beekeeping.

**Seasonal Management:**

There are generally three seasons for bee management.

**Spring Management:**

Beekeepers calendar starts with the activity of honeybee colonies after prolonged rain and winter cold. As this is a major flowering season, worker bees become busy for collecting nectar and pollen from flowers and thereby help in building the colonies. This is known as honey-flow season and in this season mostly extraction operations are done. However, there is a problem with swarming, which is acute in Indian species but less in western species. Swarm prevention can be done by frequent inspection of the colony, removing the queen cell, dividing the strong colony and helping the week one.

**Summer and Rainy season management:**

In the summer management, honeybee colonies should be kept under shade just to protect from scorching sun. As there is continuous rainfall in Assam during the rainy season, honeybee colony should be provided with artificial diet, composed of carbohydrate, protein and water. Effective artificial diet comprises of sugar (as carbohydrate); black gram or green gram or soybean powder (as protein source) and water @ 1 kg of sugar and 100 gm protein powder mixed in 1 liter of water. It should be administered as per requirement of the colony.

**Winter management:**

The honeybee colonies should be taken out from the shade and exposed to the sunlight.

**Migration of the colony:**

During winter, honeybee colonies should be migrated to the oilseed crop fields (mainly rapeseed-mustard crop) and then to spring blooms of fruit, forage crops and forest plantations. In this practice both the beekeepers and farmers will be mutually benefited in terms of honey and crop production. Migration is essential for getting substantial yields of crops as well as honey.

**Disease and Enemy Management**

Several diseases, *viz*., and fungal, bacterial, viral and protozoan diseases infect honeybees. Out of these, viral and protozoan diseases are the most serious ones in Assam.

**a). Sac-brood disease:**

This is a viral disease caused by Thai-sac strain. The symptoms of this disease are.

- The larvae become pale; then turn brownish-black and gradually dry up.
- The punctured capping with dead pupa within the cell.
- The infection is usually in worker, seldom in drone and spread by drifting nurse bees.

**Control:**

**Dequeening and Requeening:** Create broodlessness for some time by dequeening and requeening through production of new queen cell. Infected colony should be treated with antibiotics like Teramycin, 250 mg @ one tablet per 4 liters of sugar syrup.

**b). Nosema disease:**

Nosema disease is found in adult Western bee colony. The infected bees show the symptom of crawling, disjointed wings and the mid-intestine become swollen with pale colour.

The treatment with hydroxy quinoline mixed with sugar @ 250 mg/4 liters of sugar syrup will give effective control. Against fungal disease, proper aeration and exposure to the sunlight prove to be effective.

**Natural Enemy:** Almost half a dozen natural enemies such as lizard, wasp, wax moth, cockroach, birds, ants and mites infest honeybee colonies. Out of them, wax moth and predatory wasp are most serious enemies.

**c). Wax moth, *Galleria mellonella:***

Wax moth lays eggs on the stored combs or on the spare combs in the colony. Larvae develop by feeding on wax and pollen in comb cells. Wax moths are most active in summer and rainy seasons. To manage this pest; store combs by removing extra combs from the colony. Fumigate stored combs in air-tight space (in hive chambers, sealed between with mud or dung) and treated with acetic acid or formalin. But fumigation with sulphur smouldering is most effective. Removal and destruction of the infected portion of the comb is also recommended. The biocontrol practice of treatment with *Bt* formulation var. *kurstaki* @ 0.5 gm/100 ml. of water per hive gives effective control of the pest.

**d). Predatory wasp:**

Among the wasps, the burrowing (*Vespa magnifica.*) and aerial wasps (*Vespa cinta*) are two common species in Assam, which predate on honeybees. To protect the colonies from the wasp, practices such as destruction of wasp nests in the vicinity of the apiary and artificial net covering (with nylon net of 1 cm mesh size) over the bee hive colonies are found to be effective.

For controlling bee mites, honeybee colonies should be exposed to the sunlight and in acute cases chlorbenzilate fumigation gives effective control. Sulphur dusting @ 200 mg/hive on top bars of frames is also effective.

**Honey bees in crop production**

Honeybees are essential for pollination of all the cross-pollinated crops. It has been clearly demonstrated through experimentation at AICRP on Honeybee, AAU, Jorhat that honeybee colonies enhance crop yield to the tune of 1.5 to 2 times as shown below:

| Crop | Colony requirement | Yield (q/ha) | Percent yield increase over open pollination |
|---|---|---|---|
| Mustard | 5 | 12.2 | 157.65 |
| Niger | 6 | 6.1 | 146.98 |
| Buckwheat | 5-6 | 14.2 | 152.68 |
| Litchi | 5-6 | 66.7 | 142.75 |
| Assam lemon | 4 | - | - |

Hence honeybee colonies are considered as essential input for increasing productivity of cross-pollinated crops. Migration of honeybee colonies is necessary for both crop and honey yield.

**Pesticidal Poisoning to Honeybees:**

In order to protect the honeybees from pesticide poisoning eco-friendly pesticides, which are less toxic to honeybees should be recommended. Moreover, the application of pesticide on flowering crops should be done in the afternoon when the bee activity stops in the field. Some of the bee friendly pesticides like botanicals, biopesticides such as *Bt.* formulation, NPV, etc. which are having less or no residual toxicity should be incorporated in the Integrated Pest Management Packages.

**Recommendation of Bt. Formulation against Wax moth:**

Wax moth, *Galleria mellonella* is one of the most serious pests of honeybee causing severe damage to the colonies. Various remedial measures such as cultural, chemical were adopted without having any full proof effect. In order to have effective management of this pest, biocontrol experiments with *Bt*. Formulation, var. *kurstaki* @ 0.5 per cent controlled the wax moth effectively. Hence, *Bt* var. *kurstaki* 0.5 g/hive/liter of water has been recommended against wax moth.

# PACKAGE OF PRACTICES FOR RICE HISPA MANAGEMENT

To manage rice hispa, the following strategies are suggested:

**Monitoring:**

1. As the initial population buildup of hispa takes place on *boro* and early *ahu* from February onwards, appropriate protection measures must be taken when the population reaches economic threshold level (ETL) of 1 adult or 1 damaged leaf/hill.
2. Routine monitoring at weekly intervals must be taken up from April onwards to detect the initial population build up in and around rice fields followed by similar surveillance from July onwards. Monitoring is advisable for the next year in the hispa affected areas during middle of February to April.

**Cultural Control:**

3. Draining out of water at the time of population build up for 2 to 3 days helps in decreasing hispa population.
4. Weed free cultivation should be encouraged.
5. In areas where rice is not widely grown during *boro* and *ahu* seasons, small, swampy areas and roadside pits having alternate hosts serve as the breeding pockets of hispa during February to April. Such infested pockets which are not very large in rice areas are easily identifiable from a long distance because of their characteristic withered conditions and burnt appearance. Therefore, it is advisable to destroy these weeds (alternate hosts) by mechanical methods (cutting and burning). This operation will minimize further multiplication and dispersal of adults into the rice crop. Use of insecticides cannot be advocated in the swamps as these are used as grazing land and natural fisheries.
6. Early transplanting helps to escape hispa attack. Whenever and wherever possible, staggered transplanting should be discouraged.
7. After harvesting of rice, stubbles should be burnt in November-December to kill the hibernating adults.
8. Deep ploughing of hispa affected fields should be done during February - March.

**Mechanical Control**

9. Clipping of leaf tips of seedlings before transplanting must be done.
10. Clipping and destroying of affected leaves up to about 6 inches from the tip should be done as these leaves contain eggs, larvae and pupae.
11. Awareness campaign and community involvement make hispa management more effective. Collection of adults in rice field, hand picking of grubs in the roadside swamps and pits may be encouraged among school children and village youths.

**Biological Control**

12. Insecticide spraying during October may be avoided, as the natural enemies are more abundant during this period. Apply *Beauveria bassiana* impregnated RHSDRB medium @ 3 kg/ha. On preparing the solution in water containing liquid detergent, the solution will contain $10^7$ spores /ml.

**Chemical Control**

13. All the currently recommended insecticides (Table 1) are effective.
14. Spraying operation should be done during early morning and late afternoon hours as the adult activities of hispa are maximum during this period.

**Awareness Campaign**

15. Bringing people to an understanding of the rice hispa problem is the best way to deal with this problem. Hence farmers, students and youths should be trained up on hispa problem and its management. No programme is more successful than the degree of commitment made by the people involved.
16. Farmers must be educated on the importance of applying recommended insecticides at proper dose and spray volume. It should be kept in mind that application of insecticides at doses below the recommended ones are not effective, rather it produce some undesirable effects, such as development of resistance in insects, resurgence (abrupt increase of target pest) and secondary pest outbreak.

# MANAGEMENT OF *PARTHENIUM*

Parthenium (*Parthenium hysterophorous*), commonly known as carrot weed, congress weed, white top, is one of the seven most dreaded weeds of the world. It is a native of Mexico, West Indies, North and South America and believed to have been introduced in India through wheat seeds imported from the USA under PL-480 during 1956. In Assam, within a short span of time, the weed has spread in different places with varying degrees of infestation, affecting primarily the roadside vegetation, amusement parks, railway tracts, industrial areas, city dwelling habitats, etc. High seed production ability as well as efficient dispersal mechanism, absence of natural enemies, completion of 3-4 life cycles in a year, adaptability to the extremes of the climate, possessing allelopathic effects, etc. are some important factors enhancing rapid invasion and colonization of the weed. Among the districts, the degree of infestation and abundance are relatively high in Bongaigaon, Kamrup, Golaghat, Tinsukia, Dibrugarh, Jorhat and Nagaon. The abundance of the weed is comparatively less in the districts situated on the North Bank of river Brahmaputra. The weed is also observed in the two hill districts of Karbi Anglong and North Cachar of the state.

For effective control of *Parthenium hysterophorous*, metribuxin 0.2% is recommended.

**Control of *Ipopmoea carnea* (BAM-KOLMOU)**

For control of *Ipopmoea carnea* (BAM-KOLMOU), application of post-emergence herbicide Glyphosate @ 1.5 kg/ha with a spray volume of 500-600 liter/ha at active growing stage with repetition of the spray when new leaves appear.

# CULTIVATION OF MILKY MUSHROOM

(*Calocybe indica*)

**Special characters of milky mushroom**:

1. Can be grown at higher temperatures (28 to 35°C)
2. Good keeping quality
3. Good texture mushroom (texture is retained after cooking)
4. Requires casing similar to that in button mushroom
5. Requires a special type of cropping room
6. Requires more light in comparison to oyster or button mushroom
7. Possesses the highest biological efficiency, up to 140%

**Season**: May to September

**Variety:** APK-2 from TNAU, Coimbatore

**Steps in cultivation**:

**Substrate:** Wheat straw and paddy straw is the best substrates. Mushroom yield is higher when a blend of wheat straw + paddy straw (1:1 w/w) is used.

**Preparation of substrate:** Straw is cut into small pieces of 3-5 cm size. It is soaked in cold water for 4 hrs.

**Hot water treatment:** Water is boiled in a big aluminum saucepan. The wet straw is immersed in hot water (80-90°C). The boiling of straw is continued for 1 hr.

**Sun drying:** The boiled straw is spread on a clean cement floor. The straw is allowed to dry for 1 to 2 hrs depending on the prevailing weather. The moisture content of the straw should be brought down to 60 to 70%. Sometimes, the growth of the mycelium is inhibited after one week of spawning. This is the biggest problem in the cultivation of milky mushroom. About 20 to 50% of straw in the beds is colonized by the mycelium of milky mushroom. The remaining portion of straw is not colonized. A clear zone of inhibition is seen at the edge of the white patches of growth. There will be no further colonization of the straw after one week of spawning. Ink caps start emerging from the un-colonised straw.

The major reason for the stoppage of mycelium of milky mushroom is believed to be the excess moisture content in the straw at the time of spawning. Therefore, correct moisture level in the straw is the crucial factor for the mycelial run of milky mushroom.

**Spawning:** High density polypropylene bags of 30 cm × 60 cm are used. Twenty holes are made with a single punch machine. The bottom of the bag tied with a thread to give a cylindrical shape to the bed. A spawn rate of 5% of the weight of wet straw is used. An increase in spawn rate i.e. above 5% will give faster mycelial run and increase in yield. The bag is filled with alternate layers of straw and spawn. There will be 4 layers of spawn and 5 layers of straw. A ventilation pipe of PVC is placed at the center of the bed at the time of filling. The beds are incubated at 28 to 30°C for spawn run.

**Spawn running:** The spawn run is completed in 15 to 20 days. Twenty days old spawn is better for spawn running and mushroom production. Light intensity of 600 to 800 lux favours fast growth of mycelium in beds. The growth of mycelium is slow in darkness. This is a contrasting character between milky mushroom and oyster mushroom.

**Casing**: Casing means covering the top surface of bags after spawn run with pasteurized casing material in a thickness of 4 cm. Composition of casing soil:

i. One year old FYM + loam soil 1 : 1 w/w
ii. Vermicompost
iii. Soil + dried moss 1 : 1 w/w

Besides the above, spent tea waste from tea stalls or ash from burnt rice husk cab also be used for blending with casing soil. Moss is the best casing materials because of its water holding capacity. Moss grows on walls in shady places. Chalk powder @ 10% is mixed with the casing material in order to raise the pH to 8.4. The casing soil is pasteurized in an autoclave or in a pressure cooker at 15 lbs/sq.in. for one hour.

After completion of spawn run the mouth of the bag is opened and folded black. Casing soil is applied above the newly exposed surface of the beds. The beds are kept in special type of cropping room.

**Cropping room**:

The success of growing milky mushroom depends on the design of the cropping room. The Horticultural polyhouse can be converted into a cropping room for milky mushroom, with small modifications. In Assam, temperature during May to September remains below optimum for milky mushroom. The polyhouse is able to increase inside temperature above 30°C. The roof is made of UV stabilized low density polyethylene film. Netlon agro shade of 75% is placed above the UV film in order to reduce the intensity of light. The roof is supported by brick wall on four sides. Bamboo racks with 3-tiers are constructed inside the room. The size of the room is 4.5m in width and 6 to 10m in length. Milky mushroom requires light for spawn run as well as fruiting. Maximum yield of mushroom is obtained when there is 1600 lux light in the cropping room.

**Pinning**:

The pinheads appear 6 to 8 days after casing. The mushrooms attain harvesting stage one week after pinning. A second and third crop of mushroom comes up at 10 days interval. A mist of water is sprayed every day on casing soil. The beds are retained up to 50 days after casing for mushroom production. Mushrooms should be harvested before the cap is fully opened.

**Yield**: 220 g/bed of ½ kg straw (dry wt.) in two flushes.

**Shelf life**: The shelf life of the mushroom is 5-7 days at room temperature. They can be tored in a refrigerator in perforated polypropylene bags for 20 days.

**Problems**:

1. Growth of mycelium is stopped after one week. Ink caps will emerge from uncolonised

straw. This is because of high moisture content in straw. Straw should be dried on a cement floor in bright sun for 2 hrs. The moisture content should be 60-70%. The spawn rate should be increased to 6%.

2. Mother cultures die when stored in refrigerator. Cultures should be kept at room temperature and subcultured at every 2 months interval.
3. Numerous ink caps of *Coprinus sp*. sprout from casing soil. These ink caps should be nipped everyday before they open. A light spray of formaldehyde @ 5 ml/*l* is given over the surface of casing soil just after casing was done, in order to check *Coprinus* sp.

**Cooking (Recipes)**: Milky mushroom has pungency. This is the only negative point. The pieces of mushrooms should be blanched by dipping in boiling water for 2 minutes and then rinsing with cold water. The pungency is removed by this process.

**1. Curry Ingredients:**

| | |
|---|---|
| Milky mushroom – 250g | Onion - 2 Nos. big size |
| Tomato – 2 Nos. big size | Turmeric powder – 1 tsp. |
| Garlic paste – 1 tsp. | Chopped ginger – 1 tsp. |
| Cinnamon – 3 Nos. | Cardamom – 2 pieces |
| Oil – 3 tbsp. | Salt – to taste |

**Method:**

i. Chop the onion into pieces and make a paste
ii. Cut the tomato into pieces and keep aside
iii. Heat the oil in a *kadahi* and add the cinnamon and cardamom to it.
iv. Add the onion paste and fry it to golden brown colour at low flame. Pour the tomato pieces and fry till the oils come out and the contents leave the sides of the *kadahi.*
v. Now add the mushroom pieces and cook at low heat for about 15 minutes.

**2. Pickle Ingredients:**

Milky mushroom – 200 g
Onion paste – 20 g
Garlic paste – 1 tsp
Ginger paste – 1 tsp
Chili power – 1 tsp.
Cumin powder – 1 tsp
Black pepper powder – ½ tsp
Vinegar – 3 tbsp
Mustard seed powder – 20 g

Turmeric powder – 1 tsp

Oil – 50 ml

Salt – to taste

**Method:**

i. Let the mushroom pieces dry for 1 h after blanching

ii. Heat oil in a *kadahi*, fry the onion paste till light brown colour. Add garlic and ginger paste and fry for sometime.

iii. Add cumin powder, chili powder, turmeric powder and the mushroom pieces and fry by stirring.

iv. Now add mustard powder, vinegar, salt and cook for 10 minutes.

v. Remove from heat and sprinkle black pepper powder.

**Economics of milky mushroom production with a capacity of 5 kg mushroom per day for 100 days in a year**

Milky mushroom grows well in the temperature range of 28 to 35°C. It can be grown during the months of May, June, July and August. It needs a special mushroom house with plastic roof for increasing the inside temperature. One hundred days in a year are considered for production of milky mushroom. The same facilities can be utilized for growing oyster mushroom during the remaining period of a year.

| **A. Fixed capital** | **Rs** |
|---|---|
| 1. Mushroom cropping room (poly house of 3m × 6m. | |
| with a capacity to hold 250 beds at a time | 20,000.00 |
| 2. Spawn running room with tin roof, 3m × 6m with a | |
| capacity to hold 250 beds at a time | 20,000.00 |
| 3. Chaff cutter, electrically operated | 10,000.00 |
| 4. Aluminum saucepan for boiling of straw, | |
| 120 liter capacity | 3,000.00 |
| 5. Sprayers – two | 800.00 |
| 6. Fuel efficient chullah | 1,000.00 |
| Total | 54,800.00 |
| **B. Fixed cost** | |
| 1. Interest @ 106 for 4 months of crop season | 1,826.00 |
| 2. Depreciation on items 1 to 6 @ 10% for 4 months | 1,826.00 |

| | |
|---|---|
| Total | 3,652.00 |
| **C. Working capital** | |
| 1. Paddy straw, 0.5 ton | 800.00 |
| 2. Spawn, 500 packets of 200 g each | 5,000.00 |
| 3. Polythene bags | 1,200.00 |
| 4. Labour, 100 man days @ 100/- | 10,000.00 |
| 5. Fungicides | 700.00 |
| 6. Miscellaneous | 800.00 |
| **Total** | **18,500.00** |

| | |
|---|---|
| **D. Cost of mushroom production** | |
| 1. Working capital plus fixed capital | 22,152.00 |
| 2. Cost of production of 1 kg of mushroom (1600/500) | 44.30 |
| **E. Income** | |
| 1. By sale of 5 kg mushroom/day @ 80/- for 100 days | 40,000.00 |
| 2. Total cost of production | 22,152.00 |
| 3. Net income out of 4 months | 17,848.00 |

**Supply of spawn (seed)**

The spawn of milky mushroom is available from April to August in the Department of Plant Pathology, Assam Agricultural University, Jorhat – 785 013, Ph: 0376-2341393.

## Cultivation of milky mushroom, *Calocybe indica*

Chop paddy straw into pieces of 3 to 5 cm

⇩

Soak the straw in water for 6 to 8 hrs.

⇩

Boil the straw in hot water (80°C) for 1 hr.

⇩

Sun dry the straw for 1 to 2 hrs. or until 60-70% moisture level-a crucial step

⇩

Spawn the substrate in layer in cylindrical poly bags, 06 m × 30 cm

⇩

Incubate at 28°C for 15 to 18 days

⇩

Casing with soil + moss, 1:1, pH 8.5

⇩

Keep the bags for fruiting at 28-35°C in poly house with 75% shade net

⇩

Pinheads formed in 8-10 days and ready for harvest in 5-8 days

⇩

Yield, 300-400 g/ ½ kg dry straw

# CULTIVATION OF PADDY STRAW MUSHROOM

## (*Volvarella volvacea*)

**Advantages:**

i. Quickest growing mushroom – sowing to harvesting takes only 14 days

ii. Tastes better as compared to oyster

iii. Method of cultivation is easy – boiling of straw is not required.

**Disadvantages:**

i. The keeping quality of this mushroom is very poor.

ii. Huge quantity of straw is required.

iii. Bioefficiency is less – 13-15%, i.e. 100 kg of straw (dry wt.) will give only 13-15 kg of mushrooms.

**Season:** May to September. The month of August supports maximum production. The optimum range of temperature is 28° to 35°C.

**Raw material:** Paddy straw is the best substrate. If the straw is of poor quality (exposed to rain), weed mushroom (*Coprinus* sp.) will cause problem.

**Steps in cultivation:**

**Tying of straw into bundles:** Straw is tied into bundles of 85 cm length and 12 cm diameter. Each bundle weights 800 g to 1 kg. Twenty-four such bundles are required to construct one bed of mushroom. Twenty kgs of straw is required for a standard bed.

**Trimming of straw bundles:** Both the ends of the bundles are chopped with a hatchet to give sharp ends.

**Soaking of straw bundles:** The bundles are immersed in water for 12 hrs. Straw bundles are taken out. Excess water is drained off by holding the bundles for a few minutes on bamboo bars.

**Bamboo platform:** Bamboo platform of 90 cm × 90 cm × 15 cm height is constructed for each bed.

**Mushroom house:** The size is 3 m in breadth, 6 m in length and 1.5 m in height at all sides. The roof is of thatch. The floor is of cement. The walls on all the four sides are made of bamboo. There should be an inner lining of plastic sheet in order to create high humidity and warm temperature. This house can hold 8 beds at a time.

**Spawn:** Three hundred gram of spawn is required to prepare one mushroom bed of 20 kg straw. The rate of spawning is 1.4% of the dry weight of straw. The spawn is divided into 3 lots- 75 g, 75 g and 150 g.

**Red gram dal powder as supplement: Red** gram dal powder is absolutely necessary for quick growth of mycelium and optimum yield. Two hundred and forty grams of finely ground dal powder is required for each bed. Divide the powder into 3 parts – 60 g, 60 g, and 120 g.

**Construction of mushroom bed:**

- Place 6 bundles of straw in a row on the platform.
- Place small bits of spawn in rows at 5 cm distance over the bundle of straw.Seventy five gram of spawn is used for the first layer.
- Leave 7 cm margin on all sides of bed without spawning
- Sprinkle 60 g of dal powder over the bits of spawn.
- Place the 2nd layer of 6 bundles in the opposite direction of the 1st layer.
- Sow 75 g of spawn and sprinkle 60 g of dal powder.
- Put 3rd layer of 6 bundles in the opposite direction of the 2nd layer
- Place 150 g of spawn and sprinkle 120 g of dal powder.
- Cover the bed with a polythene sheet of 2 m × 2 m and tuck the edges of sheet under the frame.
- Remove the polythene cover after 10 days of sowing. Mushrooms will be ready for harvesting in 13 to 15 days of sowing. Egg stage is the best stage for picking.

**Problems**: Straw mites (*Tyrophagus dimidiatus*) are the most serious problem. Good control of mites can be achieved by spraying ethion 50EC @ 2.5 ml/*l* on straw, one hour before sowing.

**Economics:** The economics for a cottage unit with a production capacity of 20 kg mushroom in each cropping period of 20 days duration was worked out as follows:

| | | |
|---|---|---|
| **A.** | **Fixed capital** | **Cost (Rs.)** |
| 1. | Mushroom house, 3m × 6m of thatch roof and brick floor with a capacity to hold 8 beds of l sq.m at a time | 7,000.00 |
| 2. | Water tank, 1.1 m × 1.1 m × 1 m made of brick and cement | 3,000.00 |
| 3. | Hand sprayer of 5 liter | 700.00 |
| 4. | Bamboo platform – 8 | 400.00 |
| 5. | Polythene sheet of 2 sq. m – 8 | 160.00 |
| 6. | Plastic buckets – 2 | 300.00 |
| | **Total** | **11,560.00** |
| **B.** | **Fixed cost** | |
| 1. | Interest @ 10% on fixed capital for 20 days (one crop) | 63.34 |
| 2. | Depreciation @ 15% for 20 days | 95.00 |
| | **Total** | **158.34** |
| **C.** | **Working capital** | |

| | |
|---|---|
| 1. Rice straw – 160 kg @ 0.80 | 128.00 |
| 2. Spawn – 2.4 kg @ 50/- | 120.00 |

| | |
|---|---|
| 3. Pigeon pea dal powder – 2 kg @ 50/- | 100.00 |
| 4. Labour – 7 man-days @ 100/- | 700.00 |
| **Total** | **1,048.00** |

**D. Cost of cultivation**

| | | |
|---|---|---|
| 1. | Working capital including interest on fixed capital | 1206.00 |
| 2. | Cost of production of 1 kg of mushroom (Rs. 785/20 kg) | 60.00 |

**E. Income**

| | | |
|---|---|---|
| 1. | Gross return from 20 kg mushroom @ 80/- in 20 days | 1,600.00 |
| 2. | Net income for 20 days | 242.00 |
| 3. | **Net profit from 7 crops in a growing season of 5 months** | **1,694.00** |

# CULTIVATION OF WHITE BUTTON MUSHROOM

*(Agaricus bisporus)*

Button mushroom can be grown in Assam as a seasonal crop

**Season:** The cropping period is December and January

**Species:** *Agaricus bisporus*, Strain S-1 1

**Yield:** 10-14kg/100kg compost.

**Preparation of compost:** By long method, 28 days. The composting process should be started by 20$^{th}$ September.

**Compost formulae:**

Rice straw or wheat straw – 1000 kg

Wheat bran – 25 kg

Urea – 18 kg

Gypsum – 35 kg

Water – 3000 liters

The straw is chopped into pieces of 5 cm. A minimum of 3 quintals of straw should be used.

1$^{st}$ – day: Bran, urea mix is blended with the straw. The straw is stacked in a wooden mould of 1.5m height and 1.5m in width. The moisture content of straw should be 75% to 77%. The temperature inside the stack will rise up to 65$^0$ – 70$^0$C.

2$^{nd}$ - day: Wash the compost yard with 2% formalin. Spread the straw up to 9$^{//}$- 12$^{//}$ height. Pour water for soaking of straw and leave for 24 hrs. Wheat bran and urea are mixed. Water is sprinkled to make it moist. Leave the heap as such for 24 hrs.

6$^{th}$ – day: First turning is given. Break the heap, loosen the straw for exchange of air and restack.

10$^{th}$ – day : 2$^{nd}$ turning is given.

13$^{th}$ – day : 3$^{rd}$ turning is given. Gypsum is added

16$^{th}$ – day : 4$^{th}$ turning is given

19$^{th}$ – day : 5$^{th}$ turning is given

22$^{nd}$ – day : 6$^{th}$ turning is given

25$^{th}$ – day : 7$^{th}$ turning is given

26$^{th}$ – day : Fill the compost.

**Spawning:** Rate of spawning is 0.5 to 0.7% of the weight of compost. Mix spawn with compost in a clean tub. Fill the compost in polythene bags of 85 cm width and 60 cm height. The depth of compost should be 30 – 35 cm. Compress the compost by hand to drive out the air. Close

the mouth of the bag by folding the extremity of the bag. Keep the bags at a temperature of 24°C, in a room with the doors and windows closed. The spawn running will take 3 weeks.

**Preparation of casing soil:** Farmyard manure (2 yrs. old) + loam soil (1 : 1 v/v) or moss + soil (1 : 1 v/v). Casing material should have 20% moisture. Pasteurize the casing soil at 80°C for 1 ½ hr. Alternatively, the casing soil can be treated with formalin. For this, mix 1 liter of concentrated formalin with 1 ton of casing soil. Cover the heap for 4 days. The soil is raked every day to allow the fumes to go out. The casing will be free from formalin after 14 days.

**Casing:** Put a layer of 4 cm thick casing soil. Maintain a temperature of 25°C for 10 days. After complete case run, open the ventilators and let in fresh air. The temperature should be 16° – 18°C for fruiting. Water should be sprayed everyday. Fruiting will start 10 days after case run. Put fresh casing in areas from where mushrooms have been removed.

Economics for an oyster mushroom production unit for a period of 240 days (8 months, 8 crops) is worked out as follows:

| **A. Fixed capital** | **Rs.** |
|---|---|
| 1. Mushroom house of 6 m × 10 m size with thatched roof and brick floor with a capacity to hold 250 beds at time | 7,000.00 |
| 2. Spawn running room of same size without brick floor | 4,000.00 |
| 3. Chaff cutter (manual) | 10,000.00 |
| 4. Bamboo racks | 3,000.00 |
| 5. Sprayer (one) | 3,000.00 |
| 6. Sprayer (two, small) | 500.00 |
| 7. Bucket (four, large) | 2,000.00 |
| 8. Hessian cloth | 2,000.00 |
| 9. Big drum for wetting straw | 1,000.00 |
| 10. Big an for boiling straw | 1,000.00 |
| 11. Bamboo mat (Dhari), two | 300.00 |
| 12. Balance | 4,000.00 |
| **Sub total** | **37,80.00** |
| **B. Fixed cost** | |
| 1. Interest @ 10% on fixed capital for 240 days | 2,589.00 |
| 2. Depreciation @ 15% for 240 days | 3,884.00 |
| **Sub total** | **6,473.00** |

| | |
|---|---|
| **C. Working capital** | |
| 1. Rice straw (to tons) @ 1000/-per ton | 2,000.00 |
| 2. Span : 180 kg @ 50/- per kg | 9,000.00 |
| 3. Polythene bags (60 × 40 cm) | |
| @ Rs. 100/- per kg – 10 kg | 1,000.00 |
| 4. Firewood @ 1000/- per ton – 3 tons | 3,000.00 |
| 5. Jute/plastic thread @ 100/- per kg – 3 kg | 300.00 |
| 6. Fungicide/insecticide | 500.00 |
| 7. Labour : 240 mandays @ 100/- per day | 24,000.00 |
| **Sub-total** | **39,800.00** |
| **D. Cost of mushroom production** | |
| 1. Working capital + interest on fixed capital | 46,273.00 |
| 2. Cost of production of 1 kg mushroom | |
| (46273/1400) | 33.00 |
| **E. Income** | |
| 1. Gross return from 1400 kg mushroom @ Rs. 80/- per kg in 240 days | 1,12,000.00 |
| 2. Net income for 240 days | 65,727.00 |

# COMMERCIAL PRODUCTION OF BIOFERTILIZER

The two biofertilizer agents in *Azospirillum* and PSB group, *Azospirillum amazonense* A10 and *Bacillus megaterium* P5 have been found to be superior and their utility as biofertilizer in integrated nutrient management of rice has been demonstrated. The material and steps involved in production of biofertilizers of these microorganisms are presented. These two strains can be cultivated in broth at 30±2°C. Potato dextrose agar can be used for growing the organisms in broth.

**The process of biofertilizer production**:

| Steps | Material required | Source |
|---|---|---|
| **A. Cultivation of the strains in culture broth** | i. Microbial strain<br>ii. Housing for the equipment<br>iii. (Civil work)<br>iv. Equipment<br>a. A laminar flow<br>b. Shakers 4 nos. (50 nos. of flask holder platform) or 20 liter capacity fermentor*<br>c. Autoclave<br>d. Glass wares<br>e. Incubator<br>f. Environmental shaker | AAU |
| **B. Production/ Purchase of carrier**** | From vegetation biomass, rice mill-rice husk ash, rock phosphate mix or charcoal powder | |
| **C. Mixing and packaging** | a. Manual mixing or mechanical mixture<br>b. Sealing machine<br>c. Plastic packets<br>d. Carrier material | |

* Normally, fermentors are used for cultivation of microbial strains, but to reduce cost, shakers can be used to multiply the strains under ambient temperature with the following schedule

**Production schedule in shaker under ambient temperature**

| **March – Aug** | **Sept. – Nov.** |
|---|---|
| Use in ahu and sali rice and vegetables | Use in *toria*, wheat, early rabi vegetable |
| Can be produced 200 kg in 4 In 90 days, | 9000 kg can be days in 180 days 18,000 kg. produced |

Total production in a year using 6 shakers each with capacity to carry 15 flaks of 1 liter capacity is 27,000 kg.

**** Production of carrier (compost):** Most biofertilizer producers use charcoal powder as carrier material for the microbial agents. But AAU has developed technique to produce compost as carrier material. The shelf life of the biofertilizer is also longer (eight months) in compost than in charcoal. For every 1000 kg of compost (on oven dry basis), the requirement is 2105 kg dry mass of cowdung *sali* rice straw, weed biomass, legume leaf, rock phosphate, lime sludge and rice husk ash. On fresh weight basis the biomass requirement will be 3600 kg during Jan-April. This compost contains 2.3, 2.8, 4.0, 1.26, 0.019 and 0.006% total N, P, K, Ca + mg, Zn and Cu, respectively.

| **Type of materials in the composting mix** | **Jan-April** | **May-Sept** |
|---|---|---|
| *Sali* rice straw | 810 | - |
| Weed biomass | 41 | 800 |
| Legume leaf | 82 | 92 |
| Fresh compost | 3.3 | 3.3 |
| **Type of materials in the composting mix** | **Jan-April** | **May-Sept** |
| Kalmoce | (In available substitute some % rice straw) | - |
| Water hyacinth | -do- | - |
| Fresh cowdung | 16 | 16 |
| Legume stover | (If available substitute some % rice straw) | |
| Mustard straw | -do- | - |
| Lime sludge (paper mill waste if available) | 16 | 16 |
| Rock phosphate | 16 | 16 |
| Rice husk ash | 12 | 12 |
| Time of composting (approx) | 2½ months | 2 months |

Since availability of composting materials (raw materials) depends upon the season, the individual entrepreneur will choose most available materials. Although AAU used materials of Jan-April (above) for the research, the nutrient content of the compost prepared from materials

of other seasons is not likely to change.

**Pit Size for Compost Production**: The pit size is 2 m length × 1 m breadth × 20 cm depth. Depending upon availability of composting mix space, manpower and capital, pit number will vary. Each pit accommodates 700-1000 kg composting mix to a height of 100 cm from the surface. Materials in pit are packed in layers. Few bamboo sticks are inserted in the compost heap for aeration.

**Stirring, mixing and watering:** Occasional stirring is done by shaking the bamboo sticks. After one-month materials are remixed once. Watering is done regularly. Covering the heap with jute sheet is useful for conserving moistures.

**Materials and labour cost in production of compost per 1000 kg fresh weight**

| **Item** | **Amount(kg)** | **Price rate** | **Amount (Rs.)** |
|---|---|---|---|
| *Sali* rice straw | 810 | Rs. 50/100 kg | 405.00 |
| Weed biomass | 41 | (1 hr labour cost) | 8.00 |
| Legume leaf | 82 | (2 hr) | 16.00 |
| Fresh compost | | Rs. 2.00/kg | 6.60 |
| Fresh cowdung | 16 | Rs. 1.00/kg | 16.00 |
| Lime sludge | 16 | Carrying charge | 32.00 |
| Rock phosphate | 16 | Rs. 5.00/kg | 80.00 |
| Rice husk ash | 12 | Carrying charge | 2.00 |
| Labour cost in digging pit, setting, stirring, watering | 7 mandays | LS | 420.00 |
| **Total** | | **Rs.** | **985.00** |

**Cost benefit analysis of biofertilizer production:**

**Fixed cost**:

| | | |
|---|---|---|
| (1) Equipment cost | : | Rs. 4,55,000.00 |
| (2) Van for transportation | : | Rs. 5,00,000.00 |
| (3) Civil work (3 rooms) | : | Rs. 3,00,000.00 |
| (4) Water and electricity (Installation) | : | Rs. 60,000.00 |
| Total of fixed cost | = | Rs. 13,15,000.00 |

**Running cost for 1000 kg biofertilizer**

(1) Culture medium cost : Rs. 7,390.00 (if potato dextrose agar is used the cost will be half)

(2) Cost of Carrier materials : Rs. 1000.00

(3) Electricity and water supply @ Rs. 1000/month) : Rs. 444.00

**Total : Rs. 8834.00**

| **Manpower cost**: | | |
|---|---|---|
| For five year: Two skill worker @ Rs. 100.00 day | = | Rs. 3,60,000.00 |
| One manager @ Rs. 12,000.00 | = | Rs. 7,20,000.00 |
| **Total** | **=** | **Rs. 10,80,000.00** |

50% of the fixed cost is the amount accounted for five year : Rs. 6,57,500.00 Total production in five years 27,000 × 5 = 1,35,000 kg

**Total return in five year Rs. 1,35,000 × 60 = Rs. 81,00,000.00**

Total cost of production in five years Rs. 6,57,500.00 (fixed cost) + Rs. 10,80,000.00 (manpower) + Rs. 11,92,590.00 (recurrent cost) = Rs. 29,60,090.00

**In five years net return = Rs. 81,00,000.00 – 29,60,090.00 = Rs. 51,39,910.00**

If even in 25% of s*ali* rice area of Jorhat district alone uses biofertilizer based INM the annual demand is 1.35 kg and there lies a profit of ~ Rs. 50 lakhs

**Technical guidance**: The biofertilizer agents at a nominal charge and free technical guidance will be available from AAU.

# METHODS OF APPLICATION OF *Trichoderma* spp.

1. **Slurry seed treatment:** Take 5-6 g of *Trichoderma* formulation per kg of seed. Mix well with 10 ml of water. Air-dry the seeds for 30 minutes to 1 hour before sowing.
2. **Wet seed treatment/Dip method:** Dissolve 20-25 g of *Trichoderma* in 1 liter of water. Dip 1 kg seed or appropriate numbers of seedlings/sets/rhizomes in the solution for 1 hour. Dry under shade for 1 hour before sowing/planting.
3. **Soil drenching:** Dissolve 20-25 gm of *Trichoderma* in 1 liter of water for spot application by jerry cane or sprayer whenever felt necessary.
4. **Soil application:**
   - Mix 100 kg of dried cow dung with 10 kg of Mustard Oil Cake. Moisten by sprinkling water.
   - Add 1 kg commercial formulation of *Trichoderma*, mix thoroughly and cover with newspaper or banana leaf for 3 days.
   - On 4$^{th}$ day open the cover, mix thoroughly, sprinkle water and cover it again.
   - On 7$^{th}$ day mix thoroughly the final product and apply to soil before planting. For 1 ha of land, the requirement is approximately 500 kg of dried cow dung, 50 kg. of Mustard oil cake and 5 kg of *Trichoderma* formulation.

# SOME ECONOMICALLY FEASIBLE FARMING SYSTEMS

Some of the major farming systems being practiced in different agroclimatic zone of Assam have been presented below. These systems have been identified while carrying out research on Farming System (FSR) under ARIASP. Some of the economically feasible farming systems for different agroclimatic zone of Assam have been worked out by the Nodal Scientific Team and presented below.

**(I) Lower Brahmaputra Valley Zone**

Crop + Dairy Cow + Goatery + Poultry + Duckery

Crop + Dairy cow + Goatery + Piggery + Poultry +Duckery + Pigeon + Fishery Crop + Dairy cow + Piggery + Poultry

Crop + Dairy cow + Buffalo + Piggery + Poultry + Duckery + Pigeon Crop + Dairy cow + Goatery + Poultry + Duckery + Fishery

**(II) Central Brahmaputra Valley Zone**

Crop + Dairy + Goatery + Poultry + Duckery + Fishery Crop + Dairy + Goatery + Poultry + Fishery

Crop + Dairy + Goatery + Poultry + Fishery

Crop + Dairy + Goatery + Poultry + Duckery + Pigeon + Fishery

**(III) Upper Brahmaputra Valley Zone**

Crop + Sericulture + Dairy + Goatery + Duckery + Poultry + Piggery + Fishery Crop + Dairy + Goatery + Duckery + Poultry + Piggery +Pigeon + Fishery Crop + Dairy + Duckery + Fishery

Crop + Dairy + Goatery + Duckery + Pigeon + Fishery

**(IV) North Bank Plain Zone**

Crop + Dairy + Apiary + Fishery

Crop + Dairy + Goatery + Duckery + Piggery + Apiary Crop + Dairy + Poultry + Apiary + Sericulture

Crop + Dairy + Duckery + Goatery + Piggery + Pigeon + Apiary + Fishery Crop + Dairy + Goatery + Poultry + Fishery

**(V) Barak Valley Zone** Crop + Dairy + Poultry Crop + Fishery

Crop + Diary + Fishery

Crop + Diary + Poultry + Duckery + Fishery + Pigeon Crop + Diary + Poultry + Fishery

Crop + Diary + Fishery + Poultry + Pigeon

**(VI) Hills Zone**

Crop + Goatery + Poultry + Fishery + Poultry + Sericulture Crop + Dairy + Goatery + Poultry

Crop + Piggery + Poultry + Fishery + Forestry + Sericulture

# Construction of low cost vermicompost unit

**Design and specification of Low-cost vermicoposting unit**

1. Dimension: 2.5 m (L) x 0.91 m(B) x 0. 76m (D)
2. Materials to be used:

a. Bamboo structure for the shed over the tank and construction of vermicomposting unit

b. Roof: UV stable silpulin/Thatch

c. Sides: Locally available material e.g. Dried banana leaves/ agroshed Net







d. Pit: A pit connecting the vermicomposting unit. A half-inch diameter PVC pipe may be used.

e. Inside the pit: Bamboo made vermicomposting tank and Wash collection tank should be laid with Black polythene sheet (Best Quality)

f. The bottom should have a mild slope towards the Wash collection tank.

g. A layer of fine pebble and layer of sand should be placed over the plastic sheet in the vermicomposting tank.

h. The vermicomposting tank should be filled with Semi-decomposed (pre-decomposed) and subsequently, earthworm of proven species is to be released. The dried cow dung should be free of local earthworm species.

i. A side drain may be laid out around the vermicomposting unit to prevent ant problem.

| | |
|---|---|
| **Size of each tank** | : 0.91m (B) x 0.91m (D) x 2.5m (L) |
| **Annual Production capacity** | : 900 Kg/ one cycle x 3 cycle = 2700 Kg |
| **Duration of each cycle** | : 3 months during summer and 4 months during winter |
| **Duration of the structure** | : 3 years |

# FARM MECHANIZATION AND POST HARVEST TECHNOLOGY

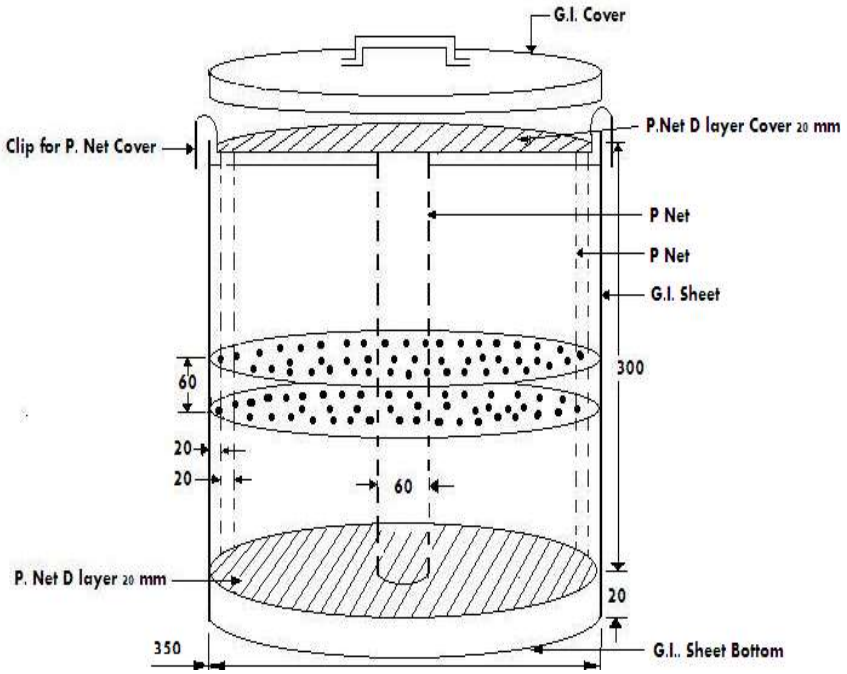
**Recommended Technologies for Package of Practices**:

1. Storage structure for green gram under high moisture environment
2. Solid state Biogas Plant (SSBGP)
3. Improved cook stove for woody biomass (Udairaj)
4. Improved Paddy Husk cook stove (*Tuh Chullah*)
5. Use of raised bed planter

1. **Storage structure for green gram under high moisture environment:** The storage structure can be used for storing green gram for a period of 12 months saving the grains from the infestation of storage grain insect-pests.

**Description of the structure:**

- The structure has one perforated cylinder at the centre.
- Four perforated sieves of same diameter as that of the drum are placed at four different elevations inside the drum.
- Insect repellent plant parts are spread over the sieves and at the bottom of the structure.
- The central cylinder filled with inert materials (ground chalk) for its moisture absorption and repellent plant parts selected on the basis of their insect-repellent activities (dried *neem* leaves).
- Cost of the structure: Rs. 1850/- at a Capacity of 12 kg



**Schematic diagram of storage structure**



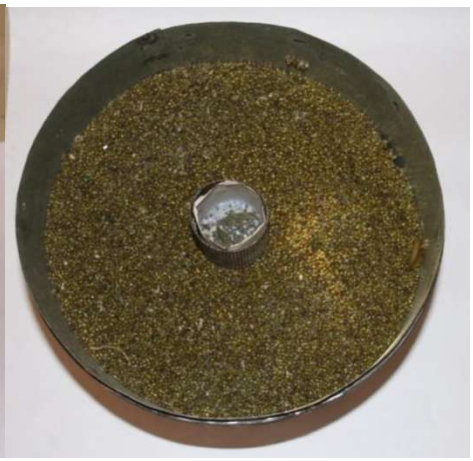
**Leaves of *neem* being dried**



**Inside of the storage structure**



**Outer cover with inner cylinder**



**Stored green gram in the structure**

**OPERATIONAL PROCEDURE:**

- Spread shed dried leaves of *Melia azadarch* on the bottom and over the top sieve of the storage structure. The selection *Melia azadarch* leaves was done on the basis of bioassay studies against pulse bruchid *Callosobruchus chinensis.*
- Put the first perforated sieve.
- Load the structure with clean and dried green gram up to near centre
- Put the second perforated sieve over the grain spread dried leaves of *Melia azadarch*
- Put the third perforated sieve at designated position
- Fill the remaining space up to the top
- Fill the central perforated cylinder with coarse ground chalk granules
- Put the fourth sieve and spread over dried leaves of *Melia azadarch*
- Cover the structure with the lid.

**FARMING SITUATIONS:** Rice based cropping system

**AGRO CLIMATIC ZONE:** Suitable for all agro climatic zones of high humid environment

**SOURCE OFTECHNOLOGY:** Dept. of Agril. Engineering. Assam Agricultural University, Jorhat-785013, Assam

**SOURCE OF CRITICAL INPUTS:**

- Metallic Portable storage structure as per specification.
- Shed dried *neem* leaves (500g).
- Coarse ground chalk granules

**OBSERVATION TO BE MONITORED:**

- Percent damaged grains
- Moisture content
- Monetary benefit by using the device in terms of quantitative and qualitative loss reduction.

2. **Solid State Bio Gas plant (SSBGP):** "Solid-state biogas plant" can be used in all the Agro Climatic zones of Assam. Digested slurry coming out of the plant has total solid concentration of 10 to 12 % and can easily be transported to the field for application. Reduction in water use while feeding (by 78%) and enhanced gas production (by 30%) can be achieved.

**Site selection** - For construction of the plant elevated sunny areas away from trees but near to the cow shed and kitchen of the house should be selected.

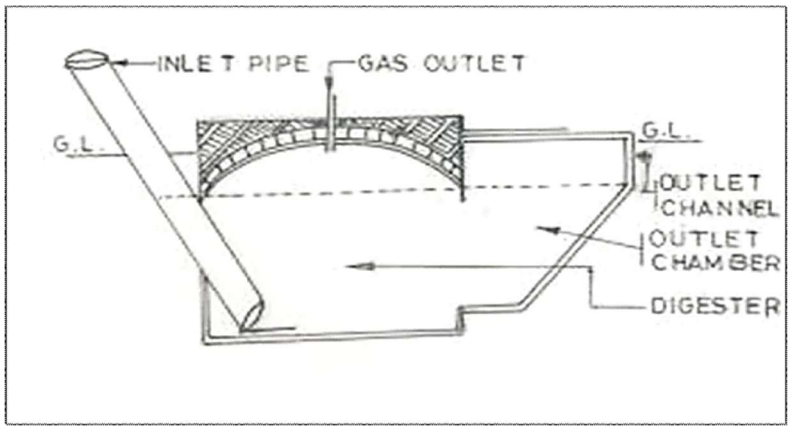
**Period of Construction**: Best period of construction is 15$^{th}$ November to 15$^{th}$ March.

**Applicability**: Both Janata type or Deenbandhu type can be modified as Solid state biogas plant.

**Cost of construction and dung requirement of different size solid state Fixed dome biogas plants**

| Sl. No. | Plant size ($m^3$) | Cost* of Construction (Rs.) | Dung requirement (kg) | No. of stall fed cattle required to supply the dung | Family member |
|---|---|---|---|---|---|
| 1 | 1 | 22,000 | 25+ | 2-3 | 3-4 |
| 2 | 2 | 28,000 | 50+ | 4-6 | 5-8 |
| 3 | 3 | 37,000 | 75+ | 7-9 | 8-12 |
| 4 | 4 | 50,000 | 100 | 10-12 | 12-16 |
| 5 | 6 | 70,000 | 150 | 13-15 | 16-20 |

**including gas stove and gas delivery pipe, cost may vary depending on fluctuation of market price*




**Schematic diagram of SSBGP (2m$^3$)**

### 3. Improved cook stove for woody biomass (Udairaj)

"Improved cook stove Udairaj" run on usual firewood can be used to meet the household cooking needs. The traditional chullhas used in Assam have thermal efficiency of 9%, resulting in high consumption of firewood. The consumption of firewood can be brought down substantially. Thermal efficiency of Udairaj is 3 times more than traditional Open hearth chullah presently being used.



**Traditional chullah**




**Udairaj improved chullha**

Udairaj improved chullha can be constructed by a semi-skilled mason. Interested person/entrepreneur can contact Head, Department of Agril. Engineering, AAU, Jorhat for skill development training.

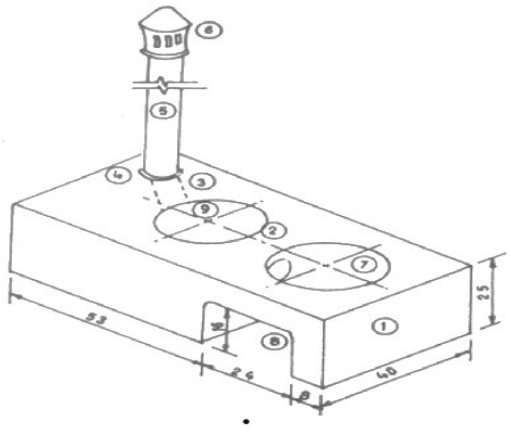
**Comparison of Traditional Vs Udairaj**

| Parameter | Traditional Chullah | Udairaj improved chullah |
|---|---|---|
| Thermal Efficiency | 7-9% | 25-26% |
| Smoke outlet | No chimney | Use of chimney (inclined, metallic) |
| Construction | Earthen | Brick, cement, earthen coating |
| Durability | 1-2years | 4years |
| Cost ,Rs. | 150/-(Approx.) | 850/-(Approx.) |
| Payback period for adopting Udairaj | | 18months |

| | | |
|---|---|---|
| over traditional in saving firewood | | |
| Number of ports | 1 | 2 (primary & secondary) |
| Inclined chimney without piercing the roof | No chimney | Max. $65^0$ Inclined chimney |
| Post flood condition (after submergence) | Un-usable | Usable within 2 days |



**Udairaj chullah in use in one of the household with the inclined chimney pipe through the sidewall**



**Isometric view of Udairaj improved cook stove**

## Items with quantity required for making one chullah

| Sl. No. | Items | Quantity |
|---|---|---|
| 1 | Bricks | 30 |
| 2 | Cement | 12 kg |
| 3 | Sand | 5 cft |
| 4 | AC pipe, cowl & tunnel (GI pipe, cowl & tunnel ) | 1 set (3m length, 7.5cm (dia) |
| 5 | Mason | ½ man day |

**Maintenance**:

1) Chimney pipes need to be cleaned periodically by removing loosely cemented cowls which can be detached easily if pulled with a turn.
2) Dual use of the port was essential in the use of these cook stove. While cooking in the

primary port, the secondary port can be used for hot water kettle. If the secondary port is not in use, it has to be covered firmly with a concave earthen lid firmly placed on the opening.

4. **Improved paddy husk cook stove ("*Tuh chullah*"):** Portable cook stove "*Tuh chullah"* can be used to meet household, outdoor and on farm cooking needs in all agroclimatic zones of Assam. Cost of running this chullah is lowest because of low cost of paddy husk (*Tuh*). This paddy husk based portable cookstove can be run continuously with the provision of intermittent filling and ash removal facility and consumes 1.4 kg husk per hour.



**Conical shaped hopper with stand**

**Fabrication:**

It consists of an outer container with a conical shaped hopper top made up of sheet metal, which is attached to the cylindrical shaped bottom portion of perforated sheet. In the centre, an inner cylinder of perforated sheet (holes of 3 mm dia) is attached at a slightly less height. This acts as the combustion chamber. The hopper and combustion chamber placed on the top of grate made of perforated sheet, is pivoted at the center of the inner cylinder. Any interested fabricator / entrepreneur can obtain the drawing of the cook stove from Head, Department of Agricultural Engineering, AAU, Jorhat.

**Maintenance**:

1) Ash is to be removed by rotating the lower movable grate of the stove while cooking.
2) Ash is to be removed by inverting the chullah 1 hour after cooking activity.
3) Nut and the bolt of the grate need periodic tightening.
4) Inner smaller cylinder is required to be replaced after one year of use.





***Tuh Chullah* in operation** **Portable cook stove in use at the backyard of the household**

5. **Raised bed planter:** Mechanical seed placement in raised bed having bed width 600-700 mm and bed height 150-220 mm by tractor drawn raised bed planter with either fluted roller type or cell type seed metering can be used for growing green gram and black gram in *Kharif* season.

Normal Pulse cultivation in *Kharif* is difficult in Assam due to water logging. In Summer season irrigation & drainage pose problem in the cultivation of pulse crop. Raised bed planting facilitates both irrigation as well as drainage.





Tractor Drawn raised bed planter irrespective of make gives much better results than manual line sowing (without bed). This technology of raised bed planting can provide a much desired solution for growing *kharif* pulse which otherwise is a risky crop because of higher rainfall and subsequent water logging.

## COMPARISON OF RAISED BED PLANTERS WITH FARMERS PRACTICE

| Parameters | Raised bed planter | | | Farmers practice |
|---|---|---|---|---|
| | Dashmesh | National | Khedut | Line sowing |
| Working width (mm) | 2000 | 1500 | - | - |
| Metering | Fluted roller | Fluted roller | Cup feed | - |
| Field capacity (ha/h) | 0.378 | 0.392 | 0.372 | - |
| Fuel consumption (l/ha) | 1.48 | 1.435 | 1.384 | - |
| Cost of operation (Rs/ha) | 1418 | 1405 | 1014 | 16000 |
| Comment | All the tested raised bed planters performed well. Dasmesh bed size were higher and better suited for water management | | | |

**List of Manufacturers and Vendors**

1. DASMESH Mechanical Works Pvt. Ltd., Raikot Road, Malerkotla, Distt. Sangrur, Pincode - 148023, Punjab, INDIA
2. National Agro Industries, Link Road, Industrial Area-A, Opp. Transport Nagar Ludhiana -

141003. Punjab. Ph. +91 8146 101 101

3. Khedut Agro Engineering Pvt.Ltd., Plot No. 6, Survey No.191, Shantidham Society Road, Near Orke Pharma, Veraval (Shaper), Rajkot, Gujarat 360024

**N.B.** For operation, repair & maintenance of small tools, implements and machinery including power tiller and Tractor, training to the interested group of personals can be given in the Department of Agricultural Engineering, Assam Agricultural University, Jorhat- 785013, Assam with prior intimation observing official formalities.

**6. Areca nut peeling machine:** AAU-developed Areca nut peeling machine with a capacity of 90-100 pieces of areca nut per hour is recommended for peeling of fermented areca nut. The peeling tool is utilized to remove the adhered skin of 8-10 numbers of fermented areca nuts within 90 seconds by subjecting nuts to shearing force against rotating blades. This technology will reduce time and the drudgery of cleaning areca nuts.

**Description of the tool:**

- The construction materials of the tool are galvanized iron and mild steel.
- The components of the tool are,
  - Cutting blade
  - 0.5 hp single phase AC motor
  - Solid shaft
  - Covering plate
  - Rigid frame
- Overall dimension of the machine is 325 mm (H) x 315 mm (D)
- The overall weight of the machine is 12 kg.
- Required manpower for operation is one.
- The tool is suitable for peeling fermented areca nuts.
- Unit cost of operation is Rs. 6.14/ per nut with betel leaf
- Fabrication cost of the machine Rs. 1800.00 per unit.

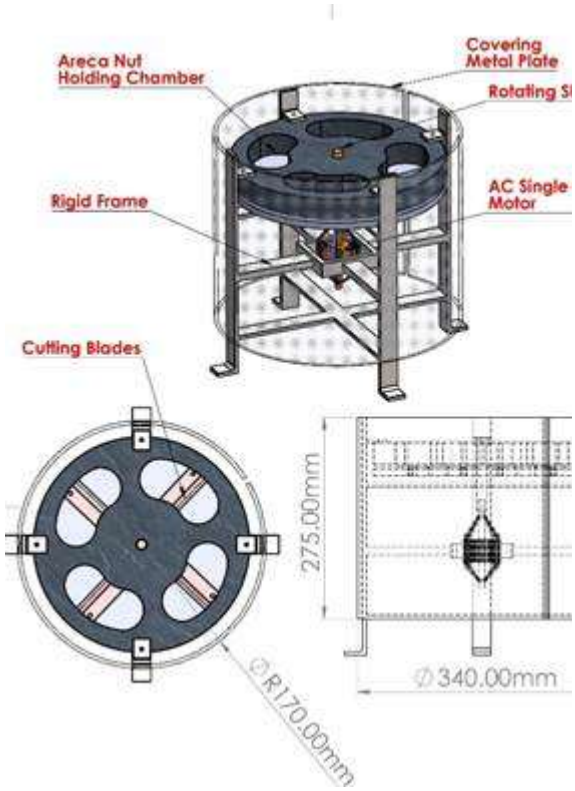
**Comparison between Manual Peeling and Areca nut peeling**

| Type of process | Peeling efficiency (%) | Capacity (No./ h) | Operating Hour |
|---|---|---|---|
| Manual | 90 | 50-60 piece | Not more than 1.5 minutes due to human fatigue |
| Mechanical | 95 | 90-100 piece | Continuous |

**OPERATIONAL PROCEDURE:**

- Select 8-10 nos. of fermented areca nut of size 4-4.5cm.
- Remove the outer layer i.e., husk of the areca nuts by hand.
- Keep the areca nuts inside the chamber of the peeler.
- Place the cover plate over the peeling chamber.
- Connect the power cable of the tool with 220-volt, 50 Hz AC power supply.

- Power on the main switch.
- Use the regulator to maintain the speed for 1-1.5 minutes.
- Reduce the speed and disconnect the power source.
- Use spring mounted brakes to stop the rotating blades.
- Take out the peeled areca nut from the chamber.




**CAD of areca nut peeling tool**




**Areca nut peeling tool**

**PRECAUTIONS:**

The involvement of electrical power in mechanical devices is prone to risk of injury. Therefore, the operator must follow the following few basic points for operating the areca nut peeling tool.

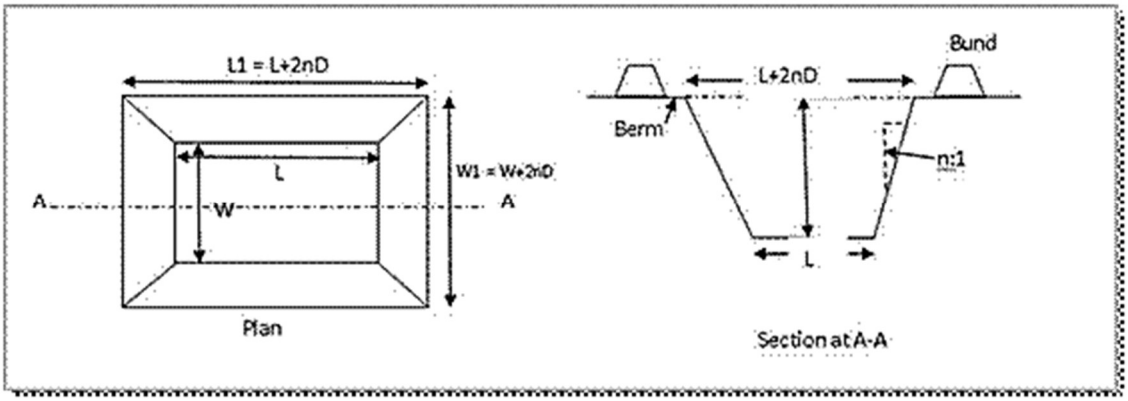
- Do not use more the 10 nos. of areca nut at one time.
- The size of the areca nut should be maintained for smooth operation.
- Always cover the areca nut chamber before electrical switch is on.
- Check the proper electrical connection before the switch is on.
- Do not open the cover until the rotation of blades is stopped.
- Do not use the brake before switching off the main switch.

**7. Lining material for rainwater storage in farm pond in sandy loam soil of Assam**

Lining of farm pond by HDPE film of 200-micron thickness may be recommended in low water table areas of Assam (eg., sandy loam soil under low water table condition of North bank Plain Zone of Assam) where percolation loss makes unlined farm ponds ineffective for rain water harvesting and subsequent utilization for life saving irrigation. The unit cost (Rs./L) for harvested rain water varies with pond water spread area (e.g. Rs 0.35/L for 1225 sqm. to Rs 0.78/L for 25 sqm. at present cost). This technology can be tested in Governmental schemes like "*Amrit Sarovars.*"

**Table 7.1: Unit cost (Rs./Litre of harvested water)**

| Pond Size | Pond Volume | HDPE Film | HDPE film cost (115 Rs./Sqm) | HDPE film laying (50 Rs./Sqm) | Pond digging (DSR 2021) | Total cost | Total Unit cost/litre of storage |
|---|---|---|---|---|---|---|---|
| $m^2$ | $m^3$ | $m^2$ | Rs | Rs./$m^2$ | Rs. | Rs. | Rs./l |
| 5 × 5 | 35.833 | 115.3 | 13257 | 5764 | 8985 | 28006 | 0.78 |
| 10 × 10 | 145.83 | 227.1 | 26111 | 11353 | 38552 | 76016 | 0.52 |
| 15 × 15 | 395.83 | 412.8 | 47466 | 20638 | 101407 | 169511 | 0.43 |
| 20 × 20 | 770.83 | 648.5 | 74572 | 32423 | 194556 | 301550 | 0.39 |
| 25 × 25 | 1270.8 | 934.2 | 107427 | 46708 | 318014 | 472149 | 0.37 |
| 30 × 30 | 1895.8 | 1270 | 146039 | 63495 | 471783 | 681316 | 0.36 |
| 35 × 35 | 2645.8 | 1656 | 190394 | 82780 | 655861 | 929035 | 0.35 |




**Cross-section of dug out farm pond**

**Advantages over unlined structures:**

§ Under low water table condition (like NBPZ, BNCA) unlined RWH structure fails due to

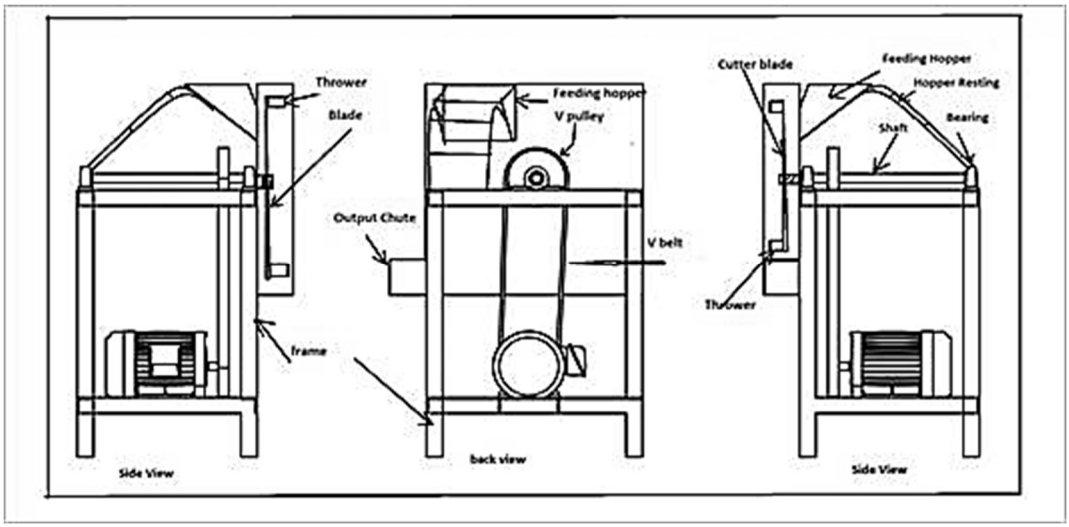
high percolation loss (dries up within 2 weeks).

§ Other lining techniques (tested) dries up within 15 weeks (Cow dung+ Soil Plaster at 2:5) and 24 weeks (Cement + Soil Plaster at 2:10).

§ The HDPE film lined farm ponds can harvest and store rainwater for longer period of time (52 weeks) and can therefore be used for supplemental and life saving irrigation in *rabi* crops during dry spell.

**8. Modified portable low-cost electrical chaff cutter**

An electric chaff cutter machine with 220-230V(1.5kw/hr) with a single phase electric motor and belt drive having dimensions of (65×65×8)cm can be used for cutting banana pseudo stem or fodder into small pieces for preparing feed stock for vermicompost production units.




**Schematic diagram of Modified Chaffcutter**



**Modified Chaffcutter**

**Table 8.1: Details of the chaffcutter:**

| Sl. No. | Components | Specifications |
|---|---|---|
| 1. | Power source | Electrical 220V (Single phase) |
| 2. | Size of box | (65×65×10) cm |
| 3. | Induction motor | 1 HP (Single Phase), 1440 rpm |

| | | |
|---|---|---|
| 4. | Transmission (Pulley and Belt drive) | V- belt, V-belt drive transmission<br>Driver pulley (4-inch diameter)<br>Driven pulley (5-inch diameter) |
| 5. | Cutter Blade | Locally Fabricated mild steel/ high carbon steel of size = (60×3.5×0.4) cm |
| 6. | Shaft and bearing | Iron Diameter = 2.54cm, Length 41cm,<br>Bearing = model P206 |
| 7. | Height of the machine | 90cm |
| 8. | Hopper size | (21×21) cm |
| 9. | No. of Operator | 1 |
| 10. | Ergonomics | User friendly (specially women), Light weight/Safety in operation |
| 11. | Total Weight | 45 kg |
| 12. | Capacity | 500-600kg/hr (Banana pseudo stem) |
| 13. | Cost of the machine (Table 2) | Rs. 20,864/-(Approx.) |

**Table 8.2: Estimate of the machine:**

| Sl. No. | Items | Quantity | Rate (Rs.) | Total (Rs.) | Remarks |
|---|---|---|---|---|---|
| 1. | Iron sheet (18 gauge) 4×8ft (27kg size) | 1 No. | 3,500.00 | 3,500.00 | Rate as per local market price |
| 2. | Angle Iron Piece | 21 kg | 84.00 | 1,764.00 | |
| 3. | Shaft 1 inch diameter (4kg size) | 1 No. | 500.00 | 500.00 | |
| 4. | Electric Motor 1hp | 1 No. | 5,000.00 | 5,000.00 | Online price |
| 5. | V Belt Pulley | 2 Nos. | 400.00 | 800.00 | Rate as per local market price |
| 6. | V Belt | 1 No. | 400.00 | 400.00 | |
| 7. | Bearing and Hub | 2 Nos. | 1,200.00 | 2,400.00 | |
| 8. | Electrical accessories (Wire/Plug/ bolts and Nut/paints) | - | 500.00 | 500.00 | |
| 9. | Cutter Blade Fabrication | 1 No. | 1000.00 | 1000.00 | Rate as per local market price |
| 10. | Work Shop Charge /Lathe machine job | - | 5,000.00 | 5000.00 | Rate as per local market price |

| | | | | | |
|---|---|---|---|---|---|
| | | | **Total** | **20,864.00 (Approx.)** | |

**Advantages over manual:**

The existing process of banana pseudo stem or fodder cutting is done either manually by using sharp knives/dao or commercially available manual or electrical chaff cutter**.** Fabrication can be done locally, Low manufacturing, repairing and maintenance cost, highly efficient, high productivity and less time consuming, can be easily transported from one place to another. Can be used for cutting fodder or banana pseudo stem into small pieces for livestock and for preparing feed stock for vermicompost.

**Production efficiency:** 3:1 (Output by chaff cutter: Manual labour)

# STANDARD OPERATING PROTOCOL (SOP) FOR DRONE APPLICATION OF NANO FERTILIZERS, MACRO AND MICRO NUTRIENTS, INSECTICIDES AND PESTICIDES IN CROPS

**SOPs for selected crops standardized by AAU**

| Parameters | Crops | | |
|---|---|---|---|
| | **Rice** | **Black gram** | **Potato** |
| Flying speed (m/s) | 4.0 – 4.5 | 4.0 – 4.5 | 3.5 – 4.5 |
| Height above crop (m) | 4.5 – 5.5 | 4.0- 5.0 | 5.0 – 5.5 |
| Water volume (l/ha) | 20 | 20 | 20 |
| Droplet size(µm) | 250 - 350 | 250 - 350 | 250 - 350 |
| Swath (m) | 3.0 – 5.0 | 3.0 – 5.0 | 3.0 – 5.0 |
| Wind speed (m/s) | <3 | <3 | <3 |
| Time of spraying | 7- 10 am | 7-10 am | 7- 9 am |

**National Standard for selected crops**

| **S.No.** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Crop name** | **Rice** | **Cotton** | **Maize** | **Ground nut** | **Pigeon pea** | **Safflower** | **Sesame** | **Soybean** | **Sugarcane** | **Wheat** |
| **1. Drone flying speed (m/s)** | | | | | | | | | | |
| During spraying | 4.5-5.0 | 4.5-5.0 | 4.5-5.0 | 4.5-5.0 | 4.0-5.0 | 4.5-5.0 | 4.5-5.0 | 3.5-4.5 | 2.0-3.0 | 4.5-5.0 |
| During launch, RTL and turning | <5.5 | <5.5 | <5.5 | <5.5 | <5.5 | <5.5 | <5.5 | <5.5 | <5.5 | <5.5 |
| **2. Drone height above crop canopy (m)** | | | | | | | | | | |
| Good crop conditions | 1.5-2.5 | 1.5-2.5 | 1.5-2.0 | 1.5-2.0 | 1.5-2.5 | 1.5-2.5 | 1.5-2.5 | 1.5-2.5 | 3.0-4.0 | 3.0-4.0 |
| Varieties/ conditions prone to lodging | 2.0-3.0 | 2.0-2.5 | 2.0-2.5 | 2.0-2.5 | 2.0-2.5 | 2.0-2.5 | 2.0-2.5 | 2.0-2.5 | - | - |
| **3. Water volume (l/ha)** | | | | | | | | | | |
| Stage-1 : Early stage | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 |
| Stage-2: Full canopy stage | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 |
| *Strictly avoid spraying during flowering stage | 6am-11am | - | 6am-11am | - | 6am-11am | 6am- 11am | 6am- 11am | 6am-11am | - | - |
| **4. Nozzles** | | | | | | | | | | |
| Type (Anti drift) | Flat fan | | Flat fan | Flat fan | Flat fan | Flat fan | Flat fan | Flat fan | Flat fan | Flat fan |
| Droplet Size (µm) for Insecticide | 250-350 | | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Droplet Size (µm) for Fungicide | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 | 250-350 |
| Discharge rate (litre/min) | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 | 0.3-0.6 |
| Angle (Degree) | 60-120 | 60-120 | 60-120 | 60-120 | 60-120 | 60-120 | 60-120 | 60-120 | 60-120 | 60-120 |
| Swath(m) | 3-6 | 3-6 | 3-6 | 3-6 | 3-6 | 3-6 | 3-6 | 3-6 | 3-6 | 3-6 |
| Number | 4-6 | 4-6 | 4-6 | 4-6 | 4-6 | 4-6 | 4-6 | 4-6 | 4-6 | 4-6 |
| Pressure (bar) | 2-3 | 2-3 | 2-3 | 2-3 | 2-3 | 2-3 | 2-3 | 2-3 | 2-3 | 2-3 |
| **5. Suitable time schedule for spray** | | | | | | | | | | |
| Summer and rainy season: | 6 am-10am | 6am-10am | 6am-10am | 6am-10am | 6am- 10am | 6am-10am | 6am-10am | 6am-10am | 6am- 10am | 6am-10am |
| | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm | 3pm-6pm |
| Winter season: | 8am-11am | 8am-11am | 8am-11am | 8am-11am | 8am-11am | 8am-11am | 8am-11am | 8am-11am | 8am-11am | 8am-11am |
| | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm | 2pm-6pm |
| *Strictly avoid spraying during flowering stage | 6am- 11am | - | 6am-11am | - | 6am- 11am | 6am-11am | 6am-11am | 6am-11am | - | - |

Do's and Don'ts for pesticides application using drone

| **Before Spraying** | **During spraying** |
|---|---|
| • Crop parameters | • Crop and site specific |
| • Crop field/ site parameters | • Drone operation |
| • Weather conditions | • Sprayer control |
| • Drone | • Drone operator and accidents |
| • Spraying system | **After spray operation** |
| • Chemicals | • Drone |
| • During storage and handling | • Spray and chemicals |
| • While preparing spray solutions. | • Drone operator |

**Before Application of pesticide using drone**

| **Do's** | **Don'ts** |
|---|---|
| **Before Application of pesticide using drone** | |
| **Crop Parameters**:<br>• The crop stage, height and canopy development should be taken into consideration before spraying using drone.<br>• The drone can be operated as per the crop condition for spot, band or elevated targeted spraying. For spot spraying the area should be marked first using appropriate marker.<br>• The drone spraying is most ideas for mono cropping pattern. | • Avoid pesticide spray during pollination stage.<br>• Avoid using drone if crop is prone to lodging.<br>• Avoid chemical spray using drone if rain is forecasted. |

| | |
|---|---|
| **Crop Field: Site parameters**<br>• Check field comes under green, yellow or red zone. If field comes under yellow zone take proper permissions from DGCA and local police.<br>• Note down the Latitude and Longitude of the field for the record. | • Do not plan for pesticide spray using drone if field comes under red zone. |
| • Check GPS connectivity and stability status.<br>• Check shape of the field whether drone can be used or not.<br>• Check terrain conditions (Slope, Plain or hills etc.).<br>• Check obstacles in the field like Trees, poles, HT line, Hill, power stations, solar systems, pumping station, fencing, agriculture machinery etc.<br>• Check Line of sight.<br>• Check the suitable Take off & Landing area | • Do not plan for spraying using drone nearby fodder crop, grazing field, water bodies like pond, river, canal etc., village, no crop field, roads, poultry house, barn yard, schools, etc. and maintain a minimum distance of 100m from these bodies.<br>• Do not plan for a large field spraying (beyond visual line of sight) using drone to avoid risk of signal loss between remote control and the drone. |
| **Do's** | **Don't** |
| **Before Application of pesticide using drone** | |
| Weather/Environmental conditions:<br>• Note down the date and time for spraying in log book.<br>• Check the temperature and plan for spray if temperature is preferably less than 35 °C preferably.<br>• Check the humidity and plan for spray if humidity is more than 50 %.<br>• Check the wind speed and plan for spray if the wind speed is less than 3 m/s.<br>• Check the wind direction and according fix the starting home point of the drone for flying.<br>• Consider the weather forecast and take the decision accordingly for drone flying. Cloudy conditions would deter GPS connectivity | • Do not plan for pesticide spray using drone if temperature is more than 35°C.<br>• Do not plan for pesticide spray using drone if humidity is less than 50%.<br>• Do not plan for pesticide spray using drone if wind speed is more than 3m/s.<br>• Do not plan for chemical spray using drone if the rain is expected within 6 hours. |

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| **Drone**:<br>• Always have motor, propeller, ESC and battery in spare.<br>• Note down the drone type and payload capacity.<br>• Always carry a proper tool kit according to Drone flying for Spraying.<br>• The drone should be capable of flying in both Manual and Auto mode interchangeably.<br>• Calibrate the drone and ensure good stability during operation.<br>• Check chemical feeder pipes to the nozzle for airlocks<br>• Ensure the nozzles are free from clogging and scaling<br>• Ensure uniformity in discharge from each nozzle<br>• Check the GPS accuracy and decide the overlap percentage.<br>• Calibrate the RTK GPS based station if auto mode flying is planned.<br>• Check and calibrate the drone sensors like Lidar, Baro meter Radar etc.<br>• Optimize the drone speed range.<br>• Check remote control range and batteries status.<br>• Check the Battery capacity and status and note down the battery age, number of batteries etc.<br>• Check the drone components for proper fitting.<br>• Check the drone propellers are in good condition.<br>• Check the drone motors are in good condition.<br>• Check the motor temperature.<br>• Check the drone propellers are in proper orientation<br>• Obtain Unique Identification Number (UIN) from DGCA for operating in controlled airspace (where the ATC services are active) and affix it on your drone.<br>• Obtain Unique Identification Number (UIN) from DGCA for drone before flying<br>• Obtain RPTO License for flying the drone of respective category (i.e. for small drone or medium drone). | • Do not fly drone if batteries are not in good condition and not properly charged or bulged.<br>• Do not fly drone if propellers are broken or cracked.<br>• Do not fly drone if motor is heating too much<br>• Don't fly a Nano drone above 50 feet from the ground level.<br>• Don't fly a micro drone above 200 feet from the ground level.<br>• Don't fly drones more than 400 feet above the ground level.<br>• Don't fly drone near other aircraft (manned or unmanned).<br>• Don't fly drone near airports and heliports.<br>• Don't fly drone over groups of people, public events, or stadiums full of people without prior permission.<br>• Don't fly drone over government facilities/ military bases or over/ near any no-drone zones.<br>• Don't fly drone over private property without permission.<br>• Don't fly drone in controlled airspace near airports without filing of flight plan or AAI/ ADC permission (at least 24 hours before actual operation).<br>• Don't drop or carry hazardous material.<br>• Don't fly drone under the influence of drugs or alcohol.<br>• Don't fly drone from a moving vehicle, ship or aircraft. |

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|---|---|
| • Obtain Unmanned Aircraft Operator Permit (UAOP), if applicable from DGCA for commercial operations and keep it handy.<br>• Fly only during daylight (after sunrise to before sunset).<br>• Fly in good weather: Good weather lets you not only to fly your drone better but also to keep track of it when it is airborne.<br>• Fly within visual line of sight (VLOS): Always be within visual range of your drone.<br>• Be aware of airspace restrictions/ no drone zones and respect privacy of people.<br>• Intimate concerned authorities (like DGCA, local police etc.) of any incidents/ accidents within the prescribed tile limit | |

| | |
|---|---|
| **Spraying system**<br>• Check the tank capacity and its cleanliness<br>• Note down the pump flow rate capacity and set the pressure as per recommendation or as mentioned in operating manual.<br>• Note down the number of nozzles and ensure that the nozzles are not blocked.<br>• Check the nozzles spacing and set as per the manufacturer manual.<br>• Note down the overall boom size of the sprayer.<br>• Check the nozzle discharge rate.<br>• Check the hose pipe and the connector for leakage.<br>• Check the leakage in the water tank and pipe line.<br>• Check the solenoid valve functioning and its control through remote control.<br>• Ensure that the tank is properly cleaned before filling the new chemical.<br>• Select right type and size of nozzles.<br>• Use separate sprayer for insecticides and weedicides.<br>• The droplet size for pesticides should in the range of 250 to 350 microns. | • Do not fly drone if the tank, nozzle, hose pipe are loose.<br>• Do not fill the chemical in the tank more than the recommended level.<br>• Do not fly drone if there is leakage in tank, hose pipe or connector.<br>• Do not fly drone if solenoid valve is not working properly.<br>• Do not fly drone if cap of the tank is missing.<br>• Do not fly drone if drone is unstable.<br>• Do not use leaky or defective equipment.<br>• Do not use defective/worn out and non-recommended nozzles.<br>• Do not blow/clean clogged nozzles with mouth. Instead use tooth brush tied with sprayer.<br>• Never use same sprayer for both weedicides and insecticides. |

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|---|---|
| **Chemical**:<br>• Purchase pesticides/biopesticides only from registered pesticide dealers having valid License.<br>• Purchase only just required quantity of pesticides for single operation in a specified area.<br>• See approved labels on the containers/packets of pesticides.<br>• See Batch No., Registration Number, Date of Manufacture/ Expiry on the labels.<br>• Purchase pesticides well packed in containers. | • Do not purchase pesticides from unregistered dealers or from un-licensed person.<br>• Do not purchase pesticide in bulk and store for whole season.<br>• Do not purchase pesticides without approved label on the containers.<br>• Never purchase expired pesticides.<br>• Do not purchase pesticides whose containers are leaking/loose/ unsealed. |

| | |
|---|---|
| **During Storage:**<br>• Store the pesticides away from home premises<br>• Keep pesticides in original containers.<br>• Pesticides/ weedicides must be stored separately.<br>• Where pesticides have been stored, area should be marked with warning signs.<br>• Pesticides should be stored away from the reach of the children and live stocks.<br>• Storage place should be well protected from direct sunlight and rain.<br>• Ensure Crop protection chemicals are stored in lock and key. | • Never store pesticide in home premises.<br>• Never transfer pesticides from original to another container.<br>• Do not store insecticides with weedicides.<br>• Do not allow children to enter the storage place.<br>• Pesticides should not be exposed to sunlight or rain water. |

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| **While handling:**<br>• Keep pesticides separate during transportation<br>• Bulk pesticides should be carried carefully to the site of application. | • Never carry/transport pesticides along with food/fodder/other eatables.<br>• Never carry bulk pesticides on head, shoulder or on the back. |
| **While preparing spray solution**: | |
| • Always use clean water.<br>• Do not allow the pesticide/its solution to fall on any body parts. Never avoid reading instructions on container's label for use.<br>• Use protective clothing *viz*., nitrile hand gloves, face masks, cap, apron, full trouser, etc. to cover whole body.<br>• Always protect your nose, eyes, ears, hands, etc. from spill of spray solution.<br>• Read instructions on pesticide container label carefully before use. | • Do not use muddy or stagnant water. Never prepare spray solution without wearing protective clothing.<br>• Never use left out spray solution after 24 hours of its preparation.<br>• Do not mix granules with water.<br>• Do not smell the spray tank.<br>• Do not use overdose which may affect plant health and environment.<br>• Do not eat, drink, smoke or chew during whole operation of pesticides spraying. |

| | |
|---|---|
| • Prepare the solution as per requirement.<br>• Avoid spilling of pesticides solutions while filling the spray tank.<br>• Always use recommended dosage of pesticide.<br>• No activities should be carried out which may affect your health. | |

**While applying spray solutions**

| **While applying spray solutions** | |
|---|---|
| **Crop and site specific**<br>• Mark the boundaries of the crop field using flag or Geotag the boundaries.<br>• Check if any other drone operator is flying the drone nearby fields.<br>• Animals should not be allowed to enter into the field during spray operation and upto 48 hours after spray. | • Do not fly drone without marking boundaries of the crop field.<br>• Do not fly drone very close to the crop canopy. It may cause crop lodging. |
| Drone operation<br>• Keep an eye on drone stability during operation.<br>• Keep drone within the visible range.<br>• Take care of obstacles in the crop field.<br>• Take care of overlap percentage of swath.<br>• Take care drone flies within the periphery of crop field.<br>• Drone should have emergency landing option (fail safe mode) when battery discharges<br>• Keep spare charged batteries for emergency. | • Drone operator should not hand over the control of drone to other unskilled person during operation<br>• Do not fly the drone very close to obstacle points.<br>• Do not fly drone out of the visible range.<br>• Do not fly drone if it is unstable.<br>• Do not keep overlap percentage more than 80 percentage or less than 60 percentage.<br>• Do not spray if the nozzles are blocked or leaking.<br>• Do not allow drone to enter adjacent crop fields. |

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|---|---|
| • Drone operator should inform to local police if drone crashed accidentally and it should be recoded in log book with proper reason of accidents. Take also the photographs of the accidents.<br>• Operator has to apply for insurance claim as soon as possible. | • Drone operator should not stand in the direction of the wind to avoid chemical exposure due to drift.<br>• Drone operator should not allow other person to stay very near the drone take-off and landing point. |

| | |
|---|---|
| **Sprayer control**<br>• Apply only recommended dose and dilution.<br>• Spray operation should be conducted on cool and calm and sunny day.<br>• Use recommended spraying combination for each mission<br>• Spray operation should be conducted in the windward direction.<br>• After spray operation, sprayer assembly and other equipment used during the operation should be washed thoroughly with clean water using detergent/soap. | • Never apply over-dose and higher concentrations than recommended.<br>• Do not spray on hot sunny day or under strong windy conditions.<br>• Do not spray just before rains and immediately after the rains.<br>• Do not spray against the wind direction.<br>• Containers and buckets used for mixing pesticides should never be used for domestic purpose even after thorough washing.<br>• Never enter in the treated field immediately after spray without wearing protective clothing |
| **Drone operator and accidents**<br>• Remote control should be in hand of the operator even if the drone is in auto mode.<br>• Drone operator should cover his eyes with goggles to avoid chemical exposure. | • Drone operator should not remove the PPEs during spraying operation<br>• Drone operator should not remove safety goggles during operation. |

**After Spray Operation**

| **After Spray Operation:** | |
|---|---|
| **General**<br>• Avoid the entry of animals/workers in the field immediately after spray. | • Do not irrigate the field just after spraying of chemicals<br>• Do not harvest the crop immediately after spraying for fodder.<br>• Do not spray fertilizer just after pesticide spray |
| **Drone**<br>• Take out the battery from the drone and put it for charging immediately.<br>• Check the motor temperature and propellers.<br>• Clean the chemical exposed surfaces of the drone.<br>•All the accessories of drone should be kept in transportation box/bag. | • Do not expose the battery terminal point to avoid short circuiting.<br>• Do not forget the drone and its accessories in the field after spraying |
| **Sprayer and chemicals**<br>• The used/empty containers should be triple rinsed, dried and punctured and handed over to disposal agency<br>• Left over spray solutions should be disposed off at safer place viz. barren isolated area. | • Empty containers of pesticides should not be re- used for storing other articles.<br>• Left over spray solution should not be drained in or near ponds or water bodies etc. |

| | |
|---|---|
| **Drone Operator**<br>• Clean the PPE after use<br>• Wash hands and face with clean water and soap before eating/smoking.<br>• On observing poisoning symptoms provide the first aid and show the patient to doctor. Also show the empty container to doctor. | • Never eat/smoke before washing clothes and taking bath.<br>• Do not take the risk by not showing the poisoning symptoms to doctor as it may endanger the life of the patient |

## SoP for application of Pesticides through drone in agriculture

**General Guidelines for pest management in crops**

- Do not spray if there is forecast of heavy rains in next 1-2 days.
- Avoid broad-spectrum insecticides when a narrow-spectrum or more specific insecticide will work. Preferably use green labelled insecticides.
- Avoid repeated usage of the same insecticide, insecticides of the same chemical class, or insecticides in different classes with similar mode of action and rotate/alternate insecticide classes and having different modes of action.
- Insecticides should be used only as a last resort when all other non-chemical management options are exhausted and P: D ratio is above 2: 1.
- Apply bio-pesticides/chemical insecticides judiciously after observing unfavourable P: D ratio and when the pests are at most vulnerable life stage. Use application rates and intervals as per label claim.
- Always use protective clothing, face mask and gloves during preparation and application of pesticides.
- Enter the field only after 48 hours of spraying pesticides. Minimum interval between application of chemical insecticide and harvest of maize corn should be 30 days.

**Critical parameters to be considered while spraying of pesticides**

**A. Pesticides/Insecticides**

- Only Central Insecticides Board and Registration Committee (CIB&RC) approved pesticides/Insecticides shall be used.
- The dose has to remain within range approved by CIB&RC.
- The pesticides/insecticides (Liquid/Solid) compatibility with the drone spray system shall be established prior to the mission for the desired dilution. This is to ensure pesticides/insecticides solubility formulation stability and ability to spray with the type of nozzles provided with the drone. In case of mixing of more than on pesticides/insecticides, CIB&RC specific guidelines must be adhered to.
- The minimum dilution shall be decided based on the fulfilment of the above-mentioned requirements and ensuring satisfactory coverage of spayed input both horizontally and vertically.

**B. Safe guarding of non-targets**

The non-targets shall be safe guarded by adhering to the following operational protocols:

- Buffer zone as per CIB&RC approved guidelines to cater to pesticides/insecticides drift, shall be maintained between the adjacent farms or different crops to avoid spray on non-targets.
- During the spray operation, operator should always maintain the distance approved by CIB&RC from the drone and avoid windward direction as much as possible.
- No human or animal movement shall be permitted within the farm during and immediately after the spray operations.
- Drone based pesticides spray operations should be conducted at a distance approved by CIB&RC from water bodies, residential areas, fodder crops, public utilities, dairy, poultry etc. and as per DGCA guidelines.

The crop wise list of pesticide molecules tested for phytotoxicity (crop safety) using drones

| **Crop** | **Insecticides, Fungicides alone & in combinations** |
|---|---|
| Rice | Chlorantraniliprole 18.5SC, Tebuconazole 50 + Trifloxystrobin 25 (75WG), Propiconazole 25EC, Azoxystrobin 18.2 + Difenoconazole 11.4SC, Picoxystrobin 7 + Propiconazole 12 SC, Chlorantraniliprole 18.5SC and (Tebuconazole 50 + Trifloxystrobin 25 - 75 WG), Chlorantraniliprole 18.5SC and (Picoxystrobin 7 + Propiconazole 12 SC), Chlorantraniliprole 18.5SC and (Flupyroxad 62.5 + Epoxiconazole 62.5EC), Chlorantraniliprole 18.5SC and (Azoxystrobin 18.2 + Difenoconazole 11.4SC), (Acephate 50 + Imidacloprid 1.8 SP), Triflumezopyrim 10SC, Pymetrozine 50WG, Dinotefuran 50SG, Cartap hydrochloride 50SP |
| Cotton | Flonicamid 50 WG, Spinetoram 11.7SC, Profenophos 50EC, Acephate 75SP, Diafenthiuron 50WP, Fipronil 5SC, Imidacloprid 17.8SL, Acetamiprid20 SP, Propiconaozole 25EC, Kresoxim methyl 44.3SC, Tebuconazole 50 + Trifloxystrobin 25 Metiram 55 + Pyraclostrobin 5, Azoxystrobin 18.2 + Difenoconazole 11.4SC |
| Red gram (Pigeon pea) | Chlorantraniliprole 18.5SC, Flubendiamide 39.35SC, Indoxacarb 14.50SC, Emamectin Benzoate 5SG, Spinosad 45SC, Novaluron5.25 + Indoxacarb 4.50SC, and Chlorantraniliprole 9.30 + Lambda-cyhalothrin 4.6 ZC |
| Groundnut | Tebuconazole 25.9EC, Chlorantraniliprole 18.5SC and Tebuconazole 25.9EC |
| Soybean | Chlorantraniliprole 18.5SC, Chlorantraniliprole 18.5SC |
| Wheat | Pyraclostrobin 133 g/l + Epoxiconazole 50 g/l SE (Opera 18.3 SE) @ 750 ml<br>Pyraclostrobin 133 g/l + Epoxivpnazole 50 g/l SE (Opera 183 SE) @1500 ml/ha |

## Critical parameters to be considered while spraying of nutrients

### A. Spraying of soil/ crop nutrients

The following aspects need to be considered when using drone for the application of soil/ crop nutrients:

1. Form/ concentration of the nutrient being sprayed/ broadcasted.
2. Sensors / spraying systems installed on the drone.
3. Both forms of essential soil / crop nutrients i.e. solid forms like powder, crystals, prills, granules, super granules, briquettes, etc. and liquid forms like water soluble powders, liquid nutrients, growth regulators etc. whether organic or inorganic can be sprayed using drone.
4. Al least 10 times more area can be sprayed per day per drone as compared to the traditional knapsack sprayers.
5. 80-90% of water can be saved in comparison to traditional spraying methods (depending on the sprayer system of the drone.)
6. Different kind of sprayer nozzles are available depending on the form and concentration of the nutrients to be applied.

### B. Procedures for application of solid (granules/powder) crop nutrients using drone.

1. Keep drone height range of 1.0 to 3.0 m above crop canopy for each crop and each stage.
2. Maintain the drone forward speed between 3-8m/sec.
3. Appropriate dispensers like rotating disc type and pneumatic dispensers can be used for spraying of granular crop nutrients.
4. Rotational speed of disc preferably be in the range of 800 to 1000 rpm or as user manual.

### C. Procedures for application of liquid crop nutrients using drone.

Liquid biofertilizers, organic (jivamrit, vermiash etc.) bio decomposers, nano fertilizers, liquid and soluble micronutrients and bio-stimulants existing formulations can be sprayed as per recommendation of approved agencies or institutions.

1. Keep the drone height in range of 1.0 to 2.0 m above the crop canopy for each crop at each stage.
2. In each flight, drone payload should not be more than 10 lit of liquid; crop nutrient can be dispensed at a forward speed between 3.0 to 6.0 m/s.
3. Preferably keep nozzle beneath the motors to minimize the drift of the droplets.
4. Restrict spraying to the intended area.

## Recommendation on spraying of nutrients through drone

| Crop | Existing recommendation at AAU | Time of application (spray) | Recommendation in AAU Through drone | Govt. Protocol using drone | Water vol (lit/ha) according to drone SOP |
|---|---|---|---|---|---|
| *Kharif* Rice (Hybrid) | Application of Zinc as basal @ 7.5 kg/ha ($ZnSO_4.7H_2O$ @ 37.5 kg/ha) and two foliar spray of 0.25 % Borax (Borax 1.5 kg in 600 liter (per spray) per ha at Panicle Initiation and Milk stage along with the recommended dose of N:P:K (80:40:40 kg/ha). | Two foliar sprays of 0.25 % Borax (Borax 1.5 kg in 600-liter (per spray) per ha at Panicle Initiation and Milk stage | Nil | Nil | Early stage-20 lit<br>Full canopy-25 lit |
| *Kharif* Maize (Hybrid) | Application of Zn @ 2.5 kg/ha and three foliar sprays of 0.50 % Zinc sulphate heptahydrate (3 kg in 600 liter water per ha per spray) at knee high, tasseling and silking stage along with the recommended dose of NPK (60:40:40) | Three foliar sprays of 0.50 % Zinc sulphate heptahydrate (3 kg in 600-liter water per ha per spray) at knee high, tasseling and silking stage | Nil | 1-2 ml nano zinc per litre | Early stage-20lit<br>Full canopy-25 lit |
| Rice HYV (Nano Urea) | Apply 1/3rd of RDN with full dose of basal at P & K as basal and 1/3rd of N as at active tillering stage and followed by FS nano urea @ 4 ml/lit at MT and PI stage | FS of nano urea @ 4 ml/lit at MT and PI stage | Nil | FS of Nano urea @ 4 ml/lit | Early stage-20lit<br>Full canopy-25 lit |
| Rice HYV (Nano DAP) | Apply 50%basal DAP (100 recommended NPK) with RDT of nano DAP @ 5ml/lit followed by FS of Nano DAP @ 4 ml/lit at 30DAT after transplanting in Kharif rice. | FS of Nano DAP @ 4ml/lit at 30DAT after transplanting in *Kharif* rice. | Nil | FS of Nano DAP @ 4 ml/lit | Early stage-20lit<br>Full canopy-25 lit |

# APPENDIX I
## Disease Reaction of Recommended Rice Varieties

| Varieties | Reaction to | | | Remarks |
|---|---|---|---|---|
| | Leaf blast | BLB | Sheath blight | |
| **Semidwarf:** | | | | |
| Prasad | R | | | |
| IR 36 | R | | | |
| Ratna | R | | | |
| Pankaj | T | T | T | Resistant to heat and cold |
| Biraj | S | | | |
| Kmj 1-19-1 | - | | | |
| Govind | R | R | | |
| IR 50 | R | | S | |
| Bala | R | | S | |
| Cauvery | T | | | |
| Krishna | S | | | |
| TTB 14-1 | S | | | |
| TTB 15-1 | S | | | |
| IET 6666 | R | T | T | |
| (Lakhimi) | | | | |
| **Tall:** | | | | |
| Monohar Sali | R | R | R | Not susceptible to brown spot and bakanae disease |
| Mahsuri | S | T | S | |
| Phulpakhri | T | | | |
| Banglami | S | | | |
| Rangadaria | S | | | |
| Ahujaha | S | | | |
| Kolasopila | | | | |
| Maibee | S | | | |
| Dimro | S | | | |

S = Susceptible, T = Tolerant, R = Resistant.

# APPENDIX II

## IPM Module for managing insect pests of rice in Assam

The IPM module recommended for insect pest management in rice crops of Assam is as follows-

**Varietal Resistance:** Any resistant/ tolerant variety

**Chemical control**:

- Need based application of pesticide based on Economic Threshold Level (ETL) (Table 1)

**Cultural control:**

- Timely planting (as per recommendation)
- Optimum plant population (as per recommendation)
- Balanced fertilizer application and split application of N (as per recommendation)
- Clean cultivation
- Regular pest monitoring (use of pheromone traps @ 8 traps/ha for YSB)

**Biological control:**

- Release of egg parasitoids *Trichogramma japonicum* @ 50,000/ha (six releases) on observing the moths of YSB
- Application of *Beauveria bassiana* impregnated Rice Husk Saw Dust Rice Bran (RHSDRB) medium @ 3 kg/ha in 600 liter of water ($10^7$ spores/ml)

**ITKs:**

- Use of bamboo perches to encourage predatory birds. Remove the perches as the crop reaches milky grain stage.
- Use of *Eupatorium odoratum* twigs and leaves in the field to repel insects like case worm
- Use of neem leaf (soil incorporation) against case worm and stem borer

# APPENDIX III

## Technical and Trades Names of Insecticides, Acaricides, Rodenticides and Biopesticide Formulations

| Technical Name | Trade name |
|---|---|
| **A. Insecticides** | |
| **1. Neonicotinoids** | |
| Imidacloprid 17.8 SL | Confidor, Seamer, Josh, Imidastar, Imiden, Courage, Midas 2000, King Dor , Jumbo |
| Imidacloprid 70 WG | Admire , Tatamida 70WS , Josh 70 , Dzire , Global 777 , Ad-Fyre , Victor Plus , Tagmyre& Tropical Magic , Pactus , Sumida |
| Clothianidin 50WDG | Dentotsu |
| Acetamiprid 20SP | Manik, Ekka, Rapid Crytal, Rekord t, Active, Award, King Prid, Acetacel, Aceta, Echo 797, Stona, Lift, Dhan Preet, Crop Pride, Sharp, Proud |
| Thiamethoxam 25WG | Actara , Kri-Oxm , Battalion , Theme , King Tara , Maxima , Wonderex , Renova , Devitara , Maestro 707 , Maxtara , Giltara , Click , Areva , Arrow , Actor |
| **2. Synthetic pyrethroids** | |
| Bifenthrin 2.5 EC | Superflex, Biflex TC |
| Bifenthrin 8 SC | Relstar Plus., Wilstar |
| Bifenthrin 10 EC | Talst, Buland |
| Lambda cyhelothrin 5EC | Karate, Lamor Pro |
| **3. Phenyl pyrazole** | |
| Fipronil 5SC | Regent SC, Ruler, Stargazette, Sonic Flo, Devigent Plus, Fipro King, Vizent , Sonic Flo, Rabid |
| **4.Avermectin** | |
| Abamectin 1.9% EC | Abacin, Cymbush, Vertimec, Tagmec |
| Milbemectin 1EC | Milbeknock |
| Emamectin benzoate5SG | Proclaim, Missile, Emstar 5, Starclaim, Robot, Spolit, Empower, Tatkal, Wegon, EM-1, Xplode, Benzer |
| **5. Diamide** | |
| Chlorantaniliprole 20SC | Rynaxypyr, Coragen |
| Chlorantaniliprole 18.5SC | Cosko, Coragen, Cover Liq |
| Flubendiamide 39.35 SC | Fame, Superzite, Voter |
| **6. Benzoylureas** | |
| Nuvaluron 10EC | Remostar, Pedestal |
| Buprofenzin 25 SC | Jawaa, Flotis, Trust, Irvy, Devifezin, Buprostar, Hillblaze, PI Bupro, Applaud, Braun 111, Apple, Phentom |
| **7. Organophosphorous** | |

| | |
|---|---|
| Ethion 50EC | Tafethion, Krithion, Mit-505, Mitkill, King Mite, Fosmite, Deviastra, Vithion, Fighter, Sumite, Hilmite, Shakti Acaron, Dhanumit |
| Profenofos 50EC | Curacron, Celcron, Jashn, Kriphos, Proven, King Cron, Carina, Devi-soldier, Profenofos 50 EC, Banjo, Maxcron, Jashn |
| **B. Acaricide** | |
| **Tetranoic acid derivatives**<br>Spiromesifen 22.9 SC | Oberon, Voltage |
| **Diphenyloxazole**<br>Etoxazole 10.00 SC | Borneo |
| **Pyrazole**<br>Tebufenpyrad 20 WP | Pyranica, Karia |
| **Sulfite ester**<br>Propargite 57.00 EC | Omite, Simbaa |
| **C. Rodenticide** | |
| Zinc Phosphide 80.00 Powder | Ratox, Ratol, Commando |
| Bromadiolone 00.25 CB | Kalrat CB, Ratcon CB |
| Bromadiolone 00.005 RB | Dr.Pest , Racumin Sure |
| Aluminum Phosphide 56.00% 3g Tablet, 10g Pouch | Celphos, Quick Phos, Phostoxin |
| **D. Biopesticide** | |
| *Bacillus thuringiensis* var. *kurstaki* | Delfin, Halt |

# APPENDIX IV
## Technical and Trade Names of Fungicides/Antibiotics/Plant Products/ Antagonists

| Technical name | Trade name and formulation |
|---|---|
| **A. Inorganic copper compounds** | |
| Copper oxychloride | Akomin, Blitox-50, Fytolan, Kilex, Fycop, Nagcoper, Dhanucop, Fycop, Blue Copper 50, COC 50WP, Copsaan, Coprex, Copsin, Cupravit, Hondacop |
| **B. Elemental inorganic sulphur compounds** | |
| Wettable sulphur | Hexasul, Thiovit, Sulfex,Akrisulf, Appu, Cosavet-DF, Microsulf, Mitex S, Pesto Sulfur 80WP, Sulfasaan, Sulphur 85WP |
| **C. Systemic fungicide** | |
| Carboxin | Vitavax, Hiltavax |
| Edifenphos | Hinosan |
| Pyroquilon | Fongoren |
| Tricyclazole | Beam, Trooper |
| Propiconazole | Tilt, Radar |
| IBP | Kitazin |
| Hexaconazole | Sitara, Hexazole, Montaf, Contaf |
| Propineb | Antracol |
| **D. Antibiotics** | |
| Streptomycin sulphate | Streptocycline +Tetracycline |
| Hydroxyquinoline | Entakon-M |
| **E. Fungal antibiotics** | |
| Validamycin | Sheathmar |
| **K. Plant derived products** | |
| Neem products | Achook, Neemazal, Neem Gold, Nimin (neem oil) |
| Cymbopogon product | Wanis |
| **L. Antagonists** | |
| *Pseudomonas fluorescens* | Biofor-PF (Jaiva Kiran), based product Pseudocon |
| *Trichoderma* spp. based | Bicure F, Trich-X-P, Viricon-L, products Bioderma, Trichostar |

# APPENDIX V
## Technical and Trade Name of Herbicides

| **Technical name** | **Trade name and formulation** |
|---|---|
| Fluchloralin | Basalin |
| Isoproturon | Graminon, Arelon, Taurus, Delron, Agrolon,Avonil, Bilron 75WP, Dararon 75WP, Dhar, Gold Medal, Haragron 75, Hilproturon 75WP, Iso 50, Isocin, Isof\guard 75WP, Isokil 75, Isomol 750, Isopar, Isoveer, Isovip, Nocilon 75 |
| Methabenzthiazuron | Tribunil |
| Pretilachlor | Rifit |
| Anilofos | Arozin,Aninoguard,Anilokil 30, Anilotaf,Anilveer, Avail 30EC, Dhanumon, Sardar Guard, Strong 30EC, Surya, Weedonil |

## APPENDIX VI
## Technical and Trades Names of Micronutrient formulation

| **Technical Name** | **Trade name and formulation** |
|---|---|
| **Micronutrient formulation** | Tracel, Agromin, Borax Sulphate, Zinc Sulphate |

# APPENDIX VII A
## Conversion Table (Nutrient-Fertilizer)

| | |
|---|---|
| 1 Kg N | 2.17 Kg Urea |
| 1 Kg $P_2O_5$ | 6.25 Kg SSP |
| 1 Kg $K_2O$ | 1.66 Kg MOP |
| 1 Kg DAP | 2.875 Kg SSP and 400 g |
| Urea 1 Kg N + 1 Kg P + 1 Kg K | 6.66 Kg Sulphala (15-15-15) |

# APPENDIX VII B
## Conversion Table (Area)

| | | |
|---|---|---|
| 1 hectare | =7.5 bigha | =10,000 sq.m |
| 1 katha | =0.2 bigha | =267 sq.m |
| 1 bigha | =0.13 hectare<br>=1333 sq.m | =5 katha |

# APPENDIX VIII

**Micro Preparation and Measurements for Agrochemicals Micro Preparation**

1 milligram of substance in 1 liter of water = 1 parts per million (ppm) solution

**Measurements for agrochemicals**

1 tea spoonful of liquid chemical = 5 ml (approx.)

1 matchbox full of powdered chemical = 7-9 g (approx.)

# APPENDIX IX

**Pesticide calculation formulae**

1) Pesticide required = $\frac{RR \times A}{\%\ a.i.} \times 100$

Where-RR = Recommended rate

A = Area in ha

% a.i. = Per cent active ingredient in the formulation

2) Milliliter of pesticide to be mixed per liter of water = $\frac{DC \times 100}{\%\ a.i.}$

Where-DC = Desired concentration (%)

% a.i. = Per cent active ingredient in the formulation

# APPENDIX X

## Ready reckoner for milliliter or grams of a commercial pesticide to be added to one liter of water in order to obtain the required concentration of spray solution

| Toxicant in the pesticide(%) | Concentration (%) of the spray solution required | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **0.01** | **0.02** | **0.03** | **0.04** | **0.05** | **0.06** | **0.07** | **0.08** | **0.09** | **0.10** | **0.20** | **0.30** | **0.40** | **0.50** | |
| **10** | 1.00 | 2.00 | 3.00 | 4.00 | 5.00 | 6.00 | 7.00 | 8.00 | 9.00 | 10.00 | 20.00 | 30.00 | 40.00 | 50.00 | |
| **15** | 0.67 | 1.33 | 2.00 | 2.67 | 3.33 | 4.00 | 4.67 | 5.33 | 6.00 | 6.67 | 13.33 | 20.00 | 26.67 | 33.33 | |
| **20** | 0.50 | 1.00 | 1.50 | 2.00 | 2.50 | 3.00 | 3.50 | 4.00 | 4.50 | 5.00 | 10.00 | 15.00 | 20.00 | 25.00 | |
| **25** | 0.40 | 0.80 | 1.20 | 1.60 | 2.00 | 2.40 | 2.80 | 3.20 | 3.60 | 4.00 | 8.00 | 12.00 | 16.00 | 20.00 | |
| **30** | 0.33 | 0.67 | 1.00 | 1.33 | 1.67 | 2.00 | 2.33 | 2.67 | 3.00 | 3.33 | 6.60 | 9.90 | 13.33 | 16.67 | |
| **45** | 0.22 | 0.44 | 0.89 | 0.88 | 1.11 | 1.33 | 1.56 | 1.78 | 2.00 | 2.22 | 4.44 | 6.66 | 8.89 | 11.11 | |
| **50** | 0.20 | 0.40 | 0.60 | 0.80 | 1.00 | 1.20 | 1.40 | 1.60 | 1.80 | 2.00 | 4.00 | 6.00 | 8.00 | 10.00 | |
| **55** | 0.18 | 0.36 | 0.55 | 0.73 | 0.93 | 1.10 | 1.27 | 1.45 | 1.64 | 1.81 | 3.64 | 5.45 | 7.27 | 9.09 | |
| **60** | 0.17 | 0.33 | 0.50 | 0.67 | 0.83 | 1.00 | 1.17 | 1.33 | 1.50 | 1.67 | 3.33 | 5.00 | 6.67 | 8.33 | |
| **65** | 0.15 | 0.31 | 0.46 | 0.61 | 0.77 | 0.92 | 1.08 | 1.23 | 1.38 | 1.54 | 3.08 | 4.62 | 6.15 | 7.69 | |
| **70** | 0.14 | 0.29 | 0.43 | 0.57 | 0.71 | 0.86 | 1.00 | 1.14 | 1.28 | 1.42 | 2.85 | 4.28 | 5.71 | 7.14 | |
| **75** | 0.13 | 0.27 | 0.41 | 0.53 | 0.67 | 0.80 | 0.93 | 1.07 | 1.20 | 1.33 | 2.67 | 4.00 | 5.33 | 6.67 | |
| **80** | 0.13 | 0.25 | 0.38 | 0.50 | 0.63 | 0.75 | 0.89 | 1.00 | 1.13 | 1.25 | 2.50 | 3.75 | 5.00 | 6.25 | |
| **85** | 0.12 | 0.24 | 0.35 | 0.47 | 0.59 | 0.71 | 0.82 | 0.94 | 1.06 | 1.18 | 2.35 | 3.53 | 4.71 | 5.88 | |
| **90** | 0.11 | 0.22 | 0.33 | 0.44 | 0.56 | 0.67 | 0.78 | 0.89 | 1.00 | 1.11 | 2.22 | 3.33 | 4.44 | 5.55 | |
| **95** | 0.11 | 0.21 | 0.32 | 0.42 | 0.53 | 0.63 | 0.74 | 0.84 | 0.95 | 1.05 | 2.11 | 3.16 | 4.21 | 5.26 | |
| **100** | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.60 | 0.70 | 0.80 | 0.90 | 1.00 | 2.00 | 3.00 | 4.00 | 5.00 | |

# APPENDIX XI
# Package of practices for Rodent management in rice

1. Synchronous planting/harvesting
2. Bund trimming (>20 cm).
3. Regular trapping with local bamboo traps preferably one month after transplanting.
4. Clean cultivation to reduce alternate source of food, particularly weeding of tall weeds on bunds such as *Eleusine indica*.
5. Two applications of bromadiolone 0.005% as cake, first at panicle initiation stage and second at milky stage of rice crop.
6. Poison baiting is also essential for rodent pest management. Prebait prepared out of 1 kg cereal + 20 g vegetable oil + 10 g dried fish may be applied @ 10 g/live burrow for 3 days. This treatment should be followed by addition of Zinc phosphide @ 25 g/kg bait on the $4^{th}$ day.

# APPENDIX XII A
## Pesticides banned for manufacture, import and use in India

**Source**: Central Insecticides Board & Registration Committee, Directorate of Plant Protection, Quarantine & Storage, Ministry of Agriculture, Faridabad
(Web site: http://www.cibrc.nic.in/list_pest_bann.htm)

**A. Insecticides/ Acaricides/ Rodenticides**

1. Aldicarb
2. Aldrin
3. Benzene Hexachloride
4. Calcium Cyanide
5. Carbaryl
6. Chlorbenzilate
7. Chlordane
8. Chlorofenvinphos
9. Copper Aceto arsenite
10. Diazinon
11. Dieldrin
12. Endosulfron (vide ad-Interim order of the Supreme Court of India in the Writ Petition (Civil) No. 213 of 2011 dated 13th May, 2011 and finally disposed of dated 10th January, 2017)
13. Endrin
14. Ethyl Parathion
15. Ethylene Dibromide
16. Fenthion
17. Heptachlor
18. Lindane (Gamma-HCH)
19. Maleic Hydrazide
20. Menazon
21. Methyl Parathion
22. Metoxuron
23. Pentachlorophenol
24. Phenyl Mercury Acetate
25. Sodium Cyanide ( banned for Insecticidal purpose only vide S.O 3951(E) dated 8th August, 2018)
26. Tetradifon
27. Thiometon
28. Toxaphene(Camphechlor)
29. Trichloro acetic acid (TCA)

**B. Fungicides**

1. Ethyl Mercury Chloride

2. Pentachloro Nitrobenzene (PCNB)
3. Fenarimol
4. Tridemorph
5. Methoxy ethyl mercuric chloride
6. Benomyl

**C. Herbicides**

1. Nitrofen
2. Paraquat di-methyl sulphate
3. Metoxuron
4. Linuron
5. Sodium Methane assonate

**D. Nematicides**

1. Aldicarb
2. Diazinon
3. Dibromochloropropane
4. Ethylene Dibromide

# APPENDIX XII B
## Pesticides/ pesticide formulations banned for use but their manufacture is allowed for export

**A. Insecticides/ Acaricides/ Rodenticides**

1. Nicotin Sulfate

**B. Fungicide**

2. Captafol 80% Powder

# APPENDIX XII C
## Pesticides withdrawn

**A. Insecticides/ Acaricides/ Rodenticides**

1. Formothion
2. Nickel Chloride
3. Paradichlorobenzene (PDCB)
4. Warfarin

**B. Fungicides**

1. Ferbam

**C. Herbicides**

1. Dalapon
2. Simazine
3. Sirmate

# APPENDIX XII D
## Pesticides restricted for use in India

**A. Insecticides/ Acaricides/ Rodenticides**

1. **Aluminium Phosphide** (The Pest Control Operations withAluminium Phosphide may be undertaken only by Govt./Govt. undertakings / Govt. Organizations / pest control operators under the strict supervision of Govt. Experts or experts whose expertise is approved by the Plant Protection Advisor to Govt. of India except [1]Aluminium Phosphide 15 % 12 g tablet and [2]Aluminum Phosphide 6 % tablet.)
2. **Cypermethrin** (Cypermethrin 3 % Smoke Generator is to be used only through Pest Control Operators and not allowed to be used by the General Public.
3. **Diazinon** (Diazinon is banned for use in agriculture except for household use.)
4. **Dichloro Diphenyl Trichloroethane (DDT)** (The use of DDT for the domestic Public Health Programme is restricted up to 10,000 Metric Tonnes per annum, except in case of any major outbreak of epidemic. M/s Hindustan Insecticides Ltd., the sole manufacturer of DDT in the country may manufacture DDT for export to other countries for use in vector control for public health purpose.)
5. F**enitrothion** (The use of Fenitrothion is banned in Agriculture except for locust control in scheduled desert area and public health.)
6. **Fenthion** (The use of Fenthion is banned in Agriculture except for locust control, household and public health.)
7. **Methyl Parathion** (Methyl Parathion 50 EC and 2% DP formulations are banned for use on fruits and vegetables.)
8. **Monocrotophos** (Monocrotophos is banned for use on vegetables.)

**B. Fungicides**

1. **Captafol** (The use of Captafol as foliar spray is banned. Captafol shall be used only as seed dresser. The manufacture of Captafol 80 % powder for dry seed treatment (DS) is banned for use in the country except manufacture for export.)
2. **Dazomet** (The use of Dazomet is not permitted on Tea.)

**3. Methoxy Ethyl Mercuric Chloride** (MEMC) (The use of MEMC is banned completely except for seed treatment of potato and sugarcane.)

**C. Herbicides**

1. **Dazomet** (the use of Dazomet is not permitted in tea)
2. **Methyl bromide** (under strict supervision of experts in non crop situation)
3. **Trifluralin** (only in wheat but not to be used near the water basin)

**D. Nematicides**

1. **Dazomet** (The use of Dazomet is not permitted on Tea)
2. **Diazinon** (Diazinon is banned for use in agriculture except for household use.)

3. **Methyl Bromide** (Methyl Bromide may be used onlyby Govt./Govt. undertakings/ Govt. Organizations / Pest control operators under the strict supervision of Govt. Experts or Experts whose expertise is approved by the Plant Protection Advisor to Govt. of India.)

# APPENDIX XII E

## MINISTRY OF AGRICULTURE AND FARMERS WELFARE

## (Department of Agriculture and Farmers Welfare)

(CG-DL-E-16022023-243654)

### NOTIFICATION

New Delhi, the 2nd February, 2023

### List of Prohibited Insecticides

| Sl No. | Name of Insecticide | Decision of Central Government |
|---|---|---|
| **1** | Dicofol | The registration, import, manufacture, formulation, transport, sale is prohibited and its use is completely banned from the date of publication of this order. |
| 2 | Dinocap | The registration, import, manufacture, formulation, transport, sale is prohibited and its use is completely banned from the date of publication of this order. |
| 3 | Methomyl | The registration, import, manufacture, formulation, transport, sale is prohibited and its use is completely banned from the date of publication of this order. |
| **Names of crops to be omitted from approved usage** | | |
| 1 | Carbofuran | All other formulations of Carbofuran except Carbofuran three percent Encapsulated granule (CG) along with the crop labels may be stopped from use. |
| 2 | Malathion | Sorghum, Pea, Soybean, Castor, Sunflower, Bhindi, Brinjal, Cauliflower, Radish, Turnip, Tomato, Apple, Mango and Grape. |
| 3 | Monocrotophos | All other formulations of this pesticide except Monocrotophos fifteen percent Water soluble granules (SG) may be stopped from use in respective label crops. |
| 4 | Quinalphos | Jute, Cardamom and Sorghum. |
| 5 | Mancozeb | Guava, Jowar and Tapioca. |
| **6** | Oxyfluorfen | Potato and Groundnut |
| 7 | Dimethoate | Label deletion of fruits and vegetables that are consumed as raw food items. |
| 8 | Chlorpyriphos | Ber, Citrus and Tobacco. |

# APPENDIX XIII

## Pest Management Rating of Commonly Used Insecticides

| Insecticide | Mammalian Toxicity rating | Non-target Toxicity rating | | | | Environmental Persistence rating | Overall Rating |
|---|---|---|---|---|---|---|---|
| | | Fish | Bird | Bee | Average | | |
| Azinphos-methyl | 4 | 3 | 2 | 4 | 3.0 | 3 | 10.0 |
| *Bt* | 1 | 1 | 1 | 1 | 1.0 | 31 | 3.0 |
| Carbaryl | 2 | 1 | 1 | 4 | 2.0 | 2 | 6.0 |
| Carbofuran | 5 | 2 | 5 | 5 | 4.0 | 3 | 12.0 |
| Carbophenothin | 4 | 2 | 4 | 4 | 3.3 | 2 | 9.3 |
| Chlorpyriphos | 3 | 3 | 3 | 5 | 3.7 | 3 | 9.7 |
| Cryolite | 1 | 1 | 1 | 2 | 1.3 | 4 | 7.3 |
| Demeton | 5 | 2 | 5 | 2 | 3.0 | 2 | 10.0 |
| Diazinon | 3 | 2 | 5 | 4 | 3.7 | 3 | 9.7 |
| Dicofol | 2 | 1 | 2 | 1 | 1.3 | 4 | 7.3 |
| Dischlorvos | - | Toxic | - | Toxic | | | |
| Diflunenzuron | 1 | 1 | 1 | 4 | 2.0 | 4 | 7.0 |
| Dimethoate | 3 | 1 | 4 | 5 | 3.3 | 2 | 8.3 |
| Endosulfan | | 4 | 4 | 2 | 22.7 | 3 | 9.7 |
| EPN | 4 | 2 | 3 | 4 | 3.0 | 4 | 11.0 |
| Ethion | 3 | 2 | 3 | - | - | 2 | 7.0 |
| Fenvalerate, | 2 | 4 | 2 | 5 | 3.7 | 2 | 7.7 |
| Permethrin | | | | | | | |
| Malathion | 2 | 2 | 1 | 4 | 2.3 | 1 | 5.3 |
| Methomyl | 4 | 4 | 3 | 4 | 3.7 | 2 | 9.7 |
| Methoprene | 1 | 1 | 1 | 2 | 1.3 | 2 | 4.3 |
| Methoxychlor | 1 | 3 | 2 | 1 | 2.0 | 2 | 5.0 |
| Mevinphos | 5 | 3 | 5 | 4 | 4.0 | 1 | 10.0 |
| Naled | 2 | 2 | 3 | 4 | 3.0 | 1 | 6.0 |
| Ovex | 1 | 2 | 1 | 1 | 1.3 | 4 | 6.3 |
| Oxydemeton | 3 | 2 | 4 | 2 | 2.7 | 2 | 7.7 |
| methyl | | | | | | | |
| Phorate | 5 | 4 | 5 | 2 | 3.7 | 3 | 11.7 |
| Quinalphos | - | Safe | - | Toxic | - | - | |
| Phosphamidon | 4 | 1 | 5 | 3 | 3.0 | 2 | 9.0 |
| Stirofos | 1 | 4 | 1 | 4 | 3.0 | 1 | 5.0 |
| TEPP | 5 | 2 | 5 | 5 | 4.0 | 1 | 10.0 |
| Trichlorfon | 2 | 1 | 2 | 1 | 1.3 | 1 | 4.3 |

**N.B.**

A. Lower the rating safer the insecticides.

B. The insecticides *viz.*, endosulfan, phorate, stirofos, fenvalerate, methomyl are highly toxic to fish; hence their use should be restricted in fish cum paddy culture.

C. Insecticides such as azinphos-methyl, carbaryl, carbufuran, carbophenothionm, chloropyriphos, diazinonm, fenvalerate, diflubenzuron, dimethoate, mevinphos, EPN, methomyl, malathion, naled, stirophos, TEPP, dichlorovos and quinalphos are highly toxic to bees, hence their use should be restricted in the oilseeds, vegetables and fruit orchards.

D. All insecticides mentioned in the list do not necessarily constitute our recommendation and the rating is based on available literature.

# APPENDIX-XIV(A)

## Ready Reckoner for Fertilizer Recommendation for HYV of Transplanted Winter Rice for Target Yield of 60q/ha

| S.No. | STV N | STV P | STV K | N | $P_2O_5$ | $K_2O$ | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|---|---|---|
| | Kg/ha | | | Kg/ha | | | Kg/bigha | | |
| 1 | 150.00 | 2.50 | 110.00 | 192.8 | 33.3 | 101.0 | 25.7 | 4.4 | 13.5 |
| 2 | 152.00 | 4.50 | 115.00 | 191.3 | 30.0 | 92.4 | 25.5 | 4.0 | 12.3 |
| 3 | 154.00 | 6.50 | 120.00 | 189.8 | 26.8 | 83.9 | 25.3 | 3.6 | 11.2 |
| 4 | 156.00 | 8.50 | 125.00 | 188.3 | 23.5 | 75.4 | 25.1 | 3.1 | 10.1 |
| 5 | 158.00 | 10.50 | 130.00 | 186.8 | 20.2 | 66.9 | 24.9 | 2.7 | 8.9 |
| 6 | 160.00 | 12.50 | 135.00 | 185.3 | 17.0 | 58.4 | 24.7 | 2.3 | 7.8 |
| 7 | 162.00 | 14.50 | 140.00 | 183.8 | 13.7 | 49.9 | 24.5 | 1.8 | 6.7 |
| 8 | 164.00 | 16.50 | 145.00 | 182.2 | 10.4 | 41.4 | 24.3 | 1.4 | 5.5 |
| 9 | 166.00 | 18.50 | 150.00 | 180.7 | 7.2 | 32.9 | 24.1 | 1.0 | 4.4 |
| 10 | 168.00 | 20.50 | 155.00 | 179.2 | 3.9 | 24.4 | 23.9 | 0.5 | 3.2 |
| 11 | 170.00 | 22.50 | 160.00 | 177.7 | 0.6 | 15.9 | 23.7 | 0.1 | 2.1 |
| 12 | 172.00 | 24.50 | 165.00 | 176.2 | 0.0 | 7.4 | 23.5 | 0.0 | 1.0 |

# APPENDIX-XIV(B)

## Ready Reckoner for Fertilizer Recommendation for HYV of Transplanted Winter Rice for Target Yield of 70q/ha

| S.No. | STV N | STV P | STV K | N | $P_2O_5$ | $K_2O$ | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|---|---|---|
| | Kg/ha | | | Kg/ha | | | Kg/bigha | | |
| 1 | 150.00 | 2.50 | 110.00 | 245.0 | 40.0 | 152.1 | 32.7 | 5.3 | 20.3 |
| 2 | 152.00 | 4.50 | 115.00 | 243.5 | 36.8 | 143.6 | 32.5 | 4.9 | 19.1 |
| 3 | 154.00 | 6.50 | 120.00 | 242.0 | 33.5 | 135.1 | 32.3 | 4.5 | 18.0 |
| 4 | 156.00 | 8.50 | 125.00 | 240.4 | 30.2 | 126.6 | 32.1 | 4.0 | 16.9 |
| 5 | 158.00 | 10.50 | 130.00 | 238.9 | 27.0 | 118.1 | 31.9 | 3.6 | 15.7 |
| 6 | 160.00 | 12.50 | 135.00 | 237.4 | 23.7 | 109.6 | 31.7 | 3.2 | 14.6 |
| 7 | 162.00 | 14.50 | 140.00 | 235.9 | 20.4 | 101.1 | 31.5 | 2.7 | 13.5 |
| 8 | 164.00 | 16.50 | 145.00 | 234.4 | 17.1 | 92.5 | 31.3 | 2.3 | 12.3 |
| 9 | 166.00 | 18.50 | 150.00 | 232.9 | 13.9 | 84.0 | 31.1 | 1.9 | 11.2 |
| 10 | 168.00 | 20.50 | 155.00 | 231.4 | 10.6 | 75.5 | 30.9 | 1.4 | 10.1 |
| 11 | 170.00 | 22.50 | 160.00 | 229.9 | 7.3 | 67.0 | 30.7 | 1.0 | 8.9 |
| 12 | 172.00 | 24.50 | 165.00 | 228.4 | 4.1 | 58.5 | 30.5 | 0.5 | 7.8 |

# APPENDIX-XIV(C)

## Ready reckoner for target = 50 q/ha, Winter Rice Numali without FYM

| Sl.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------kg/ha---------- | | |
| 1 | 180 | 12 | 70 | 55.9 | 46.0 | 81.5 |
| 2 | 182 | 14 | 75 | 53.7 | 41.2 | 75.2 |
| 3 | 184 | 16 | 80 | 51.4 | 36.5 | 68.9 |
| 4 | 186 | 18 | 85 | 49.2 | 31.7 | 62.5 |
| 5 | 188 | 20 | 90 | 47.0 | 27.0 | 56.2 |
| 6 | 190 | 22 | 95 | 44.8 | 22.2 | 49.9 |
| 7 | 192 | 24 | 100 | 42.6 | 17.5 | 43.5 |
| 8 | 194 | 26 | 105 | 40.4 | 12.7 | 37.2 |
| 9 | 196 | 28 | 110 | 38.2 | 8.0 | 30.9 |
| 10 | 198 | 30 | 115 | 36.0 | 3.2 | 24.6 |
| 11 | 200 | 32 | 120 | 33.8 | 0.0 | 18.2 |
| 12 | 202 | 34 | 125 | 31.6 | 0.0 | 11.9 |

STV-Soil Test Value for N,P,K

# APPENDIX-XIV(D)

## Ready reckoner for target = 50 q/ha, Winter Rice Numali with 5t FYM/ha

| Sl.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------kg/ha---------- | | |
| 1 | 180 | 12 | 70 | 34.4 | 41.5 | 73.9 |
| 2 | 182 | 14 | 75 | 32.1 | 36.8 | 67.5 |
| 3 | 184 | 16 | 80 | 29.9 | 32.0 | 61.2 |
| 4 | 186 | 18 | 85 | 27.7 | 27.3 | 54.9 |
| 5 | 188 | 20 | 90 | 25.5 | 22.5 | 48.5 |
| 6 | 190 | 22 | 95 | 23.3 | 17.8 | 42.2 |
| 7 | 192 | 24 | 100 | 21.1 | 13.0 | 35.9 |
| 8 | 194 | 26 | 105 | 18.9 | 8.2 | 29.6 |
| 9 | 196 | 28 | 110 | 16.7 | 3.5 | 23.2 |
| 10 | 198 | 30 | 115 | 14.5 | 0.0 | 16.9 |
| 11 | 200 | 32 | 120 | 12.3 | 0.0 | 10.6 |
| 12 | 202 | 34 | 125 | 10.1 | 0.0 | 4.2 |

STV-Soil Test Value for N,P,K

# APPENDIX-XIV(E)

## Ready reckoner for target = 60 q/ha, Winter Rice Numali without FYM

| Sl.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------kg/ha---------- | | |
| 1 | 180 | 12 | 70 | 106.8 | 60.9 | 115.5 |
| 2 | 182 | 14 | 75 | 104.6 | 56.2 | 109.2 |
| 3 | 184 | 16 | 80 | 102.4 | 51.4 | 102.9 |
| 4 | 186 | 18 | 85 | 100.2 | 46.6 | 96.6 |
| 5 | 188 | 20 | 90 | 98.0 | 41.9 | 90.2 |
| 6 | 190 | 22 | 95 | 95.8 | 37.1 | 83.9 |
| 7 | 192 | 24 | 100 | 93.5 | 32.4 | 77.6 |
| 8 | 194 | 26 | 105 | 91.3 | 27.6 | 71.2 |
| 9 | 196 | 28 | 110 | 89.1 | 22.9 | 64.9 |
| 10 | 198 | 30 | 115 | 86.9 | 18.1 | 58.6 |
| 11 | 200 | 32 | 120 | 84.7 | 13.4 | 52.3 |
| 12 | 202 | 34 | 125 | 82.5 | 8.6 | 45.9 |

STV-Soil Test Value for N,P,K

# APPENDIX-XIV(F)

## Ready reckoner for target = 60 q/ha, Winter Rice Numali with 5t FYM/ha

| Sl.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------kg/ha---------- | | |
| 1 | 180 | 12 | 70 | 85.3 | 56.4 | 107.9 |
| 2 | 182 | 14 | 75 | 83.1 | 51.7 | 101.6 |
| 3 | 184 | 16 | 80 | 80.9 | 46.9 | 95.2 |
| 4 | 186 | 18 | 85 | 78.7 | 42.2 | 88.9 |
| 5 | 188 | 20 | 90 | 76.5 | 37.4 | 82.6 |
| 6 | 190 | 22 | 95 | 74.2 | 32.7 | 76.2 |
| 7 | 192 | 24 | 100 | 72.0 | 27.9 | 69.9 |
| 8 | 194 | 26 | 105 | 69.8 | 23.2 | 63.6 |
| 9 | 196 | 28 | 110 | 67.6 | 18.4 | 57.2 |
| 10 | 198 | 30 | 115 | 65.4 | 13.6 | 50.9 |
| 11 | 200 | 32 | 120 | 63.2 | 8.9 | 44.6 |
| 12 | 202 | 34 | 125 | 61.0 | 4.1 | 38.3 |

STV-Soil Test Value for N,P,K

# APPENDIX-XV(A)

## Ready reckoner for target = 50 q/ha, Maize without FYM

| S.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------Kg/ha---------- | | |
| 1 | 250 | 15 | 150 | 101.8 | 61.4 | 66.3 |
| 2 | 252 | 17 | 155 | 100.6 | 58.6 | 63.2 |
| 3 | 254 | 19 | 160 | 99.3 | 55.7 | 60.1 |
| 4 | 256 | 21 | 165 | 98.0 | 52.8 | 57.0 |
| 5 | 258 | 23 | 170 | 96.8 | 50.0 | 53.9 |
| 6 | 260 | 25 | 175 | 95.5 | 47.1 | 50.7 |
| 7 | 262 | 27 | 180 | 94.3 | 44.2 | 47.6 |
| 8 | 264 | 29 | 185 | 93.0 | 41.4 | 44.5 |
| 9 | 266 | 31 | 190 | 91.8 | 38.5 | 41.4 |
| 10 | 268 | 33 | 195 | 90.5 | 35.6 | 38.2 |
| 11 | 270 | 35 | 200 | 89.2 | 32.8 | 35.1 |
| 12 | 272 | 37 | 205 | 88.0 | 29.9 | 32.0 |

STV-Soil Test Value for N,P,K

# APPENDIX-XV(B)

## Ready reckoner for target = 50 q/ha, Maize with 5t FYM/ha

| S.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------Kg/ha---------- | | |
| 1 | 250 | 15 | 150 | 74.5 | 55.4 | 55.4 |
| 2 | 252 | 17 | 155 | 73.3 | 52.5 | 52.3 |
| 3 | 254 | 19 | 160 | 72.0 | 49.6 | 49.1 |
| 4 | 256 | 21 | 165 | 70.8 | 46.8 | 46.0 |
| 5 | 258 | 23 | 170 | 69.5 | 43.9 | 42.9 |
| 6 | 260 | 25 | 175 | 68.2 | 41.0 | 39.8 |
| 7 | 262 | 27 | 180 | 67.0 | 38.2 | 36.6 |
| 8 | 264 | 29 | 185 | 65.7 | 35.3 | 33.5 |
| 9 | 266 | 31 | 190 | 64.5 | 32.4 | 30.4 |
| 10 | 268 | 33 | 195 | 63.2 | 29.6 | 27.3 |
| 11 | 270 | 35 | 200 | 62.0 | 26.7 | 24.2 |
| 12 | 272 | 37 | 205 | 60.7 | 23.8 | 21.0 |
| 13 | 250 | 15 | 150 | 74.5 | 55.4 | 55.4 |
| 14 | 252 | 17 | 155 | 73.3 | 52.5 | 52.3 |

STV-Soil Test Value for N,P,K

# APPENDIX-XV(C)

## Ready reckoner for target = 60 q/ha, Maize without FYM

| S.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------Kg/ha---------- | | |
| 1 | 250 | 15 | 150 | 153.6 | 78.0 | 98.4 |
| 2 | 252 | 17 | 155 | 152.4 | 75.1 | 95.2 |
| 3 | 254 | 19 | 160 | 151.1 | 72.3 | 92.1 |
| 4 | 256 | 21 | 165 | 149.9 | 69.4 | 89.0 |
| 5 | 258 | 23 | 170 | 148.6 | 66.5 | 85.9 |
| 6 | 260 | 25 | 175 | 147.3 | 63.7 | 82.7 |
| 7 | 262 | 27 | 180 | 146.1 | 60.8 | 79.6 |
| 8 | 264 | 29 | 185 | 144.8 | 57.9 | 76.5 |
| 9 | 266 | 31 | 190 | 143.6 | 55.1 | 73.4 |
| 10 | 268 | 33 | 195 | 142.3 | 52.2 | 70.2 |
| 11 | 270 | 35 | 200 | 141.0 | 49.4 | 67.1 |
| 12 | 272 | 37 | 205 | 139.8 | 46.5 | 64.0 |

STV-Soil Test Value for N,P,K

# APPENDIX-XV(D)

## Ready reckoner for target = 60 q/ha, Maize with 5t FYM/ha

| S.No. | STV N (kg/ha) | STV P (kg $P_2O_5$/ha) | STV K (kg $K_2O$/ha) | N | $P_2O_5$ | $K_2O$ |
|---|---|---|---|---|---|---|
| | | | | ----------Kg/ha---------- | | |
| 1 | 250 | 15 | 150 | 126.3 | 71.9 | 87.4 |
| 2 | 252 | 17 | 155 | 125.1 | 69.1 | 84.3 |
| 3 | 254 | 19 | 160 | 123.8 | 66.2 | 81.1 |
| 4 | 256 | 21 | 165 | 122.6 | 63.3 | 78.0 |
| 5 | 258 | 23 | 170 | 121.3 | 60.5 | 74.9 |
| 6 | 260 | 25 | 175 | 120.1 | 57.6 | 71.8 |
| 7 | 262 | 27 | 180 | 118.8 | 54.7 | 68.7 |
| 8 | 264 | 29 | 185 | 117.5 | 51.9 | 65.5 |
| 9 | 266 | 31 | 190 | 116.3 | 49.0 | 62.4 |
| 10 | 268 | 33 | 195 | 115.0 | 46.2 | 59.3 |
| 11 | 270 | 35 | 200 | 113.8 | 43.3 | 56.2 |
| 12 | 272 | 37 | 205 | 112.5 | 40.4 | 53.0 |
| 13 | 250 | 15 | 150 | 126.3 | 71.9 | 87.4 |
| 14 | 252 | 17 | 155 | 125.1 | 69.1 | 84.3 |

STV-Soil Test Value for N,P,K